

VU Medical Zone

ADMIN:

AMAAAN KHAN

**ECOLOGY, BIODIVERSITY & EVOLUTION I
BT101 MIDTERM PPT
LECTURE 1 TO 82**

What is Ecosystem?

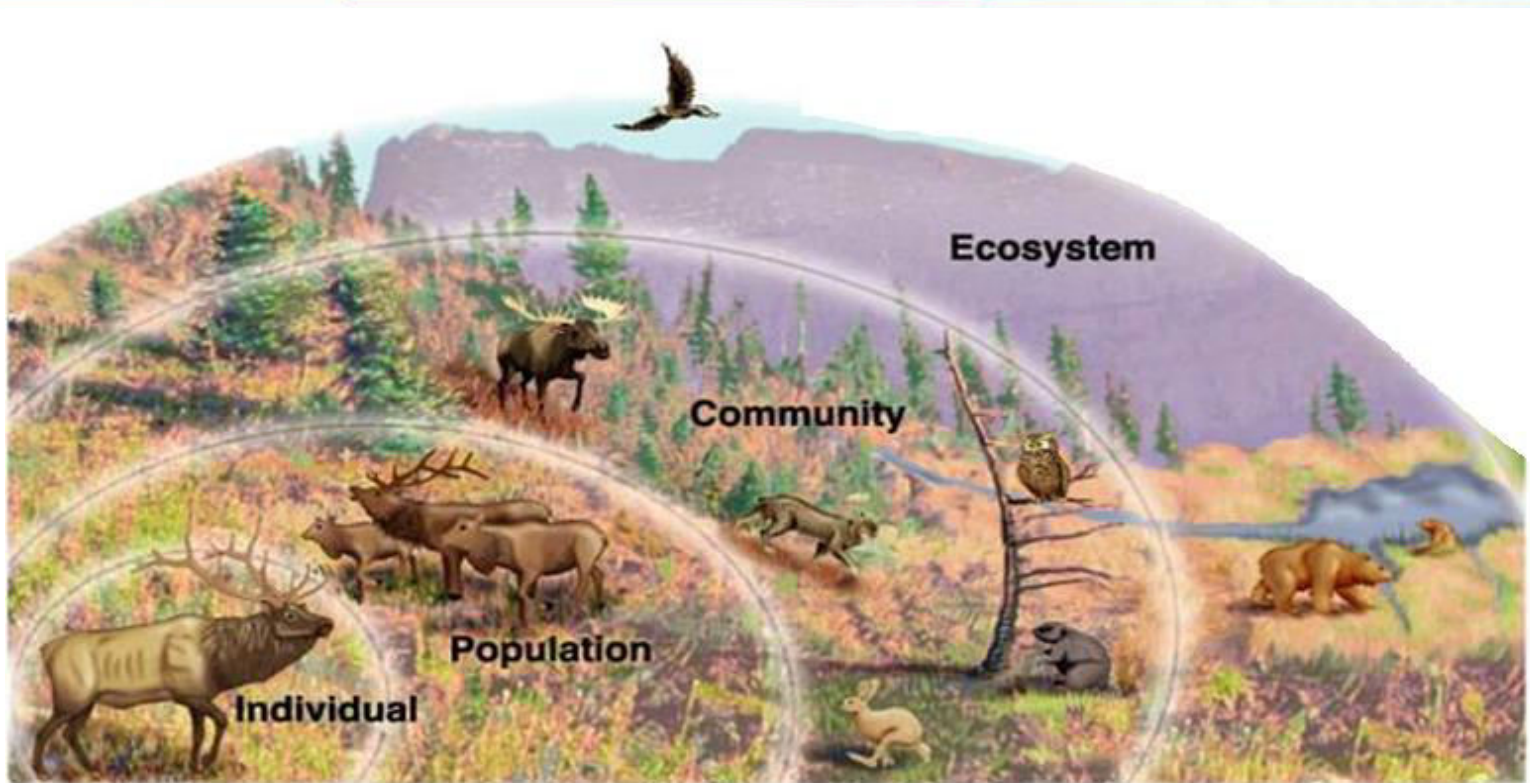
Ecosystem

- **Community of living organisms in conjunction of its non-living components**
- **Non-living components**
- **Interaction as a system between both components**

What is Ecosystem?

- **Population** is group of individuals belonging to same species occupying same space in particular time.
- **Community** is group of populations occupying same space in particular time.

What is Ecosystem?



<https://socratic.org/questions/how-does-an-ecosystem-differ-from-a-community>

What is Ecosystem?

Control of Ecosystem

- External

Climate, topography

- Internal

Types of species, population size, disease, gene pool, reproduction rate

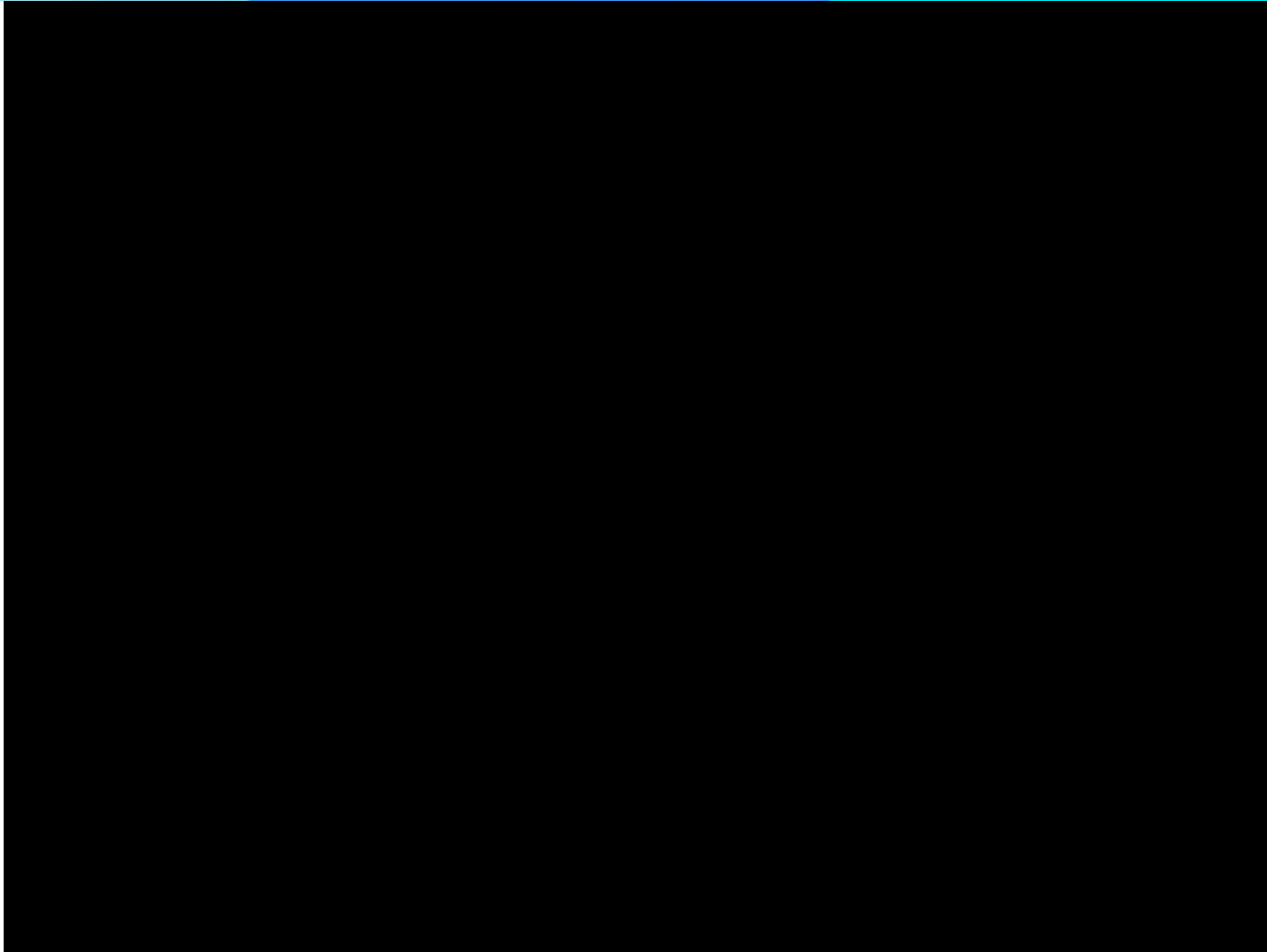
External factors are independent of internal factors.

What is Ecosystem?

Biome

- Distinct biological communities
- Formed in response to specific external factors in that environment

What is Ecosystem?



<https://www.youtube.com/watch?v=hIyoZlyPPDg>

What is Ecosystem?

- Internal factors are controlled by feedback loop

Example:

Population size is controlled by availability of resources.

- Ecosystem – a dynamic entity
- Subjected to disturbances and recovery (succession)

END

Flow of Energy in Ecosystem

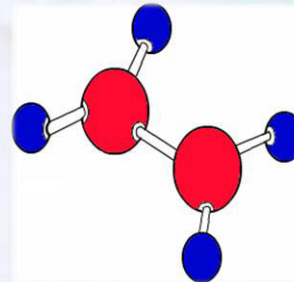
Learning Objectives

- Introduction of energy dynamics in ecosystem
- Use of energy by organisms
- Type of useable energy

Flow of Energy in Ecosystem

Energy

- Energy exists in several different forms
 - Light
 - Chemical-bond energy
 - Motion
 - Heat
- All organisms need energy to function

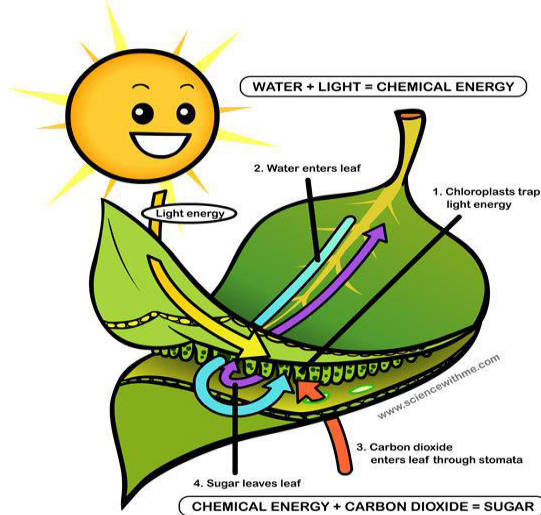


Flow of Energy in Ecosystem

Principles

- Energy is neither created nor destroyed in the biosphere.
- Organisms cannot convert heat to any of the other forms of energy.
- They only convert some chemical-bond or light energy to heat.

Flow of Energy in Ecosystem



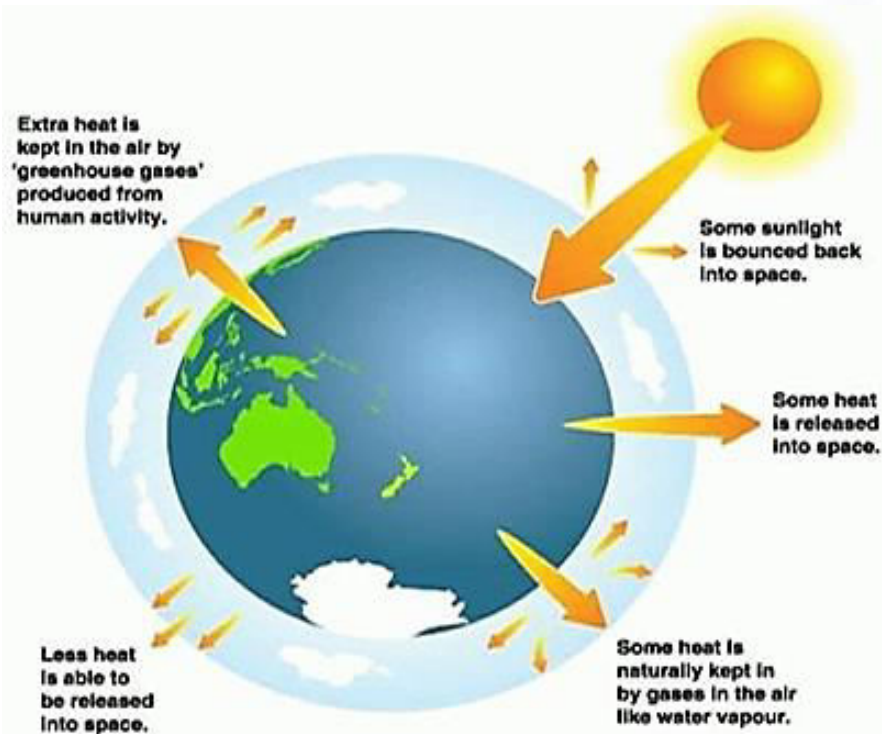
<https://mayapeacockapbiology.weebly.com/energy-transfer.html>



<https://thetechjournal.com/science/plants-become-defensive-when-attacked.xhtml>

- Earth is an open system having constant and equal exposure to light.
- Plants use light energy in photosynthesis.
- They convert light energy into chemical bond energy.
- Animals use chemical bond energy to make organic molecules.

Flow of Energy in Ecosystem



<https://climatiquechange.weebly.com/>

- Light and chemical-bond energy are partially converted to heat at every step.
- Heat is radiated into outer space at invisible infrared wavelengths.
- Earth's incoming and outgoing flows of radiant energy must be equal for global temperature to stay constant.

Producers

Learning Objectives

- Concept of producers
- Types of producers
- Photosynthesis
- Chemosynthesis

Producers

Autotrophs

- Synthesis of the organic compounds of their bodies from inorganic precursors such as:
 - CO_2 , H_2O , NO_3
- using energy from an abiotic source
 - Light
 - inorganic oxidation reactions

Producers



<https://asknature.org/strategy/photosynthesis-makes-useful-organic-compounds-out-of-co2/#.XKcbMZgzY2w>

www.designmate.com

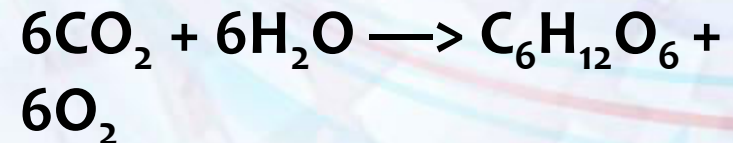


Types of Autotrophs

1. Photoautotrophs

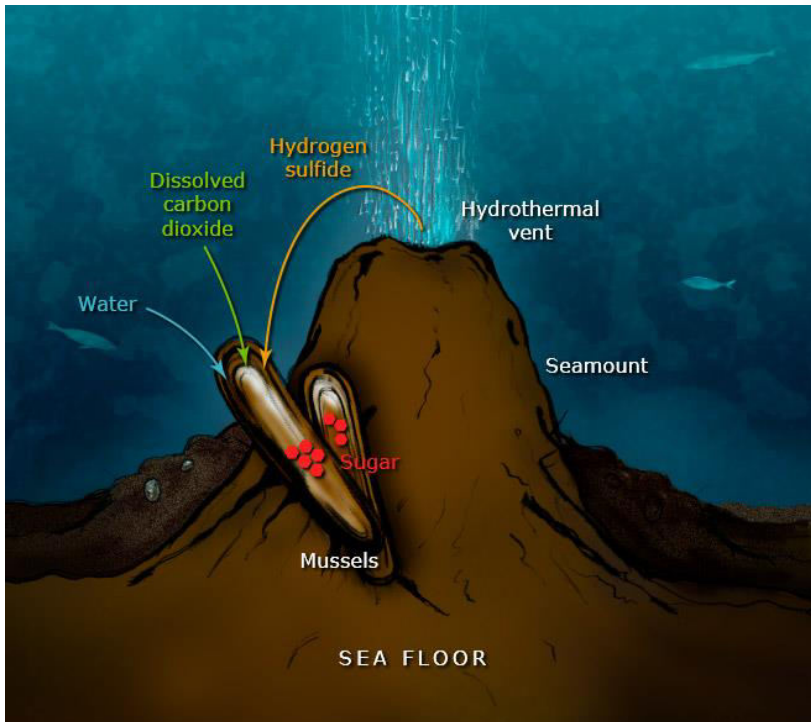
- Photosynthetic organisms
- Examples: plants, algae, and cyanobacteria

- Photosynthesis



- Main input of free energy into the biosphere

Producers



<https://teara.govt.nz/en/diagram/8960/photosynthesis-and-chemosynthesis>
<https://teara.govt.nz/en/diagram/8960/photosynthesis-and-chemosynthesis>

Types of Autotrophs

2. Chemoautotrophs

- Occurs in dark habitats
- They obtain energy by means of inorganic oxidation reactions.
- Convert toxic molecules into glucose
- Example: Microbes, mussels

Producers



<https://www.youtube.com/watch?v=jcqYhM7gMog>

Consumers

Learning Objectives

- **Concept of consumers**
- **Types of consumers**

Consumers

Heterotrophs

- Cannot synthesize organic compounds from inorganic precursors.
- Eat organisms from a different population for energy
- Liberating chemical-bond energy for metabolic use in glycolysis, Krebs's cycle and electron transport chain

Example: Animals

Consumers

Types of Consumers



<https://www.popsci.com/most-powerful-animals-ranked>

1. **Herbivore**
2. **Carnivore**
3. **Omnivore**

Consumers



<https://thevetisin.com/carnivores-omnivores-herbivores/>

1. Herbivore

- Primary food source is plant-based.
- Examples: vertebrates as well as invertebrates
- Digestive systems capable of handling large amounts of plant material.
- There are eight types of herbivory depending upon different feeding habits.

Consumers

Feeding Strategy	Diet	Example
Algivores	Algae	krill, crabs, sea snail, sea urchin, parrotfish, surgeonfish, flamingo
Frugivores	Fruit	Ruffed lemurs
Folivores	Leaves	Koalas
Nectarivores	Nectar	Honey possum
Granivores	Seeds	Hawaiian honeycreepers
Palynivores	Pollen	Bees
Mucivores	Plant fluids	Aphids
Xylophages	Wood	Termites

Consumers

2. Carnivore

- Primary food source is animal-based.
- Examples: vertebrates as well as invertebrates
- Digestive systems is more efficient in protein digestion.



<https://sco.wikipedia.org/wiki/Carnivora>

Consumers



<https://www.drcarney.com/blog/entry/omnivore>



https://en.wikipedia.org/wiki/File:Light_sussex_hen.jpg

2. Omnivore

- Primary food source is both plant and animal-based.
- Examples: human, bear, cockroach, chicken
- Digest carbohydrates, protein, fat, and fiber, and metabolize the nutrients

Consumers

3. Omnivore

Types

- **Frugivores**: wolves and orangutans
- **Insectivores**: swallows and pink fairy armadillos
- **Granivores**: large ground finches and mice

END

Decomposers

Learning Objectives

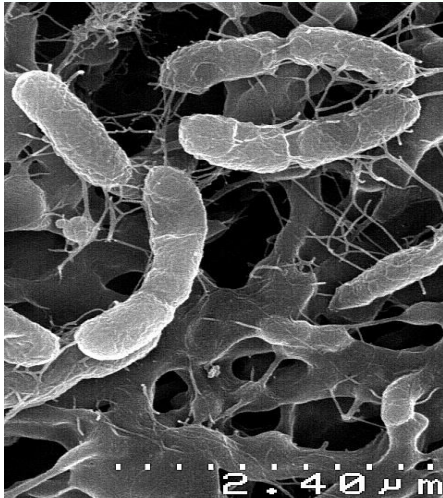
- **Concept of decomposers**
- **Types of decomposers**
- **Process of decomposition**

Decomposers

Heterotrophs

- Break down dead or decaying organisms
- Absorb nutrients directly through chemical and biological processes without ingestion.

Decomposers



<https://en.wikipedia.org/wiki/Chemosynthesis>



<http://fungiperth.org.au/>

Heterotrophs

- **Types**
 - Fungi
 - Bacteria

Decomposers

Fungi

- Primary decomposer
- Release digestive enzymes by their hyphae
- Absorb organic matter and release CO₂



<https://www.youtube.com/watch?v=i9T727tz7FA>

Decomposers

Decomposition

1. Autolysis

Breaking down of tissues by the body's own internal chemicals and enzymes

2. Putrefaction

- Breakdown of tissues by bacteria.
- release compounds such as cadaverine and putrescine causing decaying odor

Decomposers

Decomposition

- Proteolysis releases nitrogen and phosphorous
- Carbohydrate breakdown
 - **Aerobic** by fungi and bacteria
 - **Anaerobic** by bacteria
- Anerobic degradation of lipids
- Nucleic acids and bones degradation

Decomposers

Decomposition

- Plays important role in nitrogen and carbon cycles

END

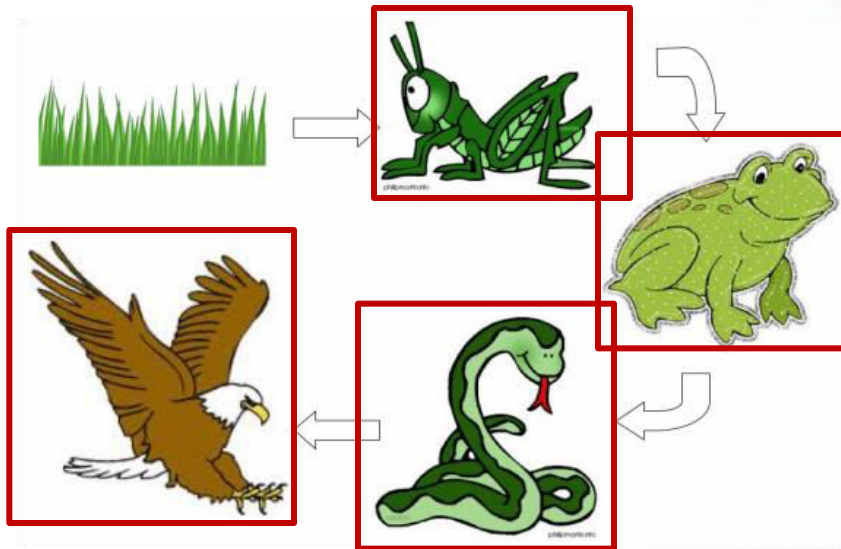
Flow of Energy in Ecosystem

Feeding Levels

The background of the slide features a large, faint, abstract graphic on the right side. It consists of several concentric circles and a series of flowing, curved lines in shades of blue, red, and green, creating a sense of dynamic movement and energy flow.

Flow of Energy in Ecosystem

Feeding Levels



<https://ecosystem300.weebly.com/food-chains.html>

- Food energy passes through different sequence.
- Example: plants → grass hopper → frog → snake → hawk
- Many plant species eaten by insects
- Many insects species eaten by frogs

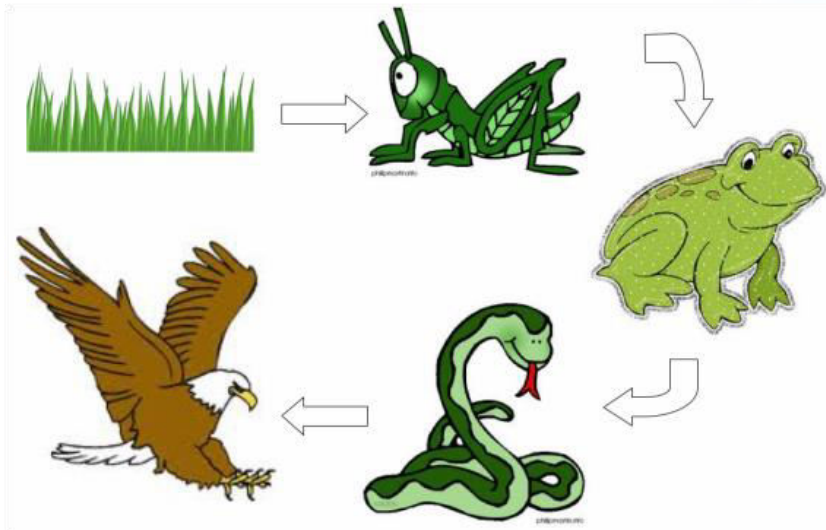
Trophic Level and Food Chain

Trophic Level

- Trophic level of an organism is position it occupies in a food chain.

Food Chain

- Linear network of links to pass energy in form of food
- Simplified abstractions of real food webs, but complex in their dynamics



<https://ecosystem300.weebly.com/food-chains.html>

Flow of Energy in Ecosystem

Feeding Levels

- This complexity is organized by ecologists.
- Recognized limited number of feeding or trophic levels

Trophic Level and Food Chain

Food Chain

- Food energy passes through different sequence.

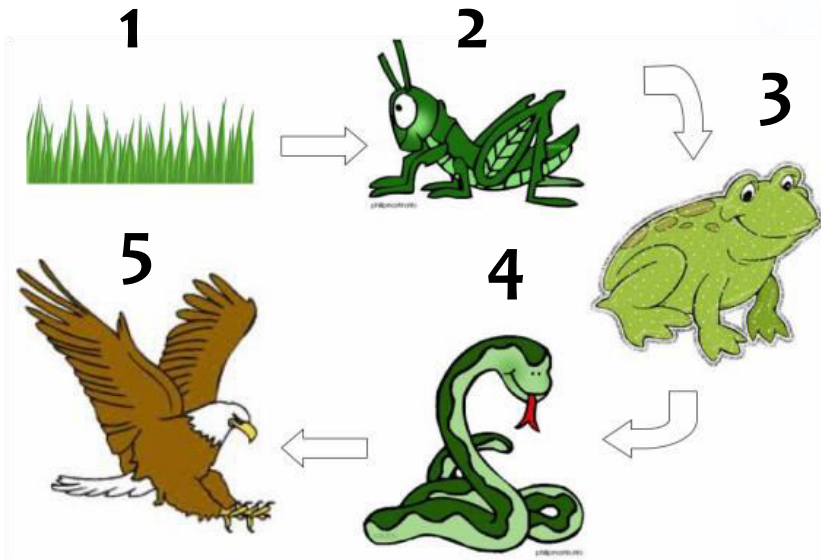
• **Level 1:** Primary producer

Level 2: Primary consumer

Level 3: Secondary consumer

Level 4: Tertiary Consumer

Level 5: Apex Predator



<https://ecosystem300.weebly.com/food-chains.html>

Trophic Level and Food Chain

A food chain also shows how the organisms are related with each other by the food they eat. Each level of a food chain represents a different trophic level. Food chain only follows a direct, linear pathway of one animal at a time.

Food Chain

- **Detritivore and decomposers can be at trophic level as well.**
- **They return nutrients back in soil to be available for producers.**

Flow of Energy in Ecosystem



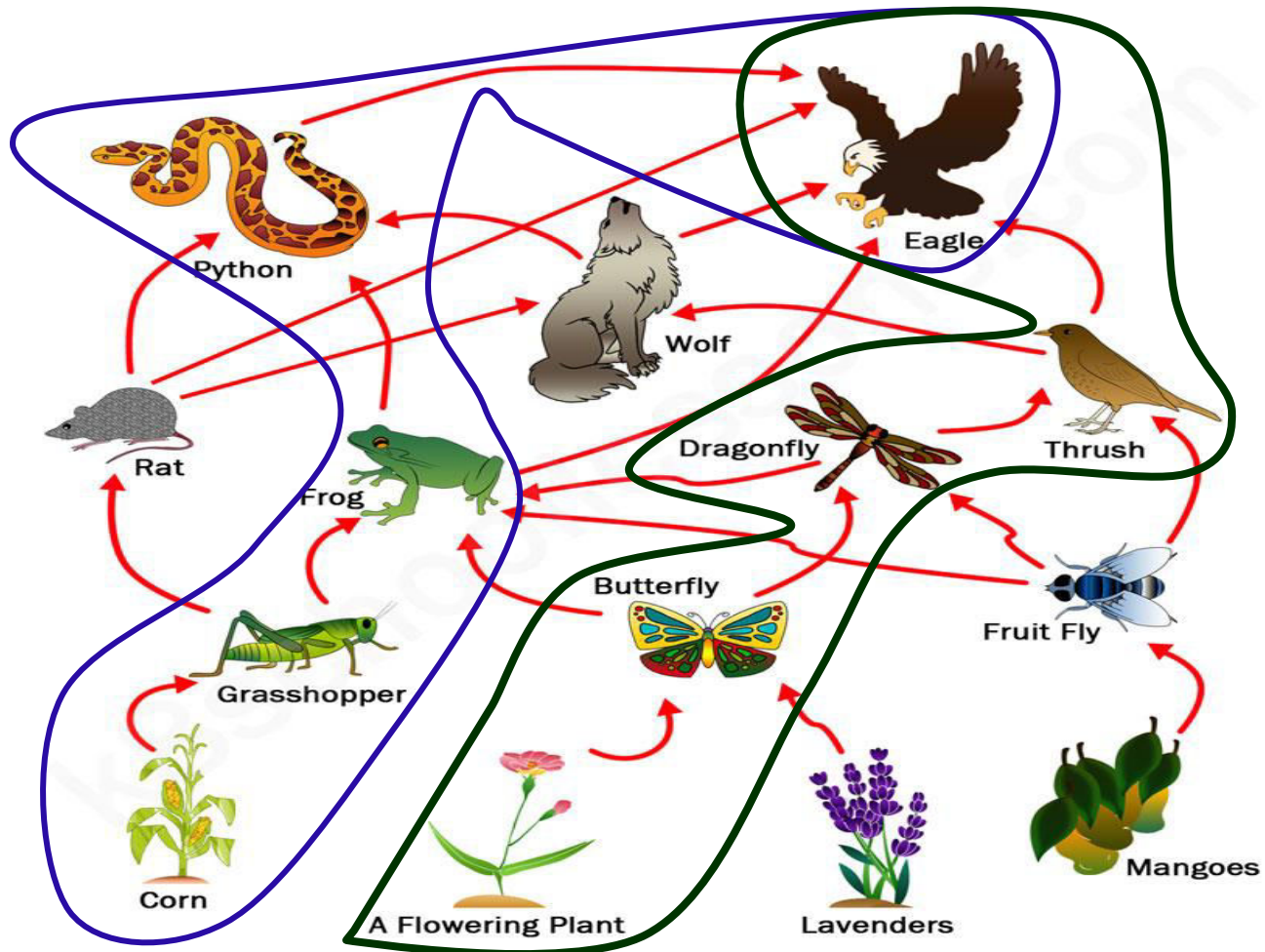
<http://youtube.com/watch?v=oZOvqYypOuo>

Food Web

Food Web

- **Complexly interconnected food chains in an ecological community**
- **Represent tiny portion of the complexity of real ecosystems**

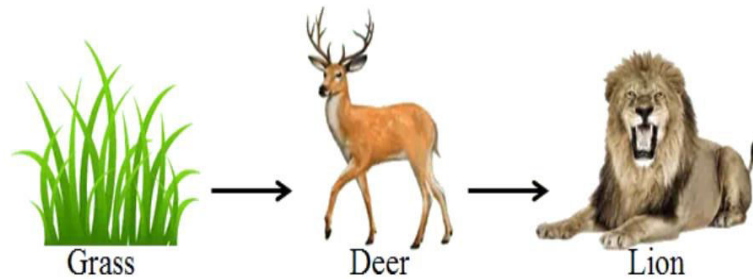
Food Web



<https://k8schoollessons.com/food-chains-food-webs/>

Food Web

Consumer-Resource system



<https://www.doorsteptutor.com/Exams/NSTSE/Class-3/Questions/Topic-General-Science-1/Subtopic-Animals-Their-Food-and-Home-2/Part-15.html>

- Direct or indirect balance between resources and consumption
- Predators indirectly increase plant growth.
- They prevent overgrazing by suppressing herbivores.
- The net effect of direct and indirect relations is called trophic cascades.

Food Web

Trophic Cascade

- Powerful indirect interactions that can control entire ecosystems
- Occurs when a trophic level in a food web is suppressed.

Food Web

Trophic Cascade

- **Example: Removal of the top predator can alter the food web dynamics**
- **Impacts of climate and human activity on ecosystem**

Food Web

Trophic Cascade

- Top-Down Effect

- Effect flows down through a trophic chain
- From apex predator to lower trophic levels

Food Web

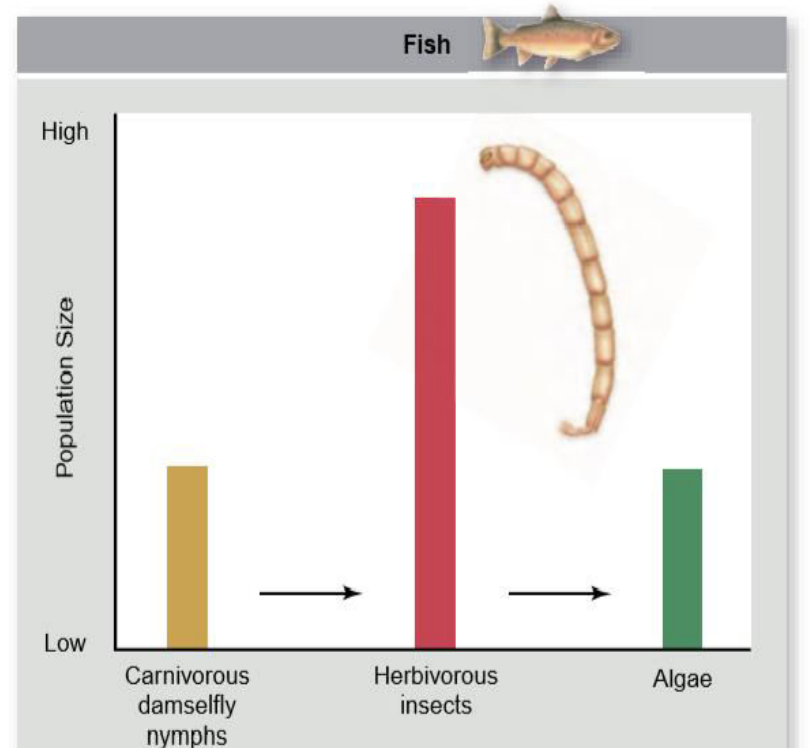
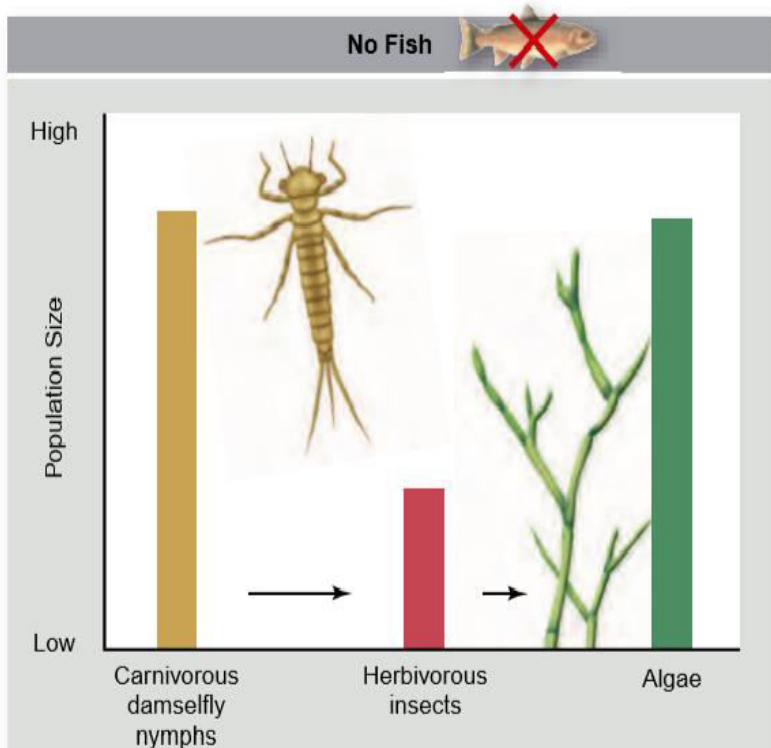
Trophic Cascade

- Bottom-Up Effect

- Effect flows up through a trophic chain
- From primary producers to higher trophic levels

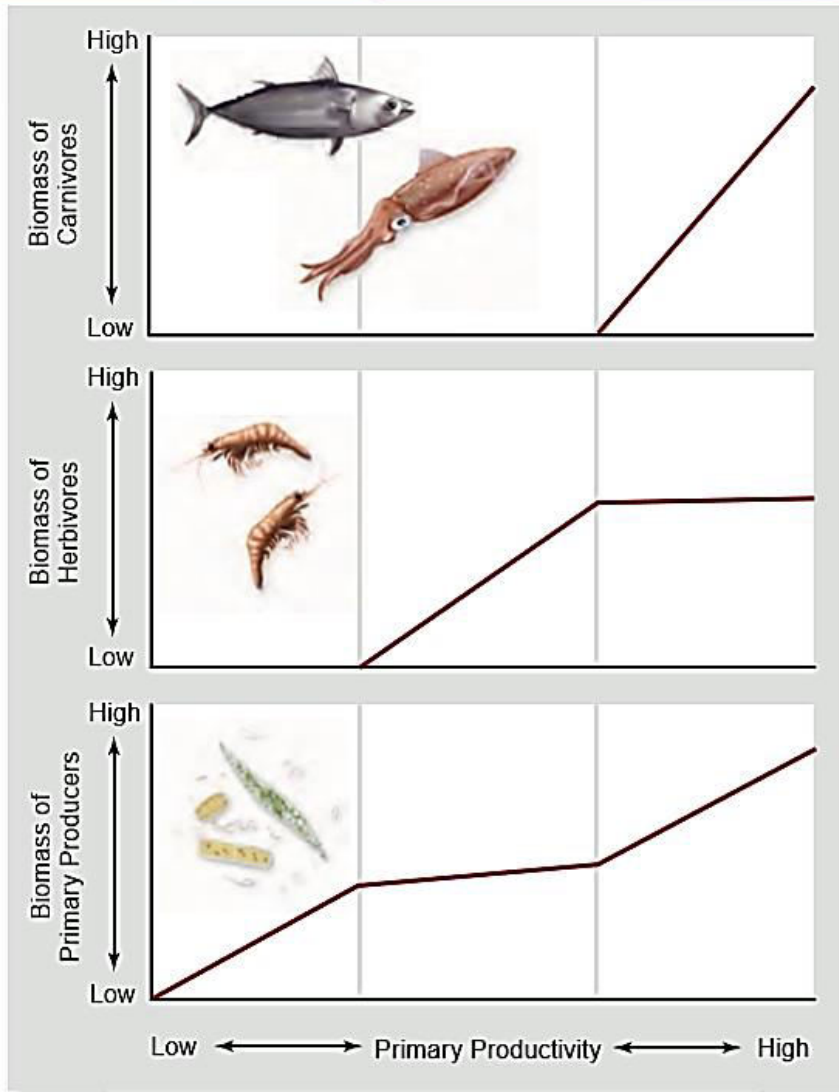
Food Web

Top-Down Effect



Biology Mason, Losos and Singer, 9th ed.

Food Web



Bottom-UP Effect

Biology Mason, Losos and Singer, 9th ed.

Food Web

Trophic Cascades

- Species Level Cascade

Subset of the food-web dynamic is impacted by a change in population numbers

Food Web

Trophic Cascades

- Community Level Cascades

Change in population numbers has a dramatic effect on the entire food-web

Example: Distribution of plant biomass

END

Trophic Level and Productivity

Productivity of trophic level

- Rate at which the organisms in the trophic level collectively synthesize new organic matter (new tissue substance)
- Energy is lost through respiration at every level.

Trophic Level and Productivity

Productivity of trophic level

1. Primary productivity (PP)

- Organic matter produced by photosynthesis
- Vascular plants; major contribution in PP
- Others; algae, mosses, liverworts
- PP is function of various factors; water, temperature, soil

Trophic Level and Productivity

Productivity of trophic level

1 a . Gross primary productivity (GPP)

- Raw rate at which the primary producers synthesize new organic matter

Gross primary production (GPP) is the amount of chemical energy as biomass that primary producers create in a given length of time.

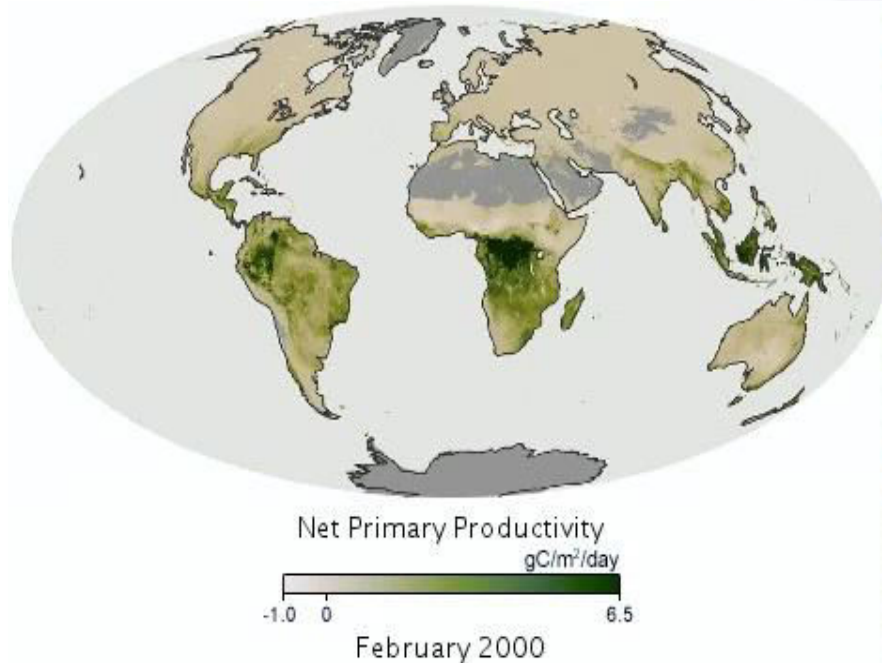
Trophic Level and Productivity

Productivity of trophic level

1 b. Net primary productivity

= GPP – Respiration

- NPP varies with seasonal changes
- Measured in mass/unit area/year



https://www.youtube.com/watch?v=URspHFp1u_k

Trophic Level and Productivity

Productivity of trophic level

1. Secondary productivity

- The rate that new organic matter is made by means of individual growth and reproduction in all the herbivores
- Each heterotroph trophic level has its own secondary productivity.

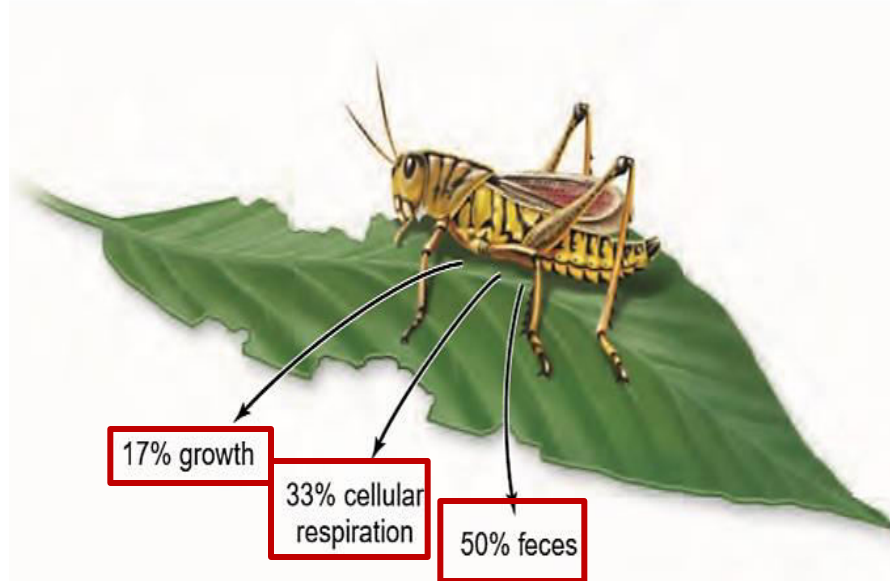
END

Some fraction of this fixed energy is used by primary producers for [cellular respiration](#) and maintenance of existing tissues (i.e., "growth respiration" and "[maintenance respiration](#)").^{[2][3]} The remaining fixed energy (i.e., mass of photosynthate) is referred to as **net primary production**. Some fraction of this fixed energy is used by primary producers for [cellular respiration](#) and maintenance of existing tissues (i.e., "growth respiration" and "[maintenance respiration](#)"). The remaining fixed energy (i.e., mass of photosynthate) is referred to as **net primary production**.

How trophic levels process energy?

- Small fraction of solar energy (1%) is consumed in photosynthesis
- Transformed into chemical-bond energy
- 99% solar energy is converted into heat.

How trophic levels process energy?



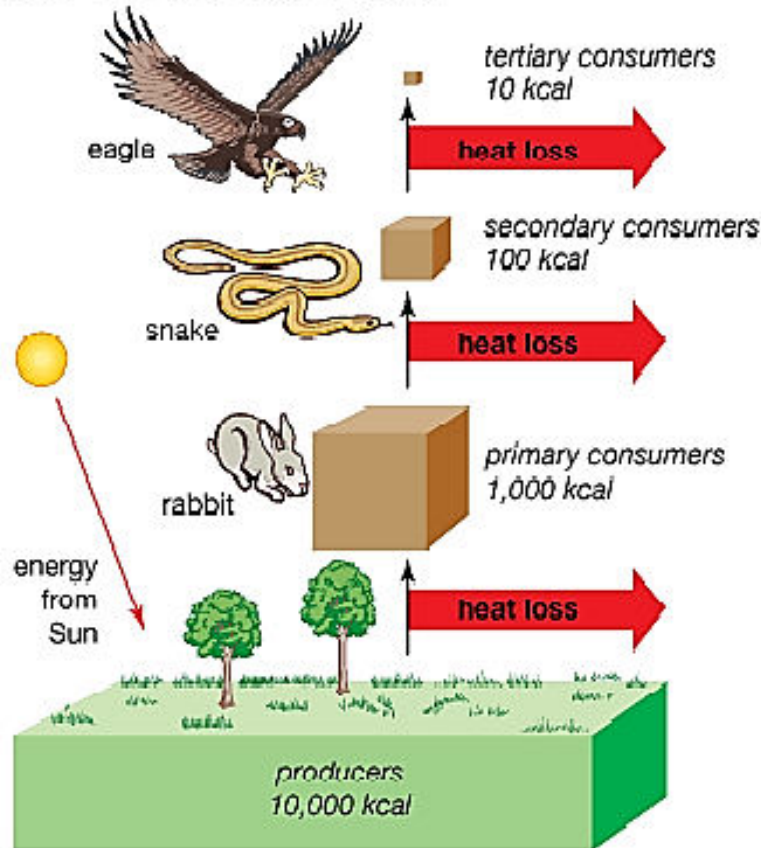
Biology Mason, Losos and Singer, 9th ed.

Synthesized chemical-bond energy is consumed:

- Absorption of nutrients
- Growth (tissue formation)
- Body functions
- **Availability to consumers**
 - Tissues / Growth

How trophic levels process energy?

Energy flow and trophic levels



Encyclopedia Britannica, 2011

- Chemical-bond energy is passed from one heterotroph trophic level to the next.
- Loss of energy at every stage
 1. Herbivore
 2. Primary carnivore
 3. Secondary carnivore
- Loss of 90% energy from one level to next

How trophic levels process energy?

Exponential decline of chemical-bond energy in a trophic chain limits:

- Lengths of trophic chains
- Number of top level carnivores

Top level carnivores have:

- large individual body sizes
- great individual energy needs

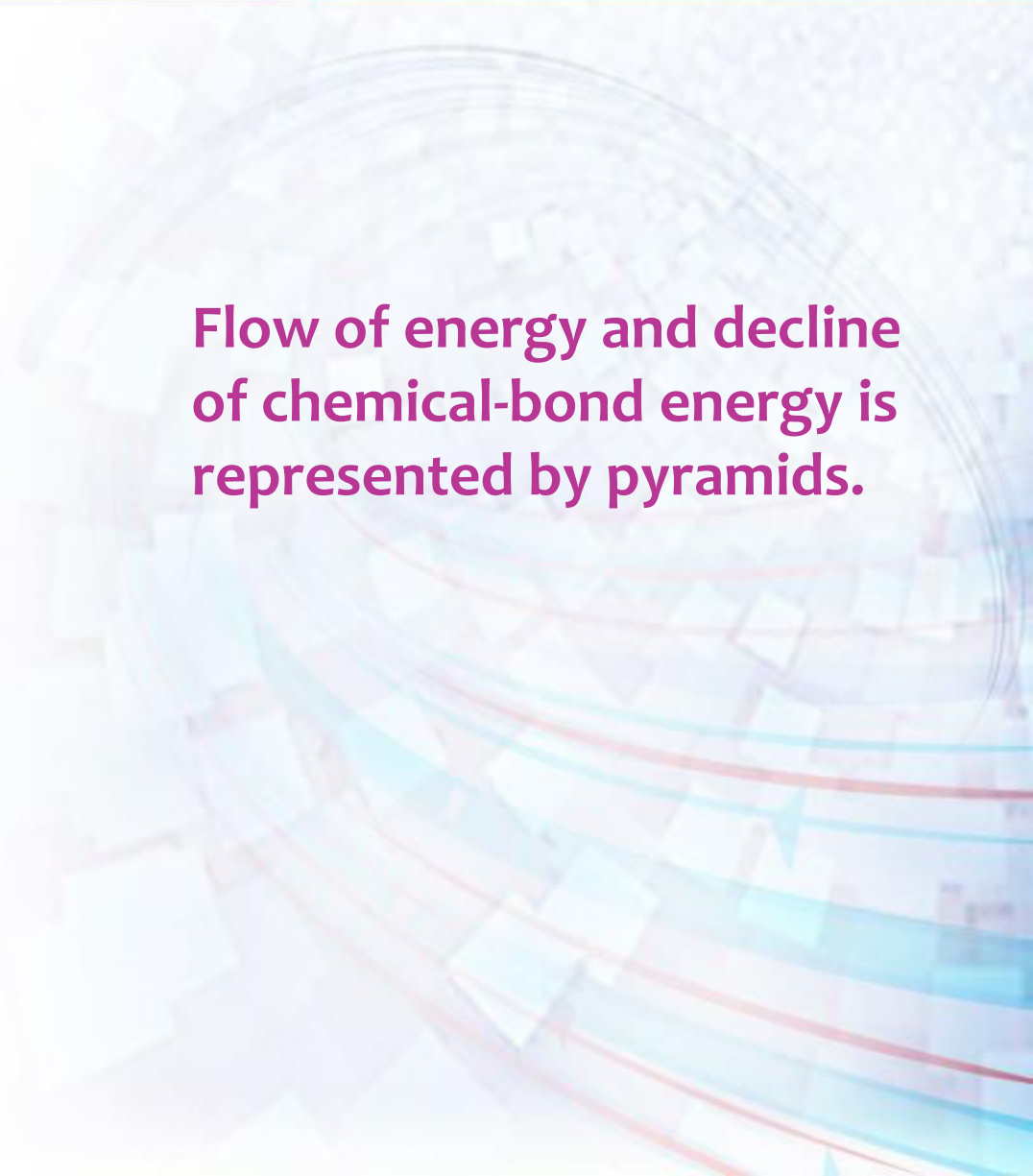
How trophic levels process energy?

- **Variability in Net Primary Productivity between ecosystems**
- **Wetlands and tropical rain forests: $2000 \text{ g/m}^2/\text{year}$**
- **Temperate forests: $1200 - 1300 \text{ g/m}^2/\text{year}$**
- **Savanna: $900 \text{ g/m}^2/\text{year}$**
- **Deserts: $90 \text{ g/m}^2/\text{year}$**

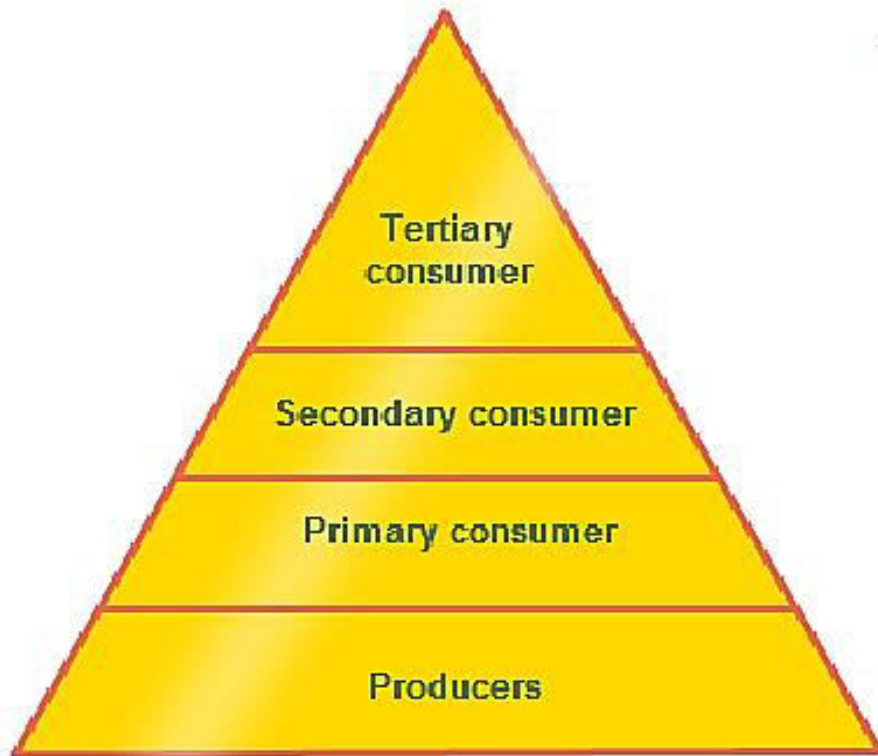
END

Ecological Pyramids

Flow of energy and decline of chemical-bond energy is represented by pyramids.



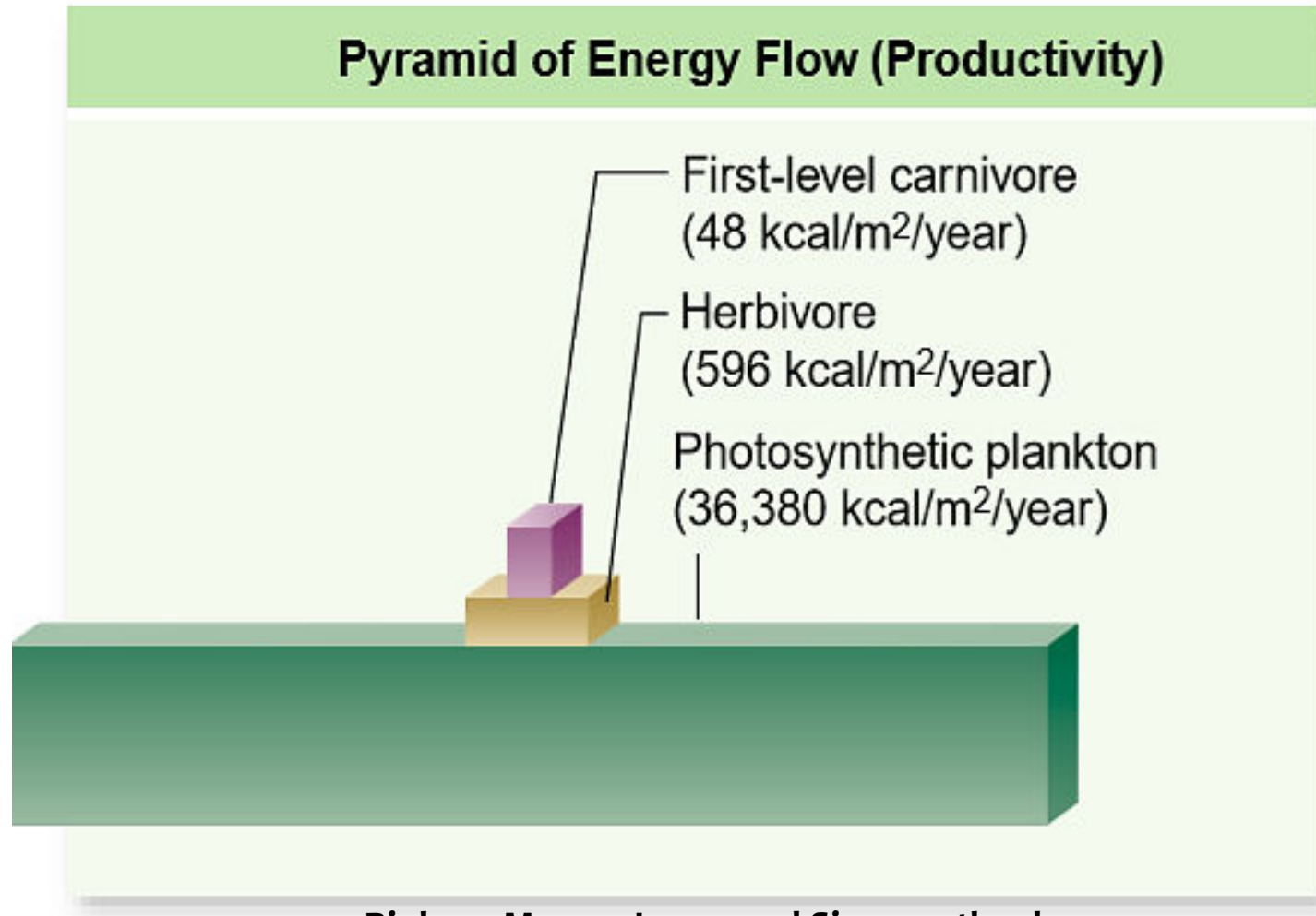
Ecological Pyramids



Flow of energy and decline of chemical-bond energy is represented by pyramids.

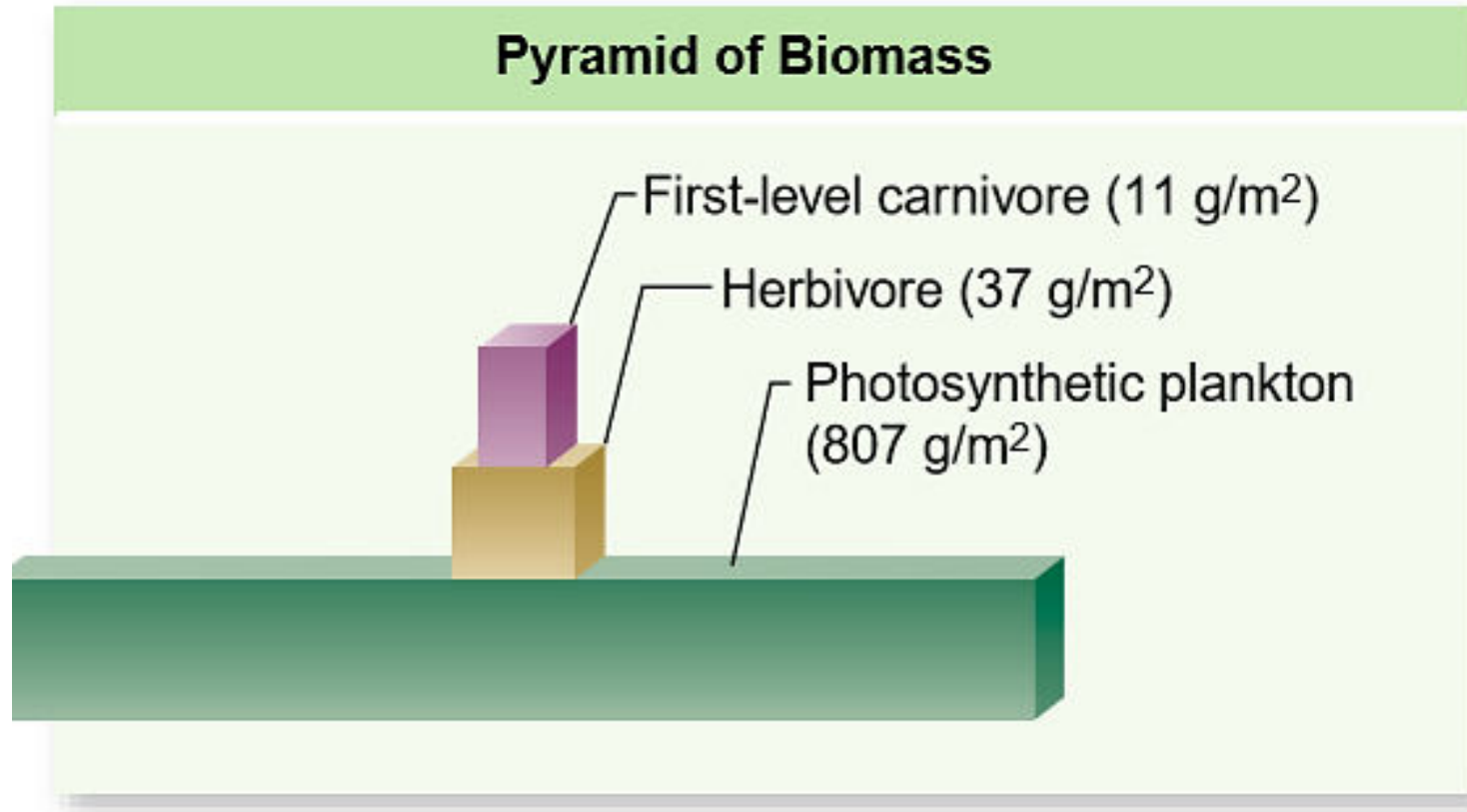
- Trophic levels are boxes stacked on top of each other.
- Width of box is proportional to productivity of that level.

Ecological Pyramids



Biology Mason, Losos and Singer, 9th ed.

Ecological Pyramids

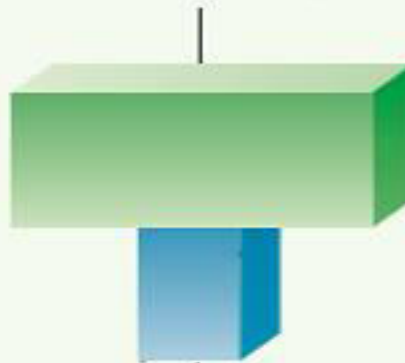


Biology Mason, Losos and Singer, 9th ed.

Ecological Pyramids

Inverted Pyramid of Biomass

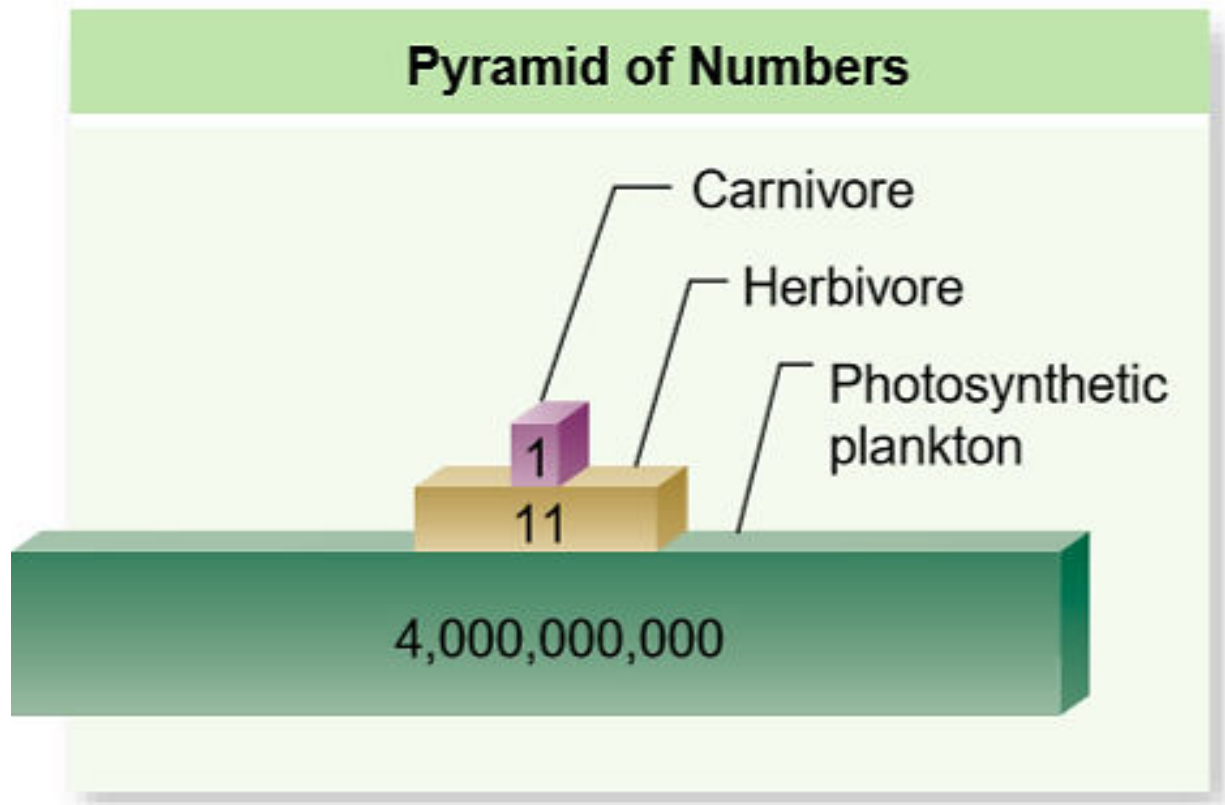
Herbivorous zooplankton and
bottom fauna (21 g/m^2)



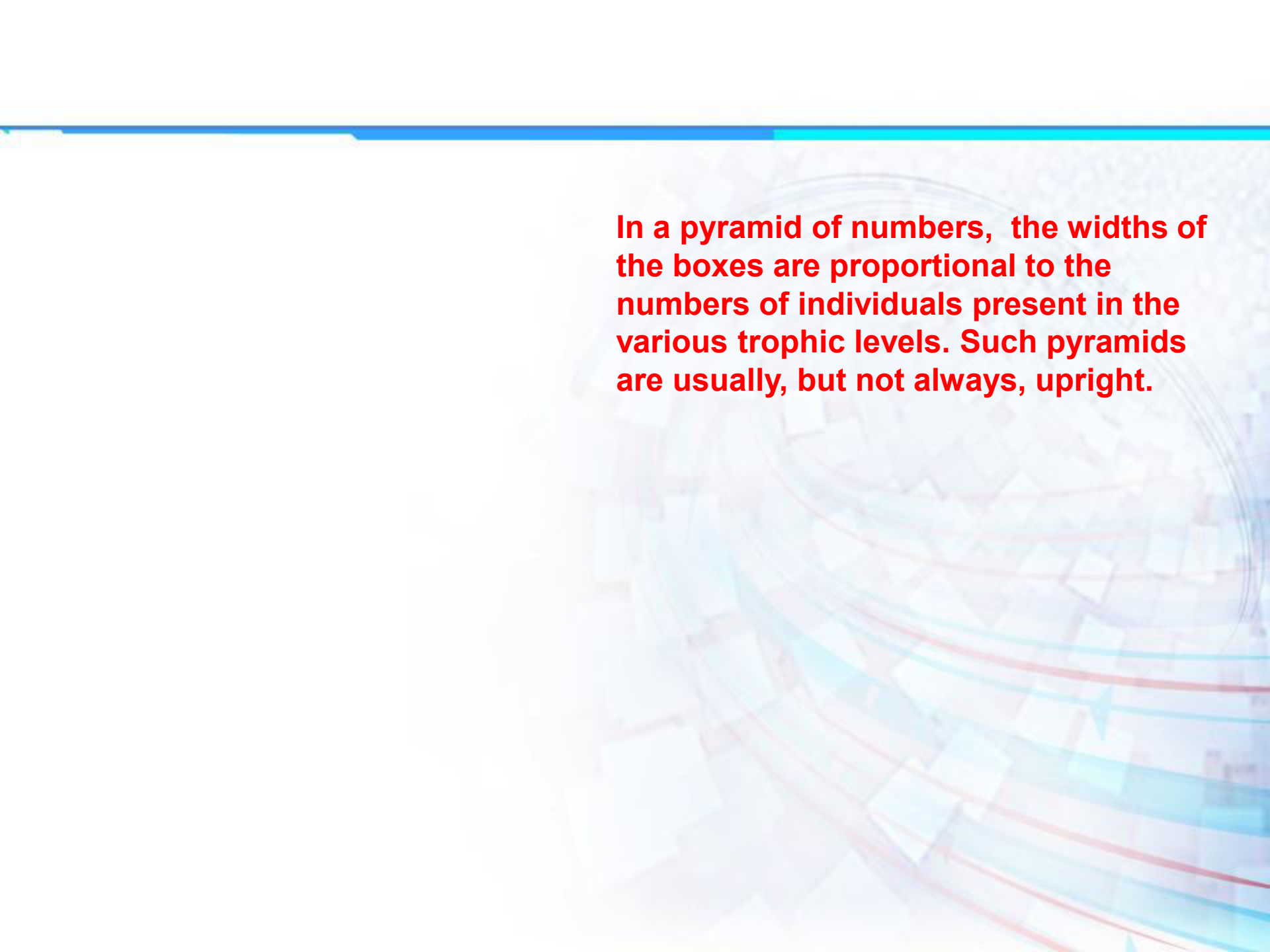
Phytoplankton
(4 g/m^2)

Biology Mason, Losos and Singer, 9th ed.

Ecological Pyramids



Biology Mason, Losos and Singer, 9th ed.



In a pyramid of numbers, the widths of the boxes are proportional to the numbers of individuals present in the various trophic levels. Such pyramids are usually, but not always, upright.

Abiotic Factors in Environment

Habitat

- Natural environment where a species/organism lives
- Where they find:
 - Food
 - Shelter
 - Protection
 - Mates

Abiotic Factors in Environment

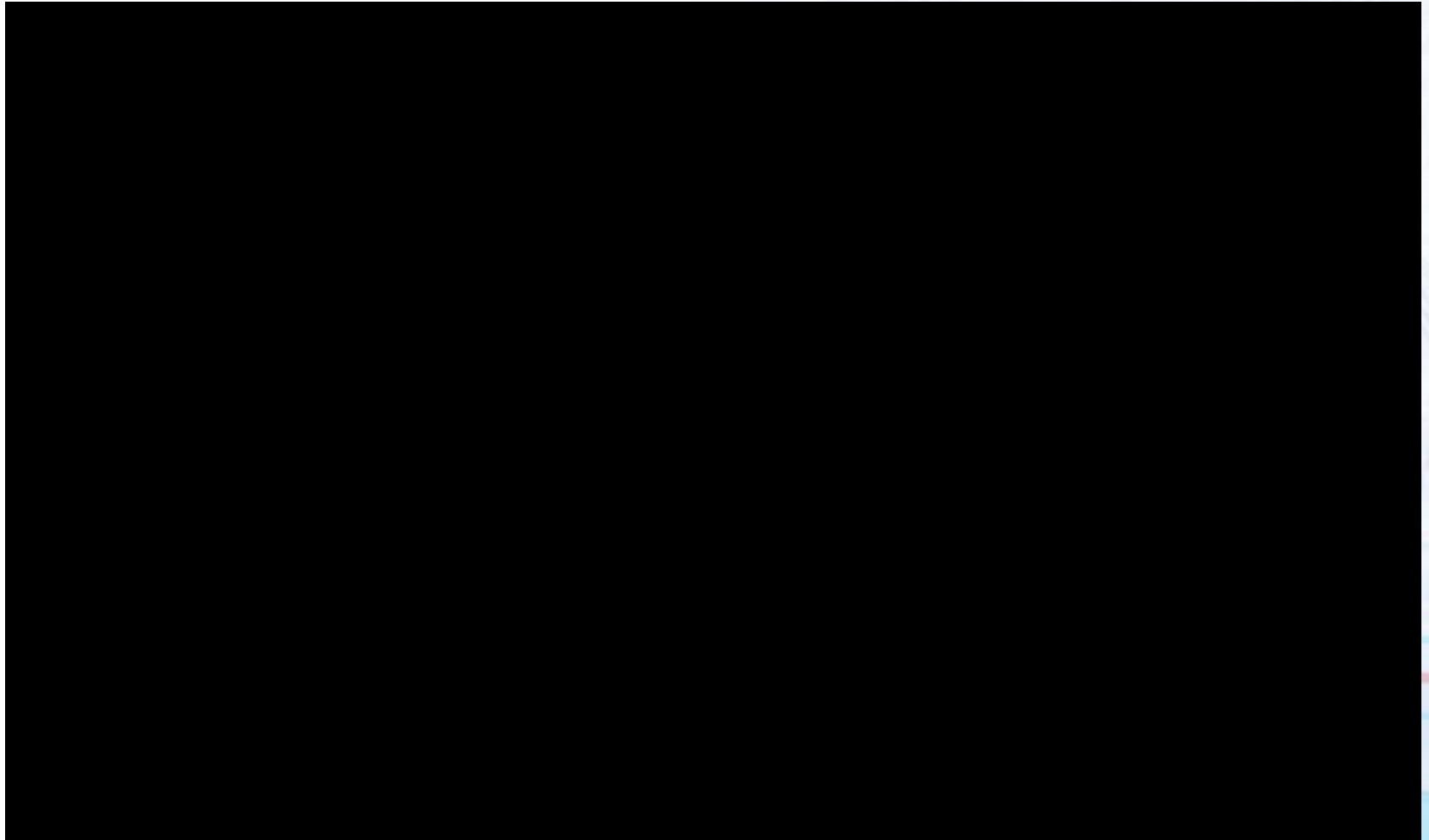
Habitat has two types of characteristics:

- Abiotic/Physical
- Biotic/Biological

Abiotic Factors

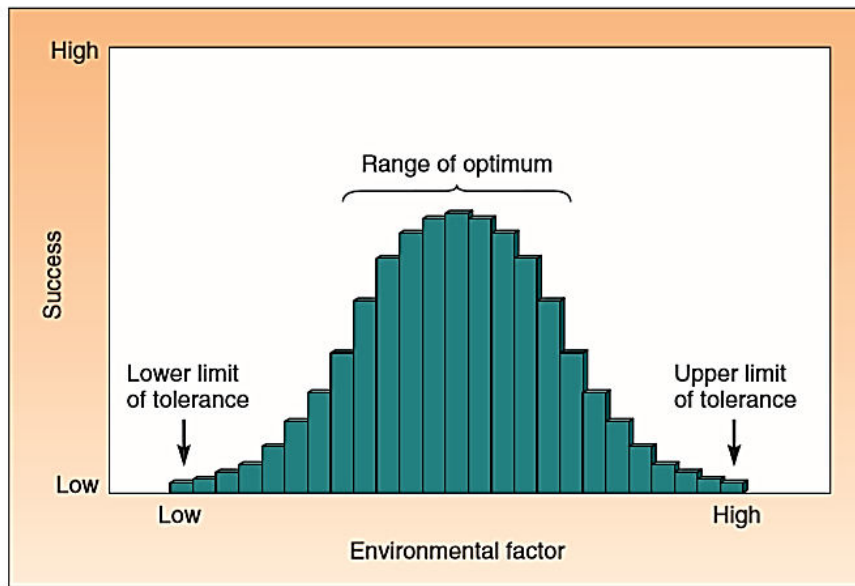
- Temperature
- Water
- Light
- Soil

Abiotic Factors in Environment



<https://www.youtube.com/watch?v=hIyoZlyPPDg>

Abiotic Factors in Environment



- Tolerance range

- At either limit, one or more essential functions cease

• - Range of optimum

- Highest success of animal

• - Limiting factor

- Factor out of tolerance range

- Taxis

Orientation towards a factor

Temperature

- **Thermoregulation**
- **Inequality between heat loss and gain**
- **Harsh condition require adaptations**
- **Both ectothermic and endothermic animals have adaptations.**

Temperature



<https://www.sciencenewsforstudents.org/article/explainer-how-brief-can-hibernation-be>



<https://smbasblog.com/2016/11/15/don't-mother-hummingbirds-sleep/>

Torpor

- Decreased metabolism
- Lowered body temperature
- Examples:
Bats, Humming birds
- Help in survival when not feeding

Temperature



<http://thewildlife.wbur.org/2015/01/20/hibernating-animals-may-hold-secrets-for-fighting-alzheimers/>

Hibernation

- Decreased metabolism
- Lowered body temperature
- May last for weeks or months
- Examples:

Rodents, Shrews

Temperature



<https://bear.org/do-black-bears-hibernate/>

Winter Sleep

- Occurs in large animals
- Large energy reserves
- Quick restoration of activity
- Examples:

Bears

Temperature



<https://www.thinglink.com/scene/630237835143675906>



<https://www.worldatlas.com/articles/what-is-estivation-animals-that-estivate.html>

Aestivation

- Survival in extended period of drying
- Avoid temperature damage and dehydration

- Examples:

Insects, frog, turtle

Temperature



<http://youtube.com/watch?v=buQalannwzo>

Abiotic Factors - Water

- Water is critical factor in homeostasis.
- Animals lose water:
 - Evaporation
 - Excretion
- Lost water must be replaced.
 - Drinking
 - Eating
 - Avoiding dehydration

Abiotic Factors - Water

Adaptations

To avoid dehydration:

- Behavioral

Nocturnal mode,
burrows/hides

- Physiological

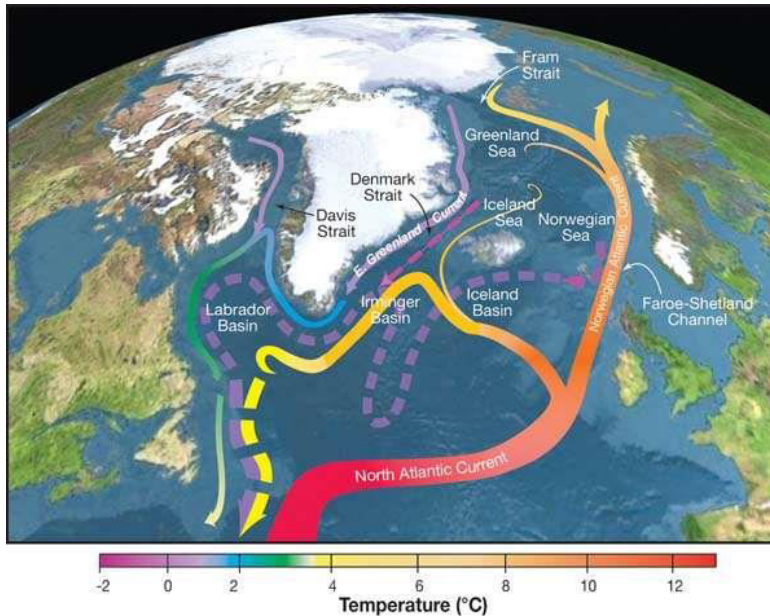
Osmoregulation



<https://www.google.com/search?q=animals+hiding+in+burrows&rlz>

Abiotic Factors - Water

Water Currents



<https://www.google.com/search?q=water+currents+in+ocean&rlz>

- Affect animal distribution in water body.
- Generated by wind
- Convection currents
- Determine biodiversity and niche
- Movement of minerals and rocks

Abiotic Factors - Water



<https://www.dailymail.co.uk/sciencetech/article-6728749/Protein-razor-sharp-teeth-inside-suckers-squid-replace-plastics.html>

Adaptations

To cope with currents:

- Suckers
- Hooks / claws
- Body flattening
- Streamlining

Abiotic Factors - Light

- Discussed in “energy”
- Animal’s response to light intensity
- Phototactic movement
 - positive (Euglena)
 - negative (Planaria, Earthworms)

Abiotic Factors - Light

Effects of Photoperiod

- Biological Clocks

- Diurnal

Distribution, sleep-wake cycle

- Annual

Reproduction, migration



[http://idahoptv.org/sciencetrek/to
pics/bird_migration/facts.cfm](http://idahoptv.org/sciencetrek/to pics/bird_migration/facts.cfm)

Abiotic Factors - Light

Effects of Light

- Increase metabolism
- Pigmentation

Sexual dimorphism,
camouflage

- Sun as reference point in navigation during daily movement/migration

END

Population Growth

- Increase in the number of individuals in a population
- Reproduction

Population growth rate

- Rate at which the number of individuals in a population increases in a given time period.
- Species specific

Population Growth

Species	Rate of Reproduction
Bacteria	Every 20 minutes 3 million / 7 hours
House Fly	600 eggs / life time
Common Carp	300, 000 / year
Mouse	30 – 140 / year
Elephants	one / 5 – 6 years

Population Growth

Limiting factors regulate population size

- Rate of Birth
- Rate of Death/mortality
- Dispersal
- Food Resources
- Space to grow
- Environmental factors

Population Growth

Carrying Capacity (K)

- The maximum population size of the species that the environment can sustain indefinitely.
- Environment's maximal load
- Population size decreases above carrying capacity

Population Growth

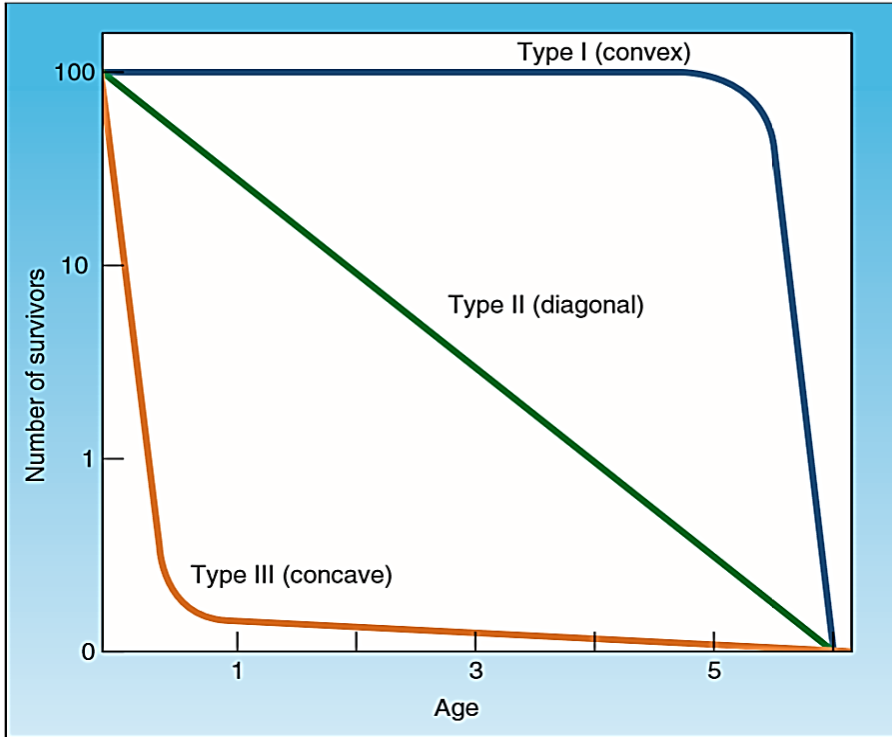
Carrying Capacity

- Variable, may vary for different species
- May change over time due to a variety of factors including:
- food availability, water supply, environmental conditions and living space

Population Growth

- **Variety of tools to track the size of population**
- **Survivorship curve is one tool.**
- **It measures the proportion of individuals in a given species that are alive at different ages.**

Population Growth

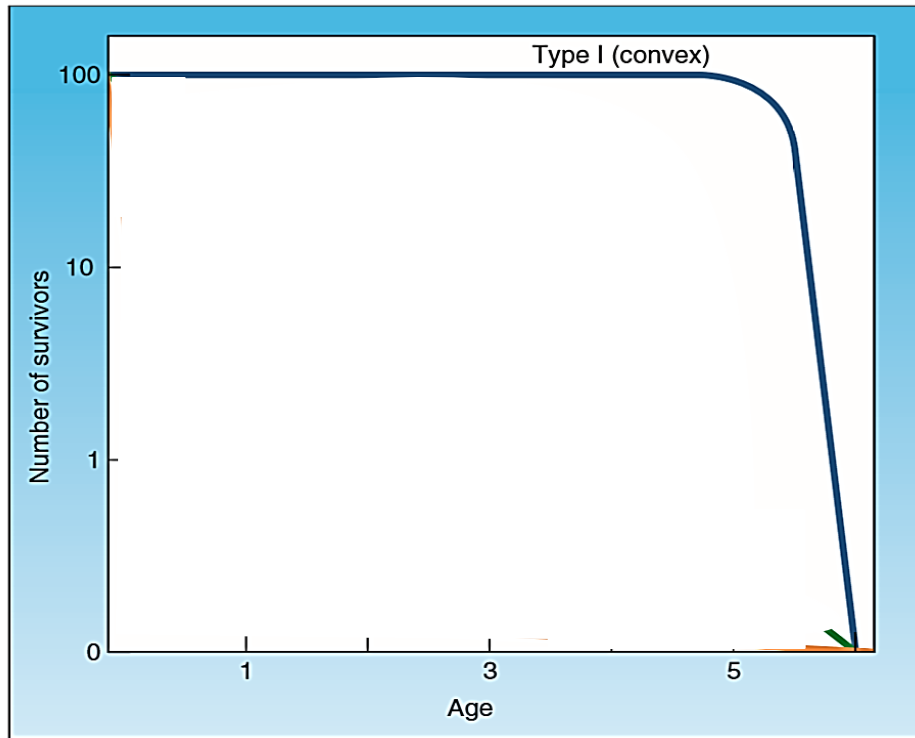


Zoology by Miller and Harley, 5th ed.

Survivorship Curve

- Y-axis is a logarithmic plot of numbers of survivors
- X-axis is a linear plot of age

Population Growth



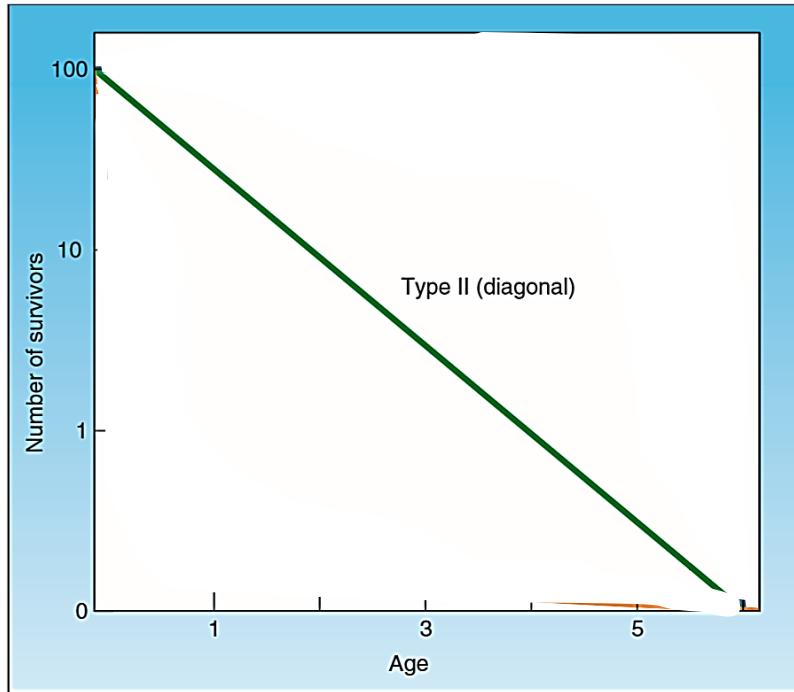
Zoology by Miller and Harley, 5th ed.

Kinds of Survivorship Curve

Type 1 (Convex)

- Populations survive to an old age, then die rapidly.
- Unimportance of environmental factors in influencing mortality
- Potential life span
- Example: Humans

Population Growth



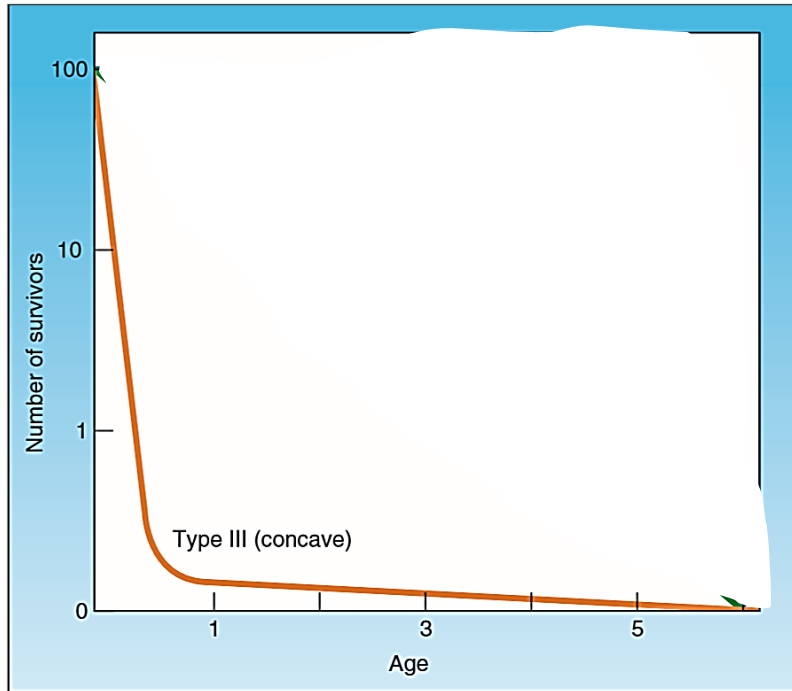
Zoology by Miller and Harley, 5th ed.

Kinds of Survivorship Curve

Type II (Diagonal)

- Constant probability of death throughout lives
- Influence of environmental factors on mortality
- Mortality independent of age
- Example: birds, rodents

Population Growth



Zoology by Miller and Harley, 5th ed.

Kinds of Survivorship Curve

Type III (Concave)

- High juvenile mortality
- Influence of environmental factors and less resistance of juveniles on mortality
- Lower mortality in adults
- Example: Fish

Exponential Population Growth

Types of Population growth

1. Exponential
2. Logistic

Exponential Population Growth

Exponential Growth

- Resources are abundant.
- Remarkable potential to increase in number
- Exponential addition in population size

Exponential Population Growth

The number of births minus the number of deaths per generation time

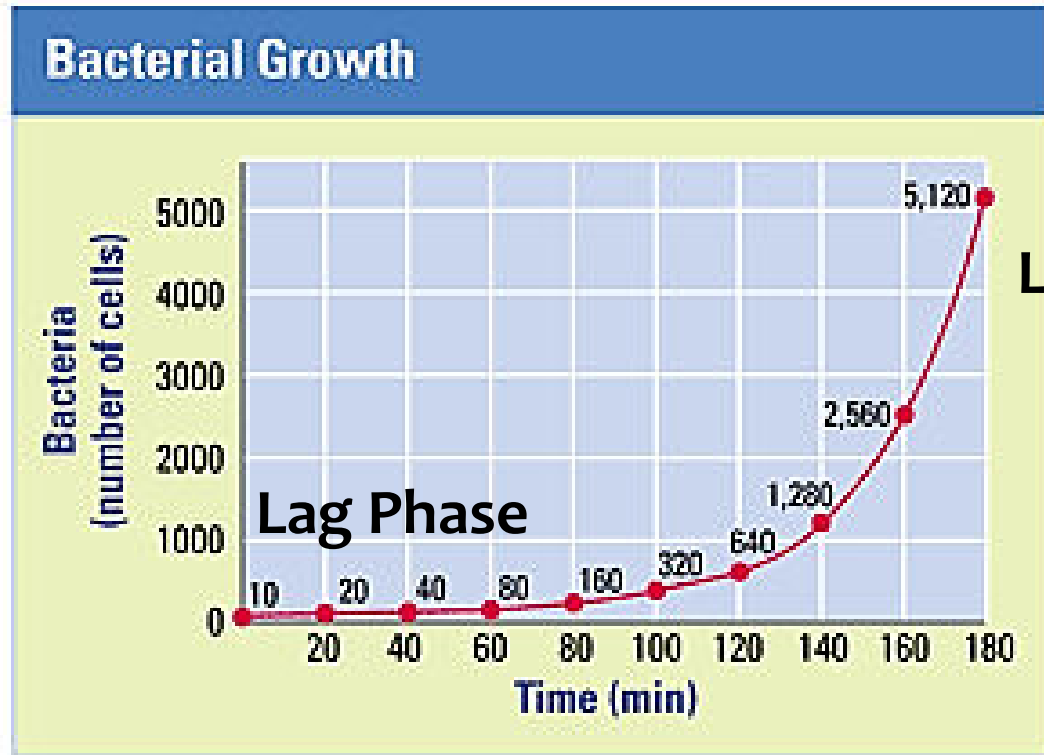
- Population may grow exponentially to certain time
- Ultimately limited by resource availability
- Formula of exponential growth

$$\frac{dN}{dt} = rN \quad N = \text{Population size}$$

r = intrinsic rate of natural increase

Exponential Population Growth

J – Shaped Curve



Log Phase

Lag Phase

<http://bodell.mtchs.org/OnlineBio/BIOCD/text/chapter16/review16.html>

Exponential Population Growth

Factors affecting reproductive rate

- Number of offspring produced
- Likelihood of survival to reproductive age
- Duration of the reproductive period
- Length of time to reach maturity
- Environmental factors

END

Logistic Population Growth

- Carrying capacity of environment affect population growth.
- Environmental resistance
- Growth rate gets smaller as approaches the carrying capacity

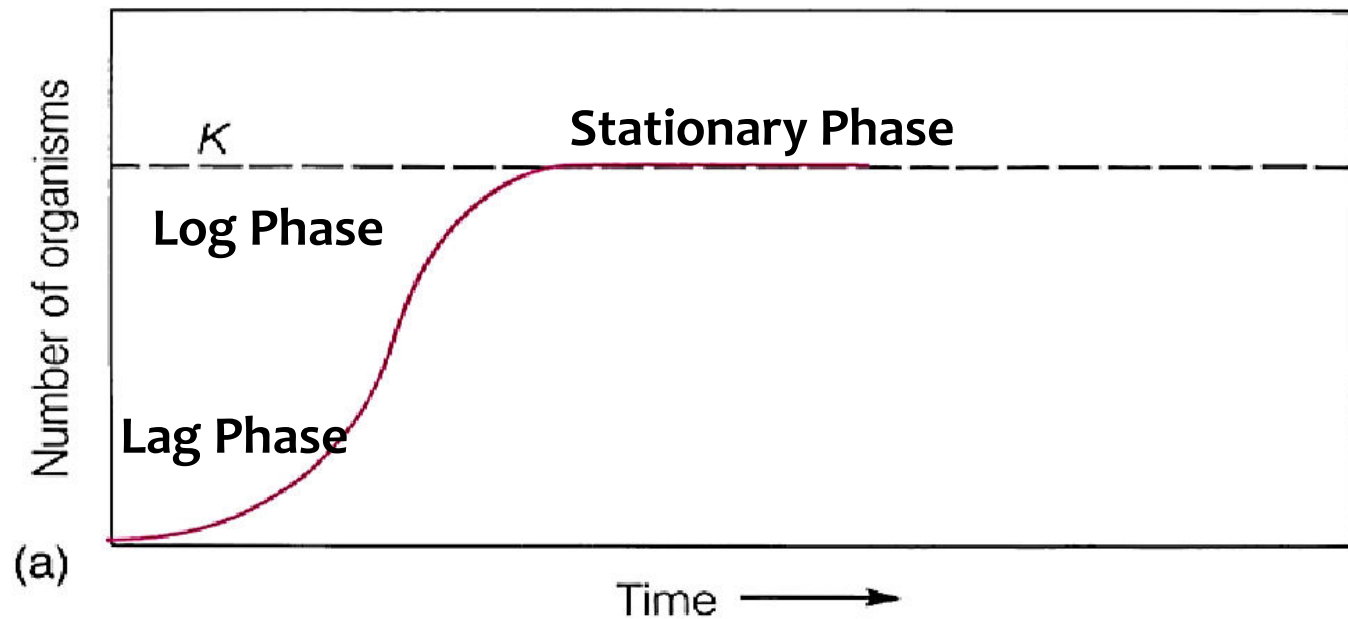
Logistic Population Growth

**Environmental factors
affect population growth.**

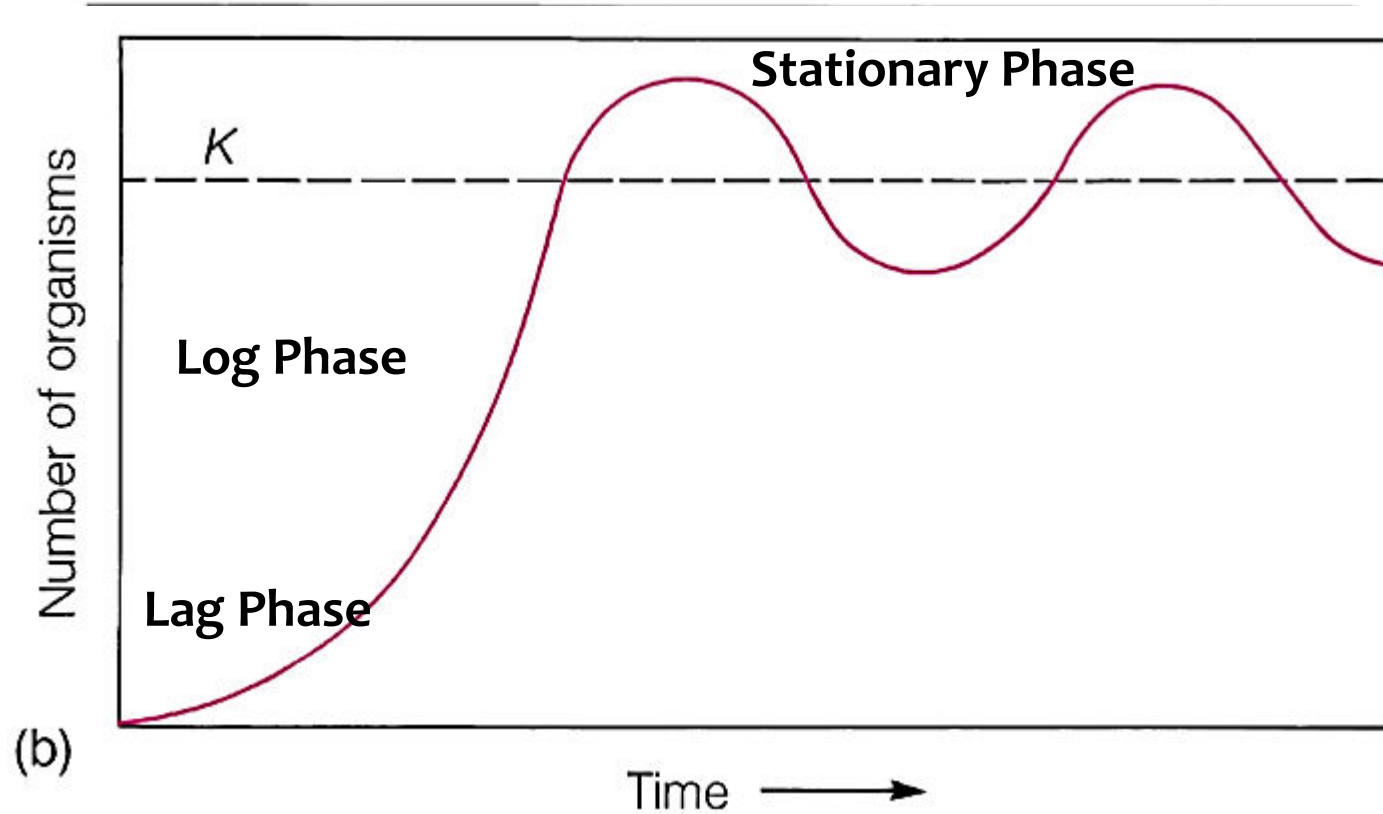
- **Climate**
- **Food**
- **Space**
- **Competition**

Logistic Population Growth

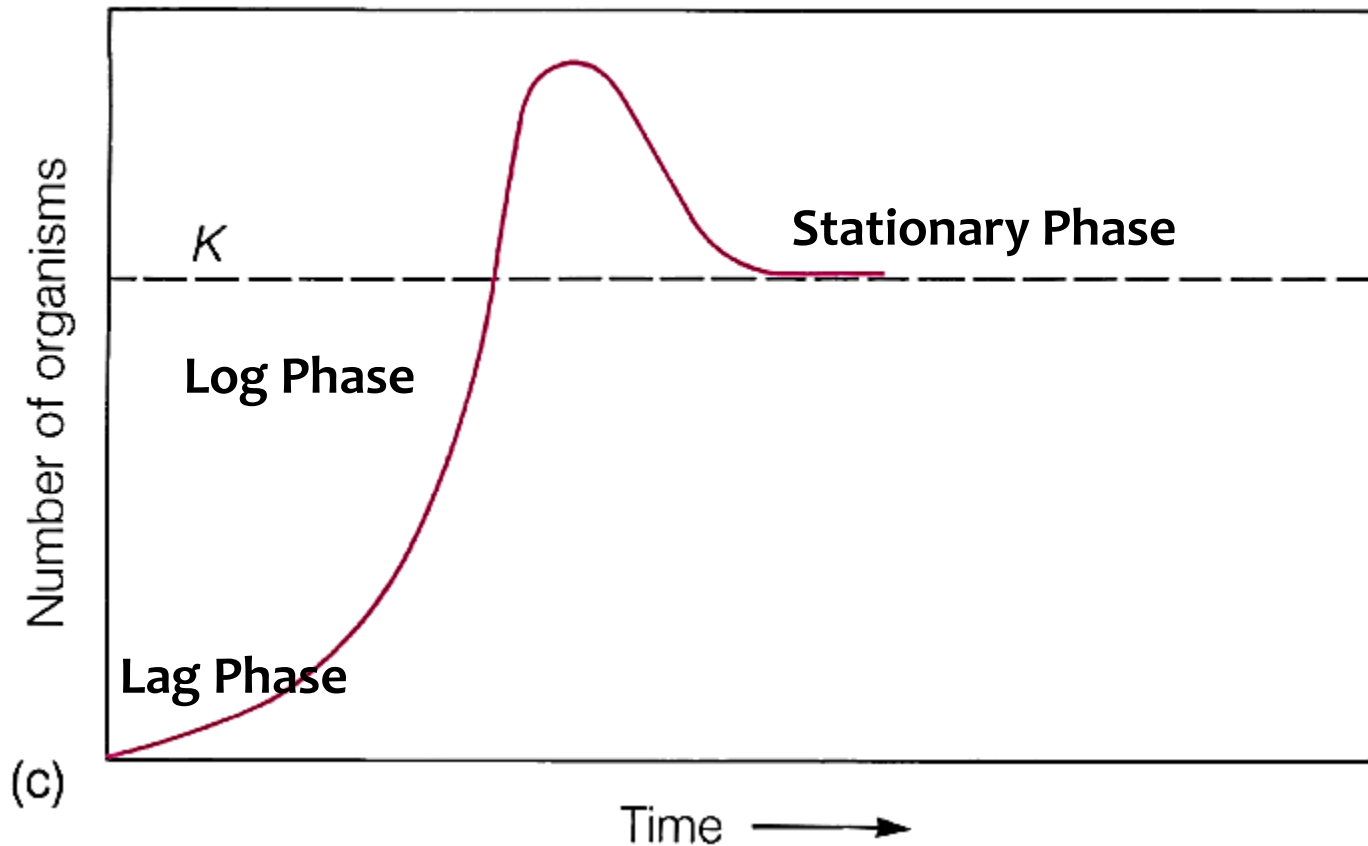
S - Shaped or Sigmoidal Curve



Logistic Population Growth



Logistic Population Growth



Logistic Population Growth

Formula

$$\frac{dN}{dt} = \frac{rN (K-N)}{K}$$

N = population size

dN / dt = change in population size per unit time

K = carrying capacity

r = intrinsic rate of natural increase

Logistic Population Growth

EXPONENTIAL GROWTH	LOGISTIC GROWTH
It occurs when resources are abundant.	It occurs when resources are limited.
Population passes well beyond the carrying capacity of the ecosystem.	Population doesn't grow beyond the carrying capacity of the ecosystem.
It produce J shaped curve.	It produce S shaped curve.
A stationary or stable phase is seldom achieved.	A stationary or stable phase is achieved.
It has two phases lag and log.	It has four phases lag, log, deceleration & steady .
It occurs in fewer organisms like lemmings, algal blooms, etc.	It is more common, example members of wildlife, yeast (a microscopic fungus), etc.

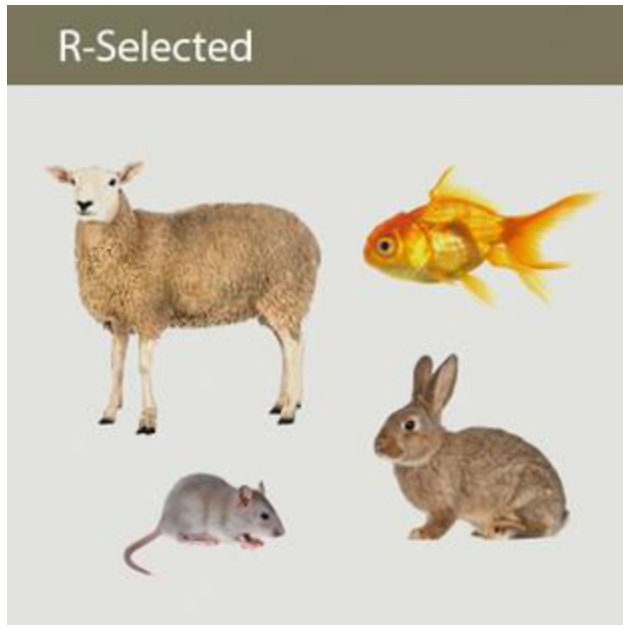
<https://www.youtube.com/watch?v=LYoguW1Haow>

r and k Selected Species

r-selected species

- High growth rates
- Typically exploit less crowded ecological habitats
- Produce many offspring
- Relatively low probability of surviving to adulthood

r and k Selected Species



r-selected species

- High reproductive rate supports their survival.
- Especially in unstable environment

<https://steemit.com/animals/@reviewsatrandonm/if-earth-was-an-mmo-growth-strats>

r and k Selected Species

Traits of r-selected species

- High rate of reproduction
- Small body size
- Early maturity onset
- Short generation time
- Ability to disperse offspring widely
- Examples: bacteria, grasses, cephalopods, rodents

r and k Selected Species

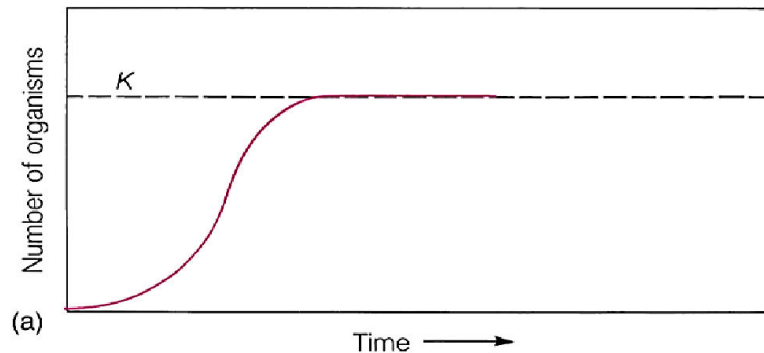


[https://steemit.com/animals/@reviewsatrando
m/if-earth-was-an-mmo-growth-strats](https://steemit.com/animals/@reviewsatrando/if-earth-was-an-mmo-growth-strats)

k-selected species

- Living at densities close to carrying capacity
- Typically are strong competitors in such crowded habitats.
- Invest more heavily in fewer offspring
- Relatively high probability of surviving to adulthood

r and k Selected Species



k-selected species

- Survive in stable environment
- Ability to compete successfully for limited resources is crucial
- Populations of k-selected organisms typically are very constant in number.
- Close to the maximum that the environment can bear

r and k Selected Species

Traits of k-selected species

- Large body size
- Long life expectancy
- Production of fewer offspring
- Often require extensive parental care until they mature
- Examples: humans, whales, parrots, eagles

END

Population Regulation

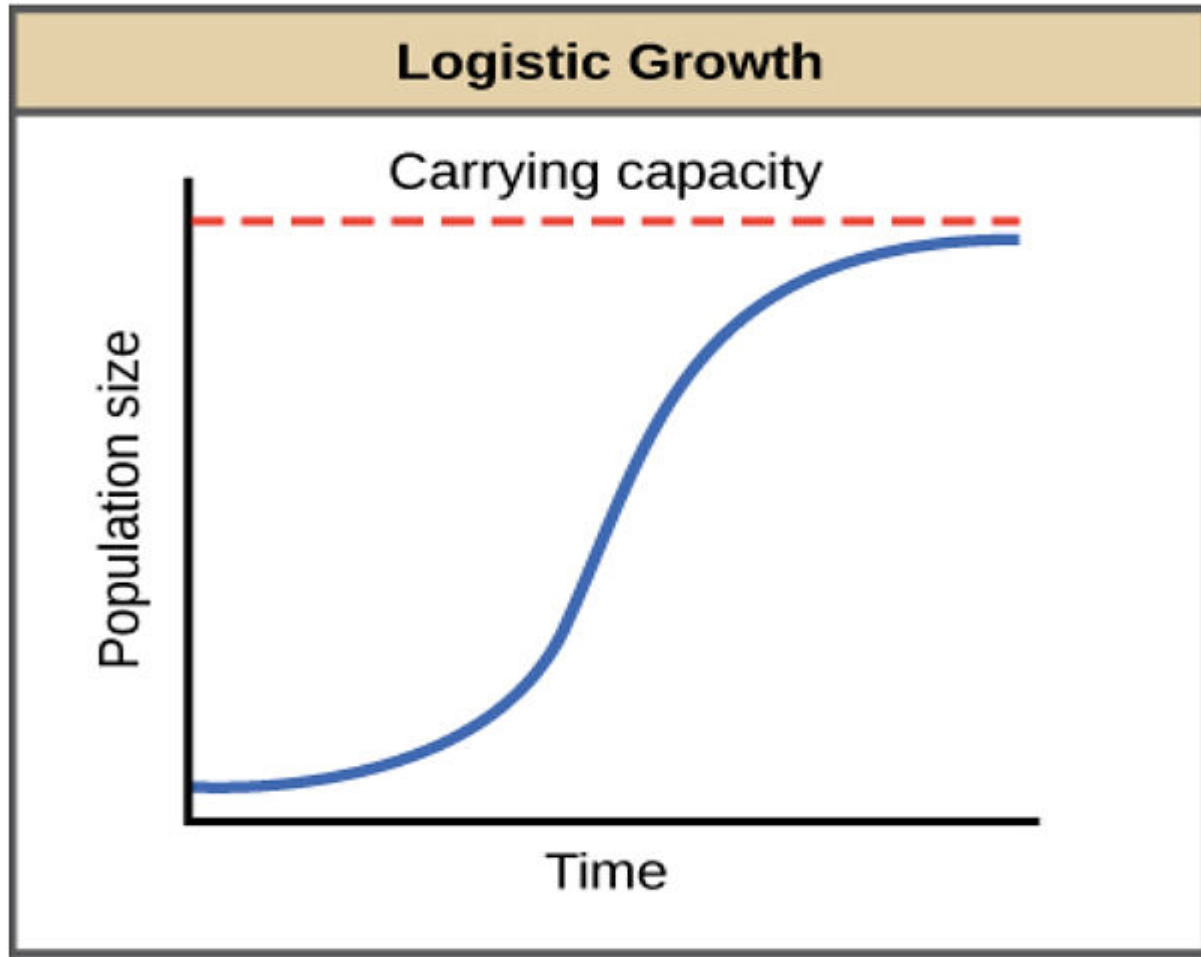
- No indefinite growth of any population
- Factors limit the population size and growth.
 1. Density dependent factors
 2. Density independent factors

Population Regulation

Density dependent factors

- **Example: Food**

Population Regulation



<https://www.khanacademy.org/science/biology/ecology/population-growth-and-regulation/a/mechanisms-of-population-regulation>

Population Regulation

Competition within the population

- When a population reaches a high density, there are more individuals trying to use the same quantity of resources.

Population Regulation

Competition within the population

- This can lead to competition for food, water, shelter, mates, light, and other resources needed for survival and reproduction.

Population Regulation

Predation

- **Higher-density populations may attract predators who wouldn't bother with a sparser population.**
- **When these predators eat individuals from the population, they decrease its numbers but may increase their own.**

Population Regulation

Disease and Parasites

- Disease is more likely to break out and result in deaths when more individuals are living together in the same place.

Population Regulation

Waste Accumulation

- **High population densities can lead to the accumulation of harmful waste products that kill individuals or impair reproduction, reducing the population's growth.**

Population Regulation

Density Independent Limiting Factors

Affect per capita growth rate independent of density of population

- **Natural disasters**
- **Severe weather**
- **Pollution**

Population Regulation

Density Independent Limiting Factors

**Cannot regulate population
at constant level**

They are accidental.

**Cause abrupt shift in
population size**

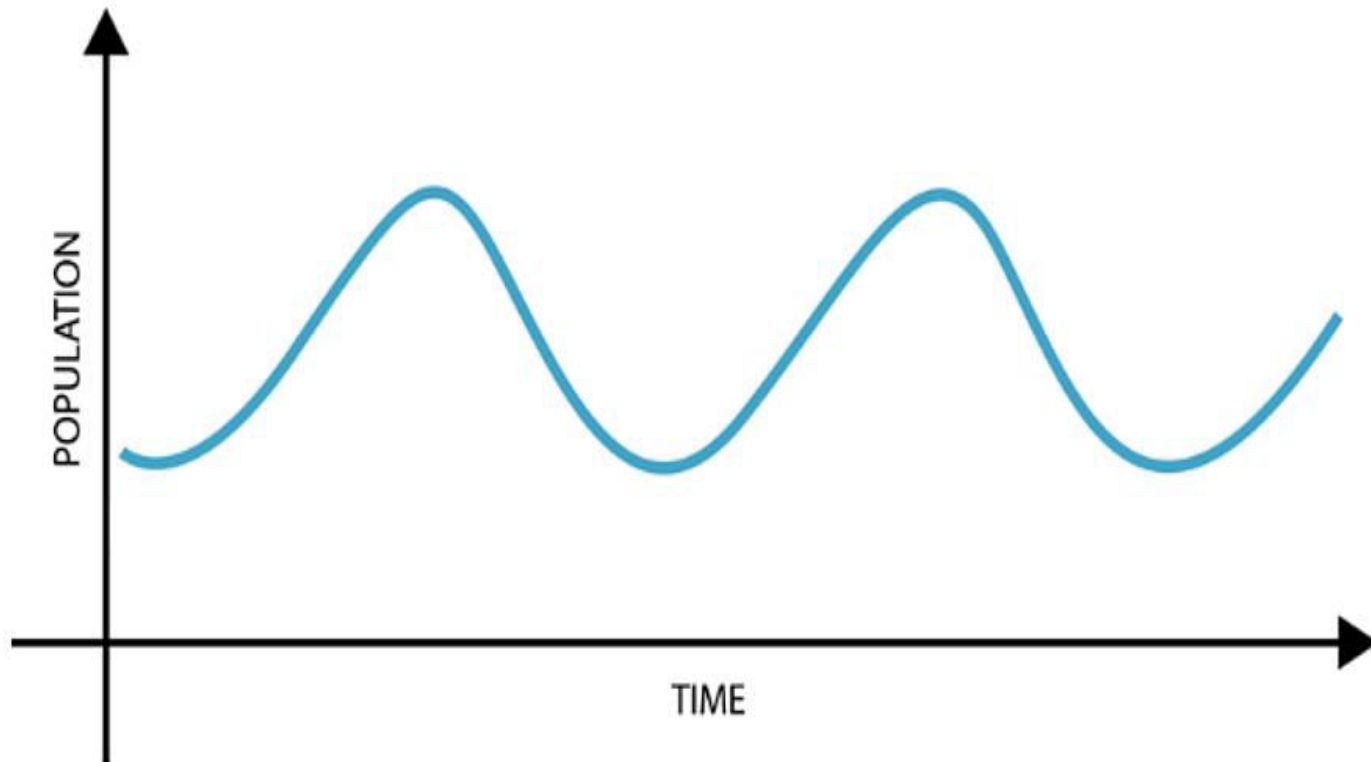
**May wipe out small
population by sporadic
event**

Population Regulation

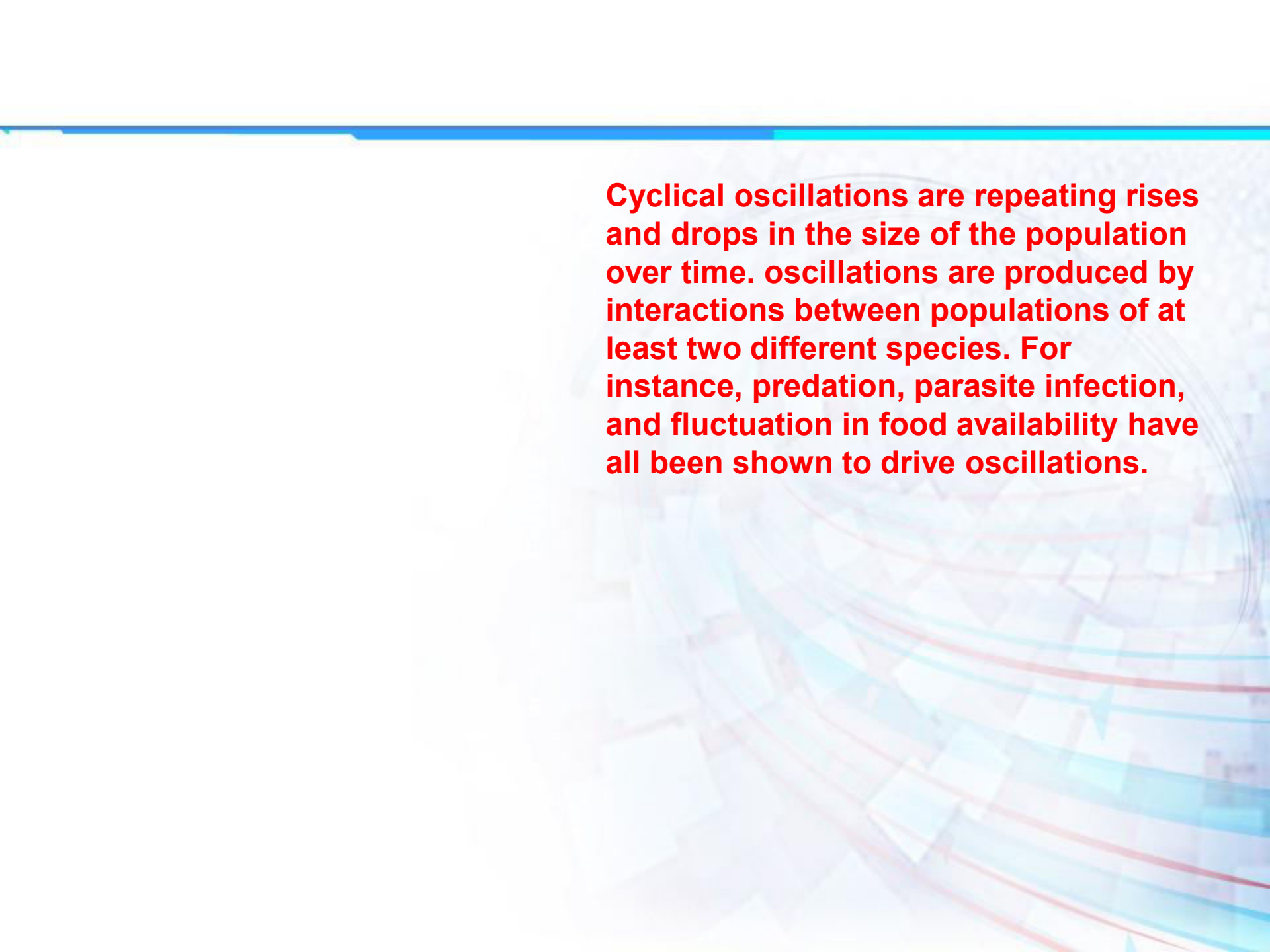
- Collectively, density-dependant and independent factors create a different population growth curve.
- Cyclic oscillations

Population Regulation

Cyclic Oscillations in Population



<https://www.khanacademy.org/science/biology/ecology/population-growth-and-regulation/a/mechanisms-of-population-regulation>



Cyclical oscillations are repeating rises and drops in the size of the population over time. oscillations are produced by interactions between populations of at least two different species. For instance, predation, parasite infection, and fluctuation in food availability have all been shown to drive oscillations.

Communities

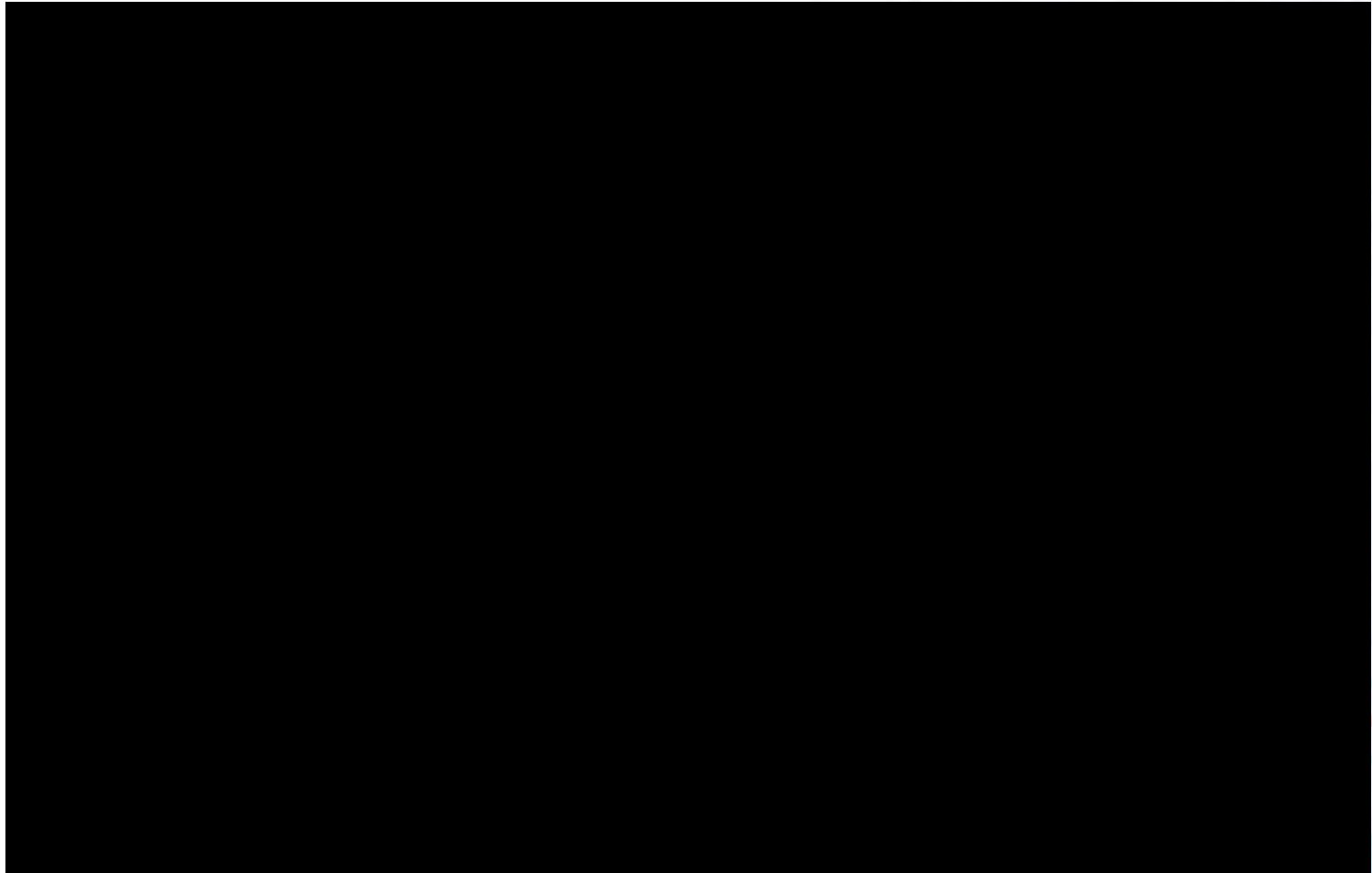
- **Group of species living in a habitat**
- **Plants, animals, fungi, protists, and prokaryotes that occur at a locality**
- **Characterized by:**
 - **Species richness**
 - **Primary productivity**

Communities



Biology Mason, Losos and Singer, 9th ed.

Communities



<https://www.youtube.com/watch?v=9bQNRVyl4l0>

Communities

Concept about complexity of communities

- Individualistic
- Holistic

Communities

Individualistic concept of communities

- Simple aggregation of species that happen to occur together at one place
- Species appear and disappear according to their ecological needs
- Structure of community varies with time

Communities

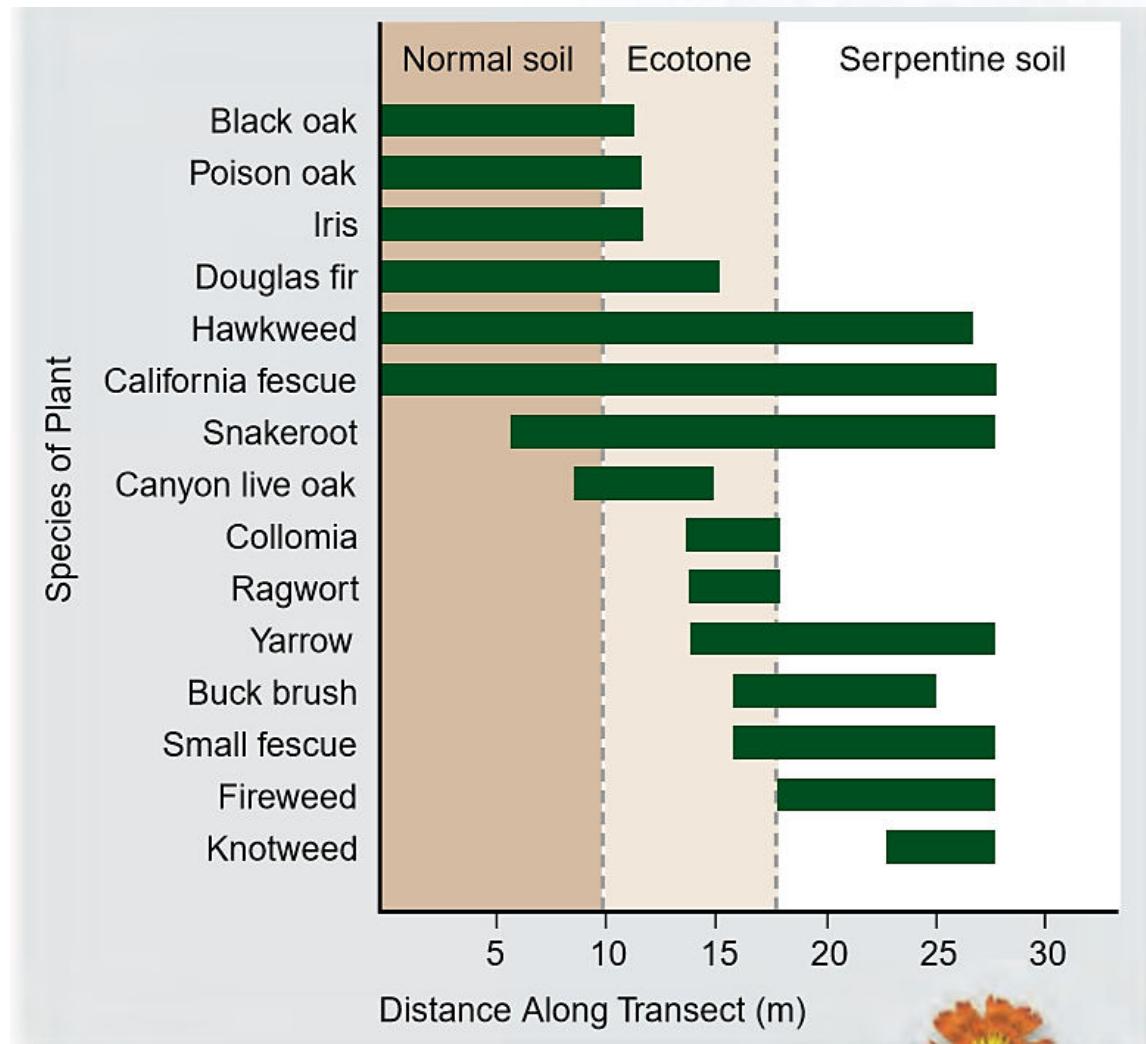
Hoslistic concept of communities

- Integrated and functioning as a whole unit
- Should stay the same through space or time
- Replaced by completely different communities with environmental changes

Communities

- **Ecologists support individualistic concept.**
- **Community composition changes gradually across landscapes**
- **Some species appear and become more abundant.**
- **Others decrease in abundance and eventually disappear.**

Communities



Biology Mason, Losos and Singer, 9th ed.



Change in community composition across an ecotone. The plant assemblages on normal and serpentine soils are greatly different, and the transition from one community to another occurs over a short distance. An ecotone is a transition area between two biomes. It is where two communities meet and integrate. Serpentine soil is an uncommon soil type produced by weathered ultramafic rock such as peridotite and its metamorphic derivatives such as serpentinite.

The Ecological Niche

- Each organism in a community confronts the challenge of survival in a different way.
- Occupies a specific functional role and place called its niche.
- The niche an organism occupies is the total of all the ways it uses the resources of its environment.

The Ecological Niche

- A niche may be described in terms of:
 - Space utilization
 - Food consumption
 - Temperature range
 - Appropriate conditions for mating
 - Requirements for moisture
 - Other factors

The Ecological Niche



<https://www.youtube.com/watch?v=p8NdASl4jY4>

The Ecological Niche

- Sometimes species cannot utilize their entire niche.
- Due to presence of other species competing for resources
- Interspecific competition, occurs between two species to use the same resources

The Ecological Niche

Fundamental Niche

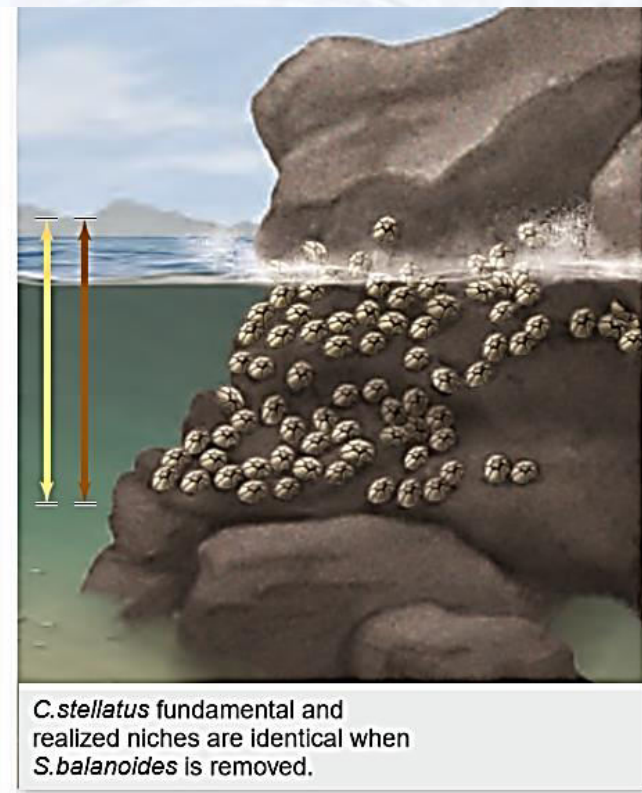
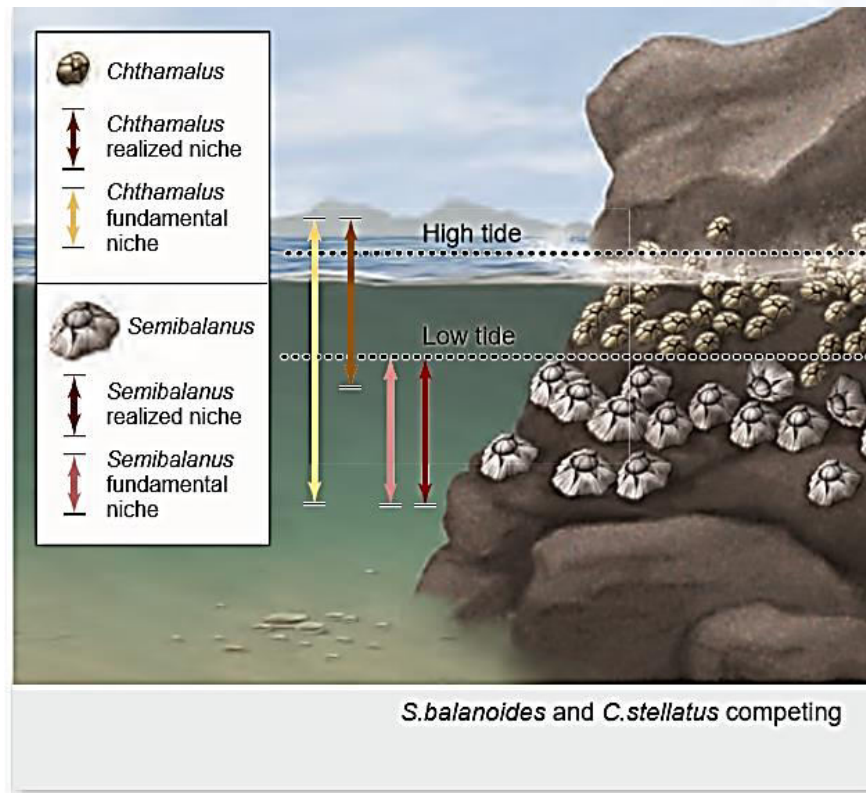
The entire niche that a species is capable of using, based on its physiological tolerance limits and resource needs.

The Ecological Niche

Realized Niche

The actual set of environmental conditions, including the presence or absence of other species, in which the species can establish a stable population.

The Ecological Niche



Biology Mason, Losos and Singer, 9th ed.

Community Stability

- Communities may begin in areas nearly devoid of life.
- The first community to become established in an area is called the **pioneer community**.
- Death, decay, new species and additional nutrients establish this community.

Community Stability

- Over thousands of years nutrients accumulate, communities established.
- Complex ecosystem becomes functional.
- Each successional stage is called a seral stage
- Entire successional sequence is a sere.

Community Stability

- Dominant life-forms of a sere gradually make the area less favorable for themselves due to consumption of resources.
- But more favorable for organisms of the next successional stage

Community Stability

- The final community is the **climax community**.
- It is different from the seral stages that preceded it because it can tolerate its own reactions.
- Complexity of ecosystem makes it more stable.

Community Stability

Types of Succession

- **Primary**

Occurring in an environment in which new substrate devoid of vegetation and other organisms usually lacking soil

- **Secondary**

An event reduces already established ecosystem.

Community Stability



<http://youtube.com/watch?v=vt3eiaduSnw>

Community Stability



<http://youtube.com/watch?v=jJozqo1opv8>

Interspecific Interactions

Herbivory

- Consumers of plants
- Anatomically and physiologically adapted to eating plant material
- Mutualistic gut flora e.g. cellulose digesting bacteria
- Important role in food chain

Interspecific Interactions

Herbivory

- Different feeding strategies

Major strategies:

1. Grazers
2. Browsers

Interspecific Interactions

Herbivory



https://en.wikipedia.org/wiki/File:Kangaroo_rudy.drs.jpg

1. Grazers

- Feeds on plants such as grasses, soft flowering plants (forbs) and algae
- Plants are not killed.
- Examples: Deer & Snails

Interspecific Interactions



<https://www.youtube.com/watch?v=7zYvIwDqhVU>

Interspecific Interactions

Herbivory

Types of Grazing

1. Graminivory

- Feeding primarily on grass
- Specifically "true" grasses in family Poaceae

Examples: Horses, cattle, hippopotamus, grasshoppers

Interspecific Interactions



<http://youtube.com/watch?v=AJbYCh3gbC8>

Interspecific Interactions

Herbivory

Types of Grazing

1. Coprophagy

- Feeding on grass, forbs, leafy weeds, fruits, tree bark
- Heavy grazing for half an hour of grazing period
- Followed by half an hour of selective feeding

Interspecific Interactions



caecotrophe

https://www.google.com/search?q=rabbits+eating+their+feces&rlz=1C1SQJL_enUS822US822&source=lnms&tbm=isch&sa=X&ved=0ahUKEwj4qtm72oviAhXHMewKHSPHAIQ_AUIDygC&biw=1280&bih=529#imgsrc=xc3YKgoG9e7kDM:

Herbivory

Types of Grazing

2. Coprophagy

- Later feeding in intervals
- Eat their or other species's soft feces (caecotrophs)

Example: Rabbit

Interspecific Interactions

Herbivory

2. Browsers

- Feed on non-grass plant material
- Leaves, soft shoots, twigs, fruits, woody plants such as shrubs

Examples: Goats, deer, giraffe, elephants

Interspecific Interactions



<https://www.youtube.com/watch?v=HXLXq-ZRT-M>

Interspecific Interactions

Herbivory

2. Browsers

- Over browsing
 - Introduction of herbivores
 - Low productivity of native plants
 - Less predation pressure

(a)



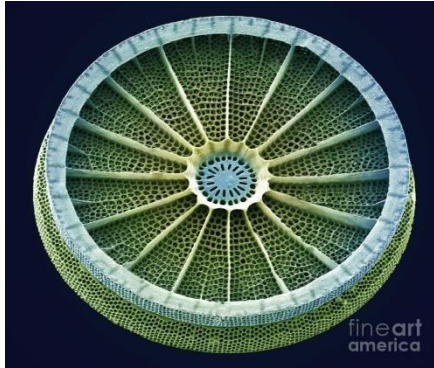
(b)



https://www.google.com/search?rlz=1C1SQJL_enUS822US822&biw=1280&bih=578&tbm=isch&sa=1&ei=cLHSXLWbBobUsAeV6JbYDQ&q=over+browsing&oq=over+browsing&gs_

Interspecific Interactions

Defense against Herbivory



https://en.wikipedia.org/wiki/Thorns,_spines,_and_prickles

1. Tolerance

2. Resistance

- Physical/Mechanical defense

Thorns, hair, resins, silica deposits in cell wall, resins make plants inedible

- Chemical defense

Interspecific Interactions

Defense against Herbivory

- Chemical defense
 - Terpenes, phenols, alkaloids
 - Give bitter taste to plants
 - Affect nervous system of herbivore

End

Physical, or mechanical, defenses are barriers or structures designed to deter herbivores or reduce intake rates, lowering overall herbivory. Thorns such as those found on roses or acacia trees are one example, as are the spines on a cactus. Smaller hairs known as trichomes may cover leaves or stems and are especially effective against invertebrate herbivores.[37] In addition, some plants have waxes or resins that alter their texture, making them difficult to eat. Also the incorporation of silica into cell walls is analogous to that of the role of lignin in that it is a compression-resistant structural component of cell walls; so that plants with their cell walls impregnated with silica are thereby afforded a measure of protection against herbivory.

Carbon-based defenses include terpenes and phenolics. Terpenes are derived from 5-carbon isoprene units and comprise essential oils, carotenoids, resins, and latex. They can have a number of functions that disrupt herbivores such as inhibiting adenosine triphosphate (ATP) formation, molting hormones, or the nervous system.[39] Phenolics combine an aromatic carbon ring with a hydroxyl group. There are a number of different phenolics such as lignins, which are found in cell walls and are very indigestible except for specialized microorganisms; tannins, which have a bitter taste and bind to proteins making them indigestible; and furanocoumarins, which produce free radicals disrupting DNA, protein, and lipids, and can cause skin irritation.

Nitrogen-based defenses are synthesized from amino acids and primarily come in the form of alkaloids and cyanogens. Alkaloids include commonly recognized substances such as caffeine, nicotine, and morphine. These compounds are often bitter and can inhibit DNA or RNA synthesis or block nervous system signal transmission.

Interspecific Interactions

Predation

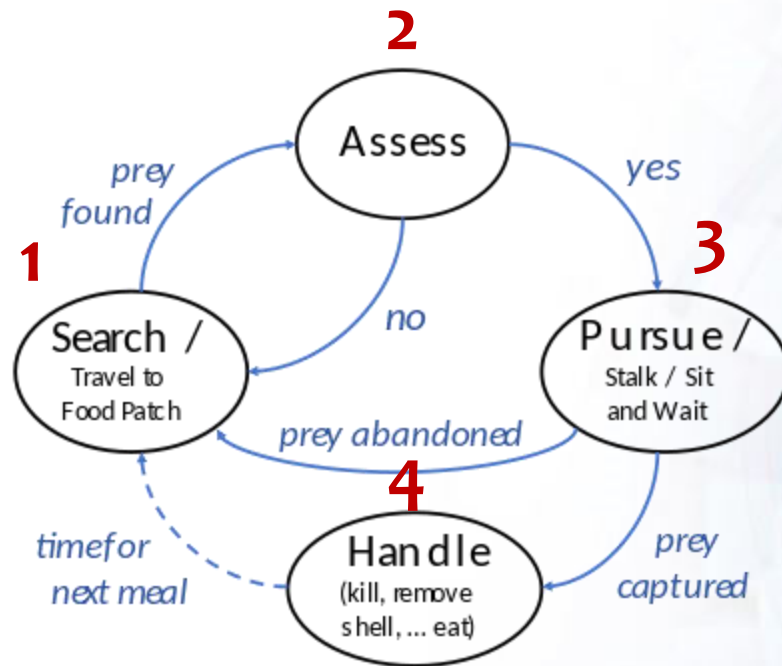
- One animal kills and eats another animal
- One is predator and other is prey
- Actively search for prey or sit and wait
- Examples: invertebrates such as jelly fish and insects
- Vertebrates such as lions

Interspecific Interactions



<https://www.youtube.com/watch?v=ffcyVpqngng>

Interspecific Interactions



<https://en.wikipedia.org/wiki/Predation>

Predation

- To feed, a predator must search for, pursue and kill its prey.
 - These stages form a foraging Cycle.
1. Search
 2. Assessment
 3. Capture
 4. Handling

Interspecific Interactions

Foraging Cycle



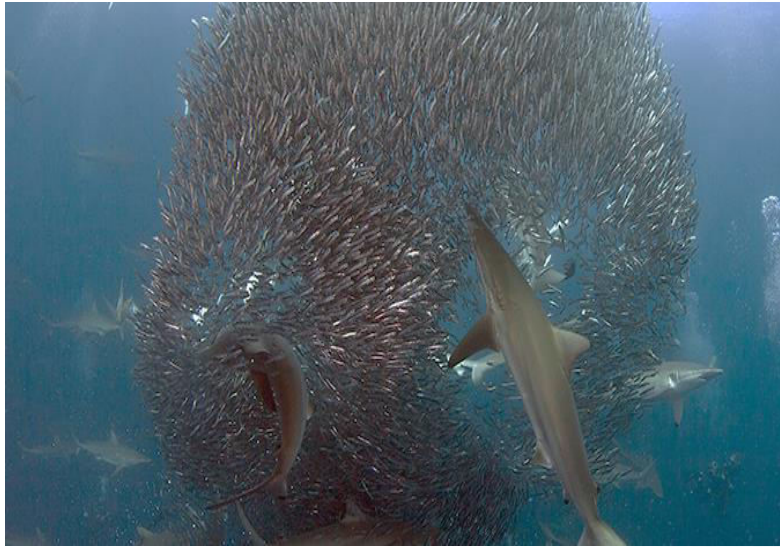
https://www.google.com/search?rlz=1C1SQJL_enUS822US822&biw=1280&bih=529&tbm=isch&sa=1&ei=DTTVXPmZM8O6aZi4vJgE&q=crocodiles+hunting&oq=crocodiles+hunting&gs

1. Search

- Sit and Wait method

- Suitable if the prey are dense and mobile
- Low energy requirement
- Example: Crocodiles, spiders, aquatic invertebrates, praying mantises

Interspecific Interactions



https://www.google.com/search?q=sharks+hunting&rlz=1C1SQJL_enUS822US822&source=lnms&tbm=isch&sa=wluoElf6s5-PLM:

Foraging Cycle

1. Search

- Active foraging

- When prey is sedentary or sparsely distributed.
- Expend more energy.
- Examples: Sharks, sunfish, Insectivorous birds

Interspecific Interactions

Foraging Cycle

- Some switch between active foraging and sit & wait method.

Example: Plovers, shore birds, crappie fish

- Prey distributions are often clumped in form of patches.

Interspecific Interactions

Foraging Cycle



https://www.google.com/search?rlz=1C1SQJL_enUS822US822&biw=1280&bih=529&tbm=isch&sa=1&ei=mTXVXIqUErCelwTWyqjIAQ&q=black-browed+albatross+hunting&oq=black-browed+albatross+hunting&gs

- Predators respond by looking for patches.
- Albatross make foraging flights within the prey range of 700 – 3000 km.

Interspecific Interactions

Foraging Cycle

2. Assessment

- Decide to kill or keep searching
- Decision depends on the costs and benefits
- Too small prey does not worth spending energy in capture.

Interspecific Interactions

Foraging Cycle



3. Capture

- **Pursuit predation**

Actively chasing fleeing prey

- **Ambush**



- **Sudden strike on nearby prey**
- **Adjusts its attack according to how the prey is moving.**

<https://en.wikipedia.org/wiki/Predation>

Interspecific Interactions



<https://en.wikipedia.org/wiki/Predation>

Foraging Cycle

3. Capture

- **Ballistic interception**
 - Predator observes the movement of a prey
 - Predicts its motion
 - Works out an interception path
 - Attacks the prey on that path

Interspecific Interactions



<https://en.wikipedia.org/wiki/Predation>

Foraging Cycle

4. Handling

- Prey can be dangerous
- Sharp and poisonous spines
- Should be handled carefully
- Mostly predators tear the prey apart
- E.g. cat fish

Interspecific Interactions

Predation Strategies

- Solitary predation

e.g. hawks

- Group predation

e.g. wolves, hyenas

- Physical adaptations

Better vision, olfaction and auditory skills

Claws, sharp teeth, beaks, venom,

END

Interspecific Interactions

Competition

- **Animals compete for same resources in ecosystem.**
- **Food**
- **Space**
- **Water**
- **Light**
- **Mates**

Interspecific Interactions



https://en.wikipedia.org/wiki/Interspecific_competition

Types of Competition

- Intraspecific

Between members of same species

- Interspecific

Between members of different species

Interspecific Interactions

Intraspecific Competition

- Same choice of resource
- Limited resources
- Reduction in fitness for individuals
- Follows logistic population growth curve

Interspecific Interactions

Types of Intraspecific Competition

1. **Direct**
 - **Results in fighting, stealing and ritualistic combats**
 - **Expelling weak competitors from territory make them vulnerable.**

Interspecific Interactions



<http://youtube.com/watch?v=d2phobqXrpg>

Interspecific Interactions



<https://www.pinterest.com/pin/231583605817014465/>

Types of Intraspecific Competition

2. Indirect

- Exploitative competition
- One depleting a shared resource
- Both suffering a loss in fitness as a result

Example: Bears catching fish near mouth of river

Interspecific Interactions

Inter-specific Competition

- **Competition between individuals of different species**
- **Comparatively variety of resources**
- **Logistic population growth curve**
- **Negative impact on each other**

Interspecific Interactions



<http://youtube.com/watch?v=M7EAKwrRXco>

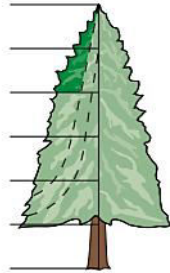
Interspecific Interactions

Types of Inter-specific Competition

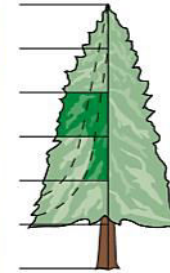
- Direct
- Example: Lions and Leopards
- Indirect

Example: Aphid- *Fordinae*
geoica out-compete *F.*
formicaria by 84%

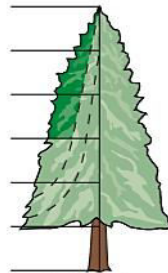
Interspecific Interactions



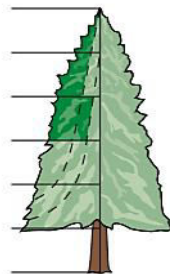
(a) Bay-breasted warbler



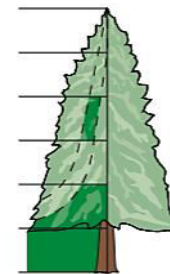
(b) Cape May warbler



(c) Blackburnian warbler



(d) Black-throated green warbler



(e) Myrtle warbler

Zoology by Miller and Harley, 5th ed.

Interspecific Interactions

Coevolution

- Two or more species living in same habitat, reciprocally affect each other's evolution.
- One exerts elective pressures on the other.

Interspecific Interactions

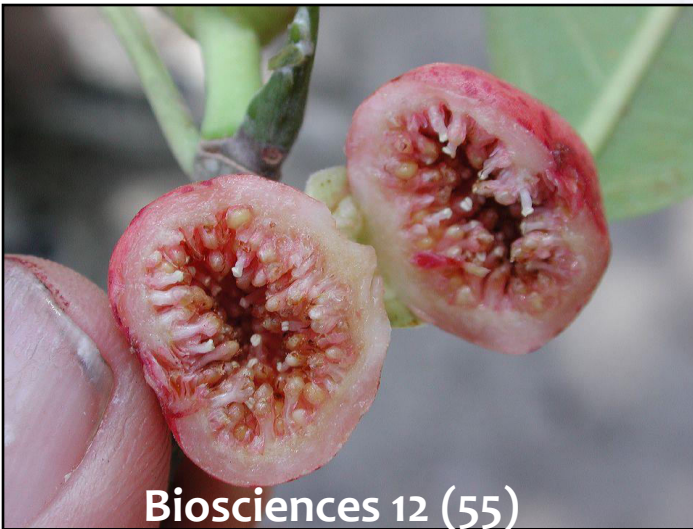
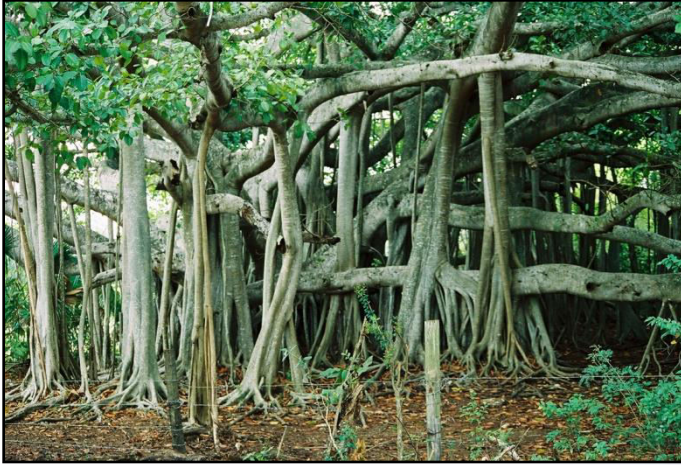
Coevolution

Mutualism

Both species coevolve to benefit each other in same habitat

- **Fig – Wasp Relationship**

Interspecific Interactions

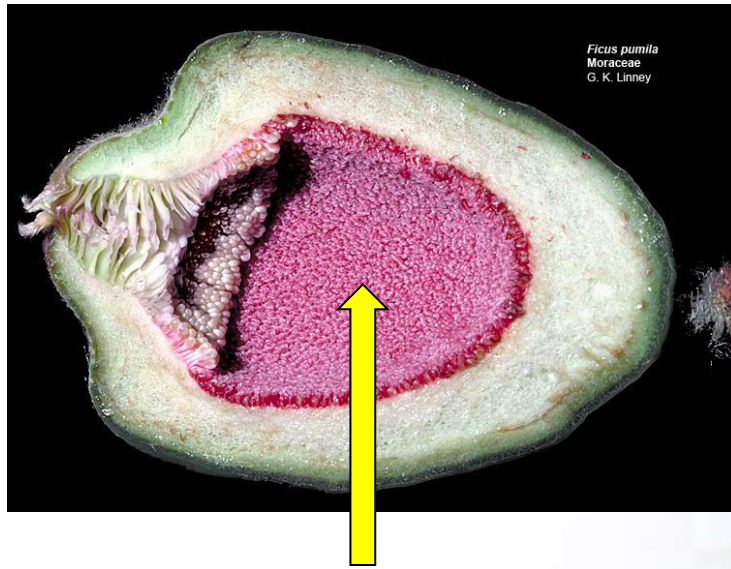


Biosciences 12 (55)

Coevolution

- ~750 tropical fig species
- All of which depend entirely on wasps for pollination
- Figs are not fruits, they are specialized inflorescences with hundreds of unisexual flowers.

Interspecific Interactions



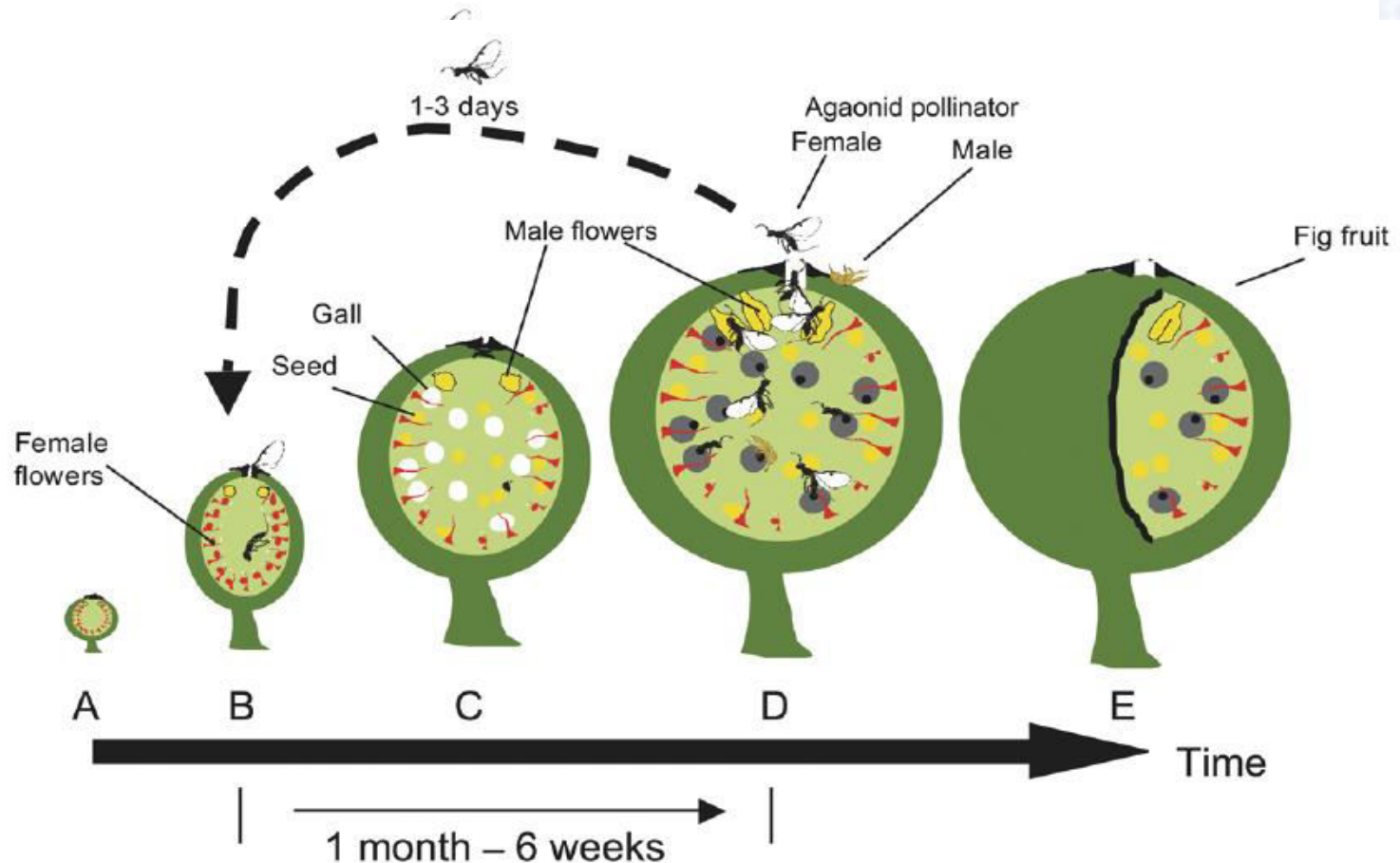
**Female wasp
enters via ostiole
and oviposits in
female flowers**

Biosciences 12 (55)

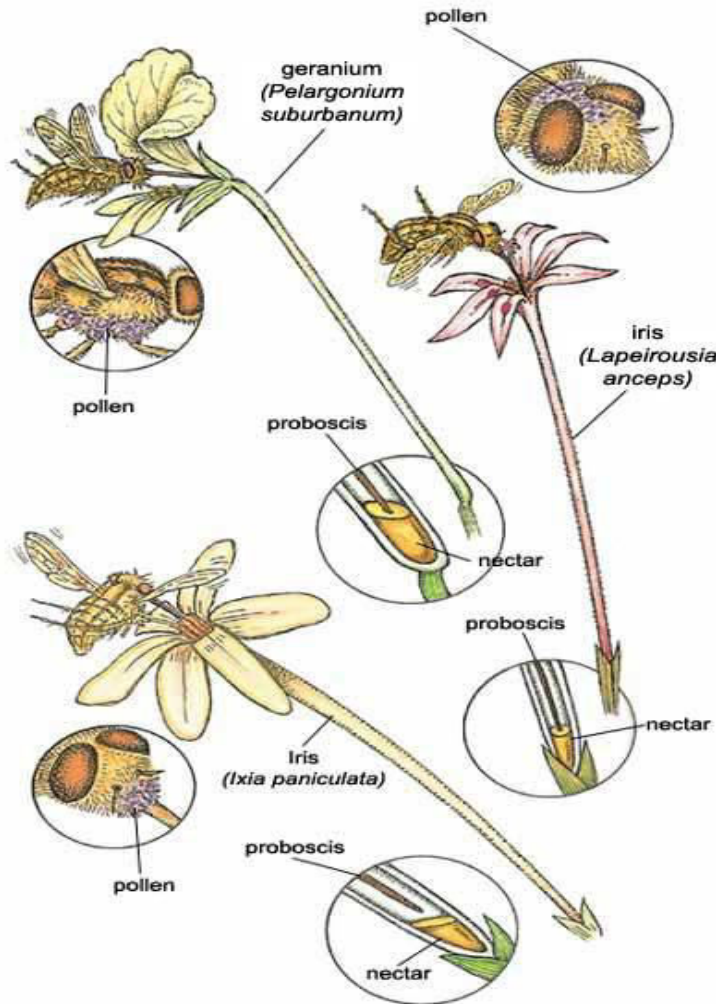
Coevolution

- Receptive figs produce scents that are specific to a particular pollinator species
- Shape of ostiole specific to head shape of particular wasp species (lock-and-key)
- Morphology of individual flowers specialized to a particular wasp species

Interspecific Interactions



Interspecific Interactions



Coevolution

- Plant-pollinator coevolution
- Flies and moths have outlandish proboscises
- Extract nectar from similarly outlandish flowers

Interspecific Interactions



Biosciences 12 (55)



Coevolution

- Attina Ants grow fungi in their colonies
- They provide leaves as food for fungi (yeasts) and eliminate competition for them.
- Fungi produce nutritional “gongylidia”
- These are hyphal swellings.

Interspecific Interactions



<https://www.picsbud.com/images/fungal-symbionts-gongylidia-ant-mycelium-26.html>

Coevolution

- Gongylidia are harvested by ants to feed their larvae.
- Coevolution exists from 50 million years.

Interspecific Interactions



<https://richsoil.com/antsandaphids/>

Coevolution

- Ants protect aphids from parasites and predators.
- Aphids produce honeydew, source of nutrition for ants.
- Specifically coevolved for ants

Interspecific Interactions

Symbiosis

Relationship between two different species living together

- **Obligate or facultative**

Types

1. **Mutualism**
2. **Commensalism**
3. **Parasitism**

Interspecific Interactions

Mutualism

- Long-term relationship between individuals of different species
- Both individuals get benefit

Example: Hermit crab and sea anemone

Interspecific Interactions



<https://www.youtube.com/watch?v=fzIDEoippf4>

Interspecific Interactions

Commensalism

- One species gain benefits while those of the other species is neither benefited nor harmed.

Example: Remora fish and Sharks

Interspecific Interactions



<https://www.youtube.com/watch?v=fzIDEoippf4>

Interspecific Interactions



<https://en.wikipedia.org/wiki/Parasitism>



<https://extension.umd.edu/hgic/topics/parasitoid-wasps-hymenoptera>

Parasitism

- One species gain benefits other species is harmed
- Intimate contact (feed off host)
- Usually do not kill host (parasitoids do)
- Enhances fitness of parasite but reduces fitness of host

Interspecific Interactions

Types of Parasites

1. Microparasites: reproduce inside host

Bacteria, viruses

2. Macroparasites: release juvenile outside host

E.g. trematodes

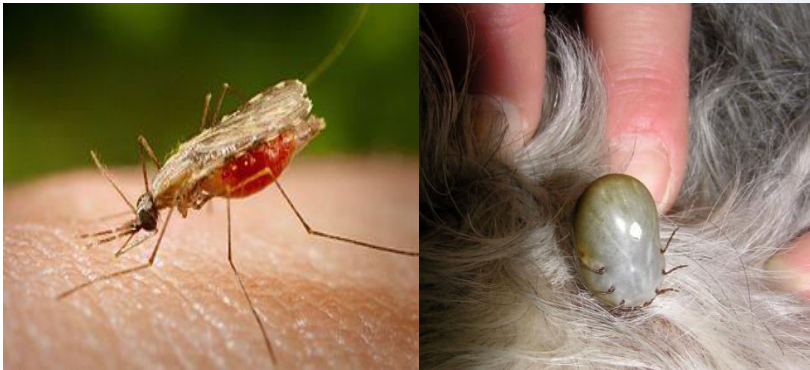
Interspecific Interactions



Types of Parasites

3. Endo parasites

Examples: Liver fluke, tape worm, nematodes



4. Ectoparasites

Examples: Ticks, mites, fleas, mosquito, lice

https://en.wikipedia.org/wiki/Parasitism#Parasitic_castrators

Interspecific Interactions



<https://en.wikipedia.org/wiki/Parasitism>



<https://extension.umd.edu/hgic/topics/parasitoid-wasps-hymenoptera>

Parasitism

- One species gain benefits other species is harmed
- Host and Parasite
- Intimate contact (feed off host)
- Usually do not kill host (parasitoids do)
- Enhances fitness of parasite but reduces fitness of host

Interspecific Interactions



The parasitic castrator
Sacculina carcini attached
to its crab host.

<https://en.wikipedia.org/wiki/Parasitism>

Evolutionary strategies of Parasitism

1. Parasitic castrators

- Partly or completely damages the reproductive system of host
- Other systems are left intact
- Host's energy used for growth of parasites

Interspecific Interactions



Human head lice are directly-transmitted obligate ectoparasites.

<https://en.wikipedia.org/wiki/Parasitism>

Evolutionary strategies of Parasitism

2. Directly transmitted

- Do not need vector for transmission to host
- Single host life cycle

Interspecific Interactions



**Trypanosoma among
human red blood cells**

<https://en.wikipedia.org/wiki/Parasitism>

Evolutionary strategies of Parasitism

3. Vector transmitted

- Need a vector for transmission to final host
- Life cycle involves an intermediate host.
- Do not sexually reproduce in intermediate hosts

Interspecific Interactions

Evolutionary strategies of Parasitism

4. Parasitoids

- Mostly are insects
- Sooner or later kill their host
- Types:
 1. Idiobiont
 2. Koinobiont

Interspecific Interactions



Idiobiont parasitoid wasps immediately paralyse their hosts for their larvae to eat.

<https://en.wikipedia.org/wiki/Parasitism>

Evolutionary strategies of Parasitism

- **Idiobiont parasitoids**
- **Sting their prey on capture**
- **Either killing them outright or paralysing them immediately**

Interspecific Interactions



Koinobiont parasitoid wasps like this braconid lay their eggs inside their hosts, which continue to grow and moult.

<https://en.wikipedia.org/wiki/Parasitism>

Evolutionary strategies of Parasitism

- Koinobiont parasitoids
- Flies as well as wasps
- Lay their eggs inside young hosts, usually larvae.
- Allowed hosts to grow
- Parasitoids emerge as adults, leaving the prey dead, eaten from inside

Interspecific Interactions



Mosquitoes are micropredators and important vectors of disease.

<https://en.wikipedia.org/wiki/Parasitism>

Types of Parasites

5. Micropredators

- Attacks more than one host
- Reducing each host's fitness
- Most micropredators are hematophagous, feeding on blood.

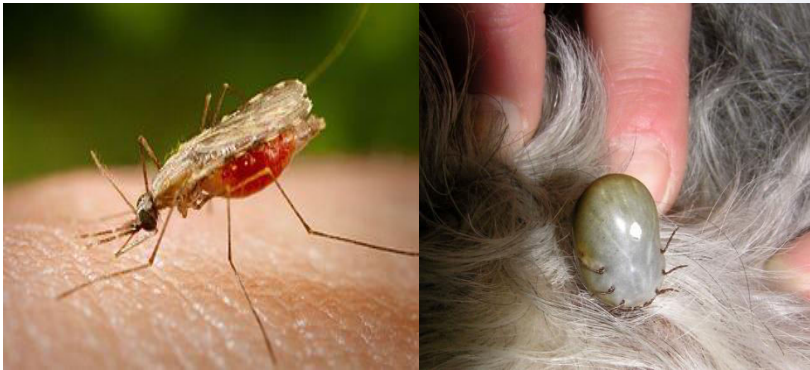
Interspecific Interactions



Types of Parasites

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Examples: Liver fluke, tape worm, nematodes



4. Ectoparasites

Examples: Ticks, mites, fleas, mosquito, lice

https://en.wikipedia.org/wiki/Parasitism#Parasitic_castrators

Interspecific Interactions

Camouflage

- Use of any combination of materials, coloration, or illumination
- Concealment from predators
- Also known as cryptic coloration

Interspecific Interactions

Principles of Camouflage

1. Crypsis

Objects hide and hard to see

2. Mimicry

Disguising them as something else

3. Motion dazzle

Using visual illusions

- Protect without hiding

Interspecific Interactions



http://youtube.com/watch?v=__ZvtiRVlpk

Interspecific Interactions



https://en.wikipedia.org/wiki/Camouflage#Resemblance_to_surroundings

Crypsis

1. **Resemblance to surroundings**
 - Colors and patterns resemble a particular natural background
 - Change their color if environment changes
 - Example: Egyptian nightjar, Parrot

Interspecific Interactions



https://en.wikipedia.org/wiki/Camouflage#Resemblance_to_surroundings

Crypsis

2. Disruptive coloration

- Strongly contrasting, non-repeating markings such as spots or stripes
- It breaks up the outlines of an animal.
- Need other strategies for survival
- Example: Leopard, Giraffe

Interspecific Interactions



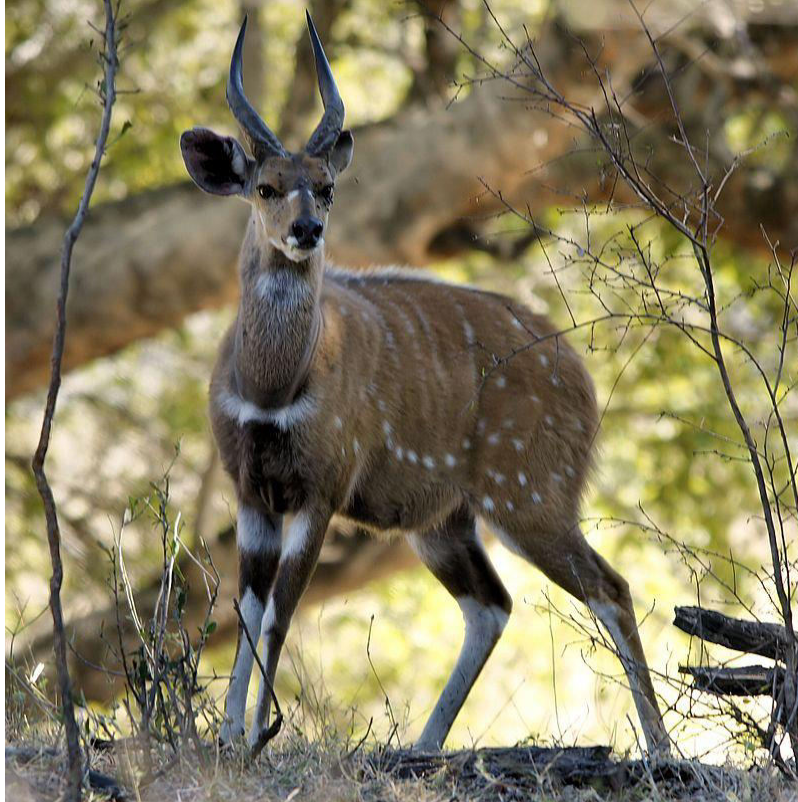
https://en.wikipedia.org/wiki/Camouflage#Resemblance_to_surroundings

Crypsis

3. Eliminating shadow

- Flattened bodies
- Sides thinning to an edge
- Animals habitually press their bodies to the ground
- Sides are fringed with white scales
- Example: Horned Lizard

Interspecific Interactions



https://en.wikipedia.org/wiki/Camouflage#Resemblance_to_surroundings

Crypsis

4. Distraction

- High-contrast markings which attract the predator's gaze
- Distract predator from recognizing whole prey
- Distract from identifying prey's outline
- Example: Bushbuck

Interspecific Interactions



https://en.wikipedia.org/wiki/Camouflage#Resemblance_to_surroundings

Crypsis

4. Self Decoration

- Decorating with materials such as twigs, sand, or pieces of shell
- Break up their outlines
- Conceal the features of their bodies
- Match their backgrounds
- Example: Decorator crabs

Interspecific Interactions



https://en.wikipedia.org/wiki/Camouflage#Resemblance_to_surroundings

Crypsis

5. Changeable skin coloration

- Actively change their skin patterns and colours
- Change can be rapid or seasonal
- Chameleons rapidly change their color.

Interspecific Interactions



<http://youtube.com/watch?v=nHZNY5DSYeQ>

Interspecific Interactions



https://en.wikipedia.org/wiki/Camouflage#Resemblance_to_surroundings

Crypsis

5. Changeable skin coloration

- Arctic fox changes from brown or grey in the summer to white in the winter
- Use special chromatophore cells, fur molting to resemble their current background.

Interspecific Interactions



https://en.wikipedia.org/wiki/Camouflage#Resemblance_to_surroundings

Crypsis

6. Countershading

- Shadow makes upper side darker and underside lighter
- Countershading graded color to counteract the effect of self-shadowing

Interspecific Interactions



http://youtube.com/watch?v=2yD5TDJV_no

Interspecific Interactions

Crypsis

6. Countershading

- Creating an illusion of flatness
- Makes animal darker below and lighter above

Example: gazelles, grasshoppers, sharks, dolphins

Interspecific Interactions

Crypsis

7. Counter-illumination

- Producing light to match a background that is brighter than an animal's body
- Light-producing organs (photophores) scattered all over its underside
- Example: hatchet fish, Firefly squid

Interspecific Interactions

Crypsis

7. Counter-illumination

- Illumination makes them lighter in color from below
- Eliminates shadow
- Protect them from predator

Interspecific Interactions



http://youtube.com/watch?v=__ZvtiRVlpk

Interspecific Interactions



https://en.wikipedia.org/wiki/Camouflage#Resemblance_to_surroundings

Crypsis

8. Transparency

- Few marine animals float near the surface are highly transparent
- Makes them invisible to predators at certain distance
- More efficient in deeper water
- Example: Jellyfish

Interspecific Interactions



https://en.wikipedia.org/wiki/Camouflage#Resemblance_to_surroundings

Crypsis

9. Silvering

- Transparency can be imitated by silvering
- It makes body reflective
- Light comes from above reflected by mirror like body
- Make fish invisible from the side
- Example: Herring fish

Interspecific Interactions



https://en.wikipedia.org/wiki/Camouflage#Resemblance_to_surroundings

Motion dazzle

- Crypsis might be ineffective when animal moves
- Motion makes it visible to predator
- Rapidly moving bold patterns of contrasting stripes counteracts this risk
- Example: Zebra

Interspecific Interactions

Motion dazzle

- Due to optical illusion
- Motion dazzle may degrade predators' ability to estimate the prey's speed and direction accurately
- Giving the prey an improved chance of escape

Interspecific Interactions



<https://www.youtube.com/watch?v=IISNFkRkSxl>

Mimicry

- The resemblance of one species to another for protective or aggressive purposes
- Different from camouflage
- They imitate instead of hiding by camouflage strategies

Mimicry

Types of Mimicry

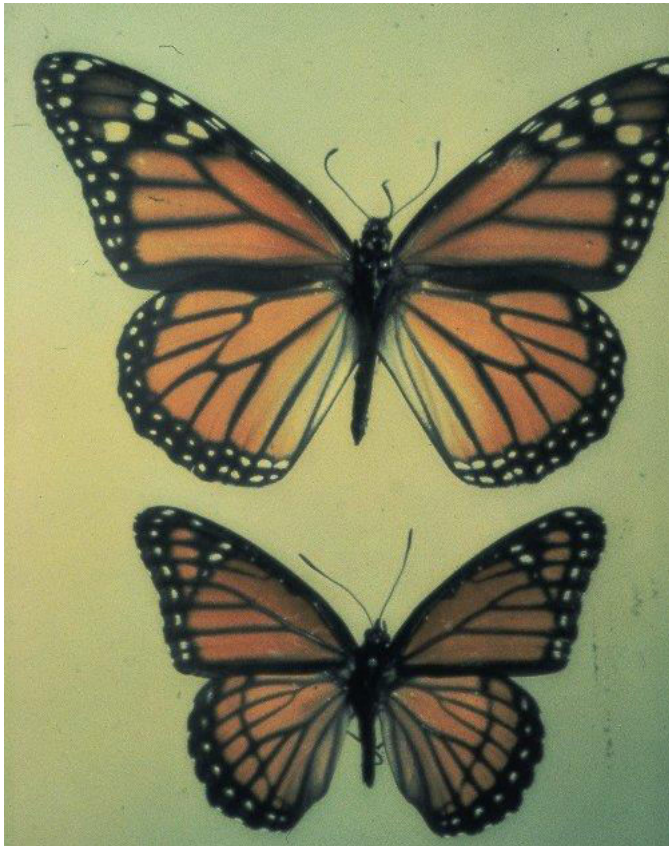
- On basis of adopting different models to mimic:
 1. Batesian Mimicry
 2. Mullerian Mimicry

Mimicry

Batesian Mimicry

- Association in which the members play two different roles
 1. Model
 2. Mimic
- Model – a species which possesses the inherent protection, e.g. poisonous chemicals in insects e.g. Butterflies

Mimicry



<https://en.wikipedia.org/wiki/Mimicry>

Batesian Mimicry

- The Mimic – a species which lacks the basic protection but through resembling the model superficially gains protection from predators
- Model: Monarch butterfly
- Mimic: Viceroy butterfly

Mimicry

Batesian Mimicry

- Mimics usually smaller than models
- Mimicry restricted to females
- One sex must stay the same to be recognizable
- Females do the choosing so males must be the original form

Mimicry

Batesian Mimicry

- Abundance of the mimic is limited by its effectiveness
- If too common then predators learn the wrong signal.
- Numbers increase to point of selective neutrality where benefit equals cost.

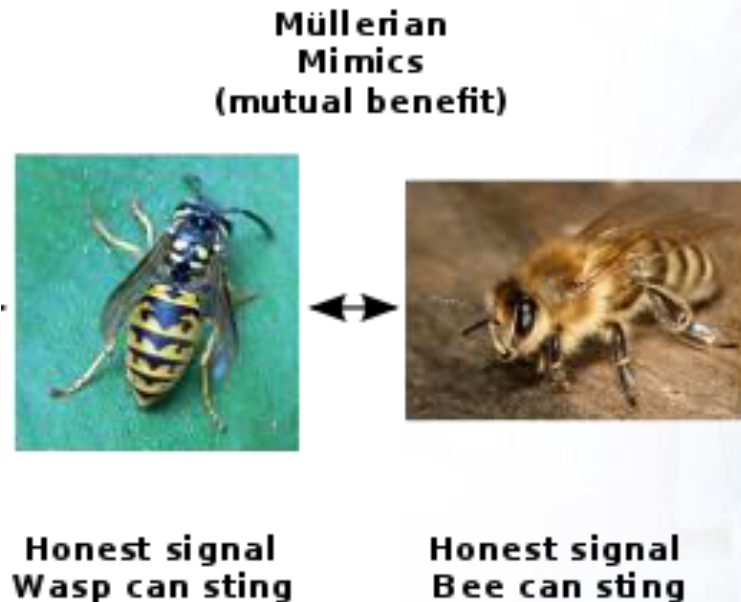
Mimicry

Batesian Mimicry

- Polymorphism is common
- If morph A is at selective neutrality
- New morph B, mimicking a different model, will be at a selective advantage

Mimicry

Mullerian Mimicry



<https://en.wikipedia.org/wiki/Mimicry>

- Distasteful species, that may or may not be closely related
- Share one or more common predators
- Mimic each other's honest warning signals, to their mutual benefit
- Predators learn to avoid all of them

Mimicry

Mullerian Mimicry



- Polymorphism is infrequent
- Models tend to stay the same because there is a disadvantage in being different
- No limit to number of species participating

<https://en.wikipedia.org/wiki/Mimicry>

Mimicry

Mullerian Mimicry

- Different colors depend on location and visibility
- Groups at different levels in tropical forest have different colors due to different light levels

Aposematism

- **Appearance of an animal that warns predators it is toxic, distasteful or dangerous.**
- **Advertisement of signals**
 1. **Color**
 2. **Odours**
 3. **Sound**
 4. **Aggression**

Aposematism

- Function is to prevent attack.
- Beneficial for both predator and prey, saving them from harm.
- Types of Signals
 - Primary: Easily detectable (colors)
 - Secondary: invisible (ultrasonic noise for specific predator)

Aposematism



- Primary Signals
- Bright and high contrast colors
- Patterns such as stripes
- Effective colors are red, yellow, black and white
- Colors resist changes in shadow and lighting.

https://en.wikipedia.org/wiki/Aposematism#Defence_mechanism

Aposematism

Warning coloration evolves in response to:

- Background
- Light conditions
- Predator vision

Other visible signals

- Odour
- Sound
- Aggressive behavior

Aposematism



Examples:

Tiger moth: Bitter taste and ultrasonic sounds

Skunk: Noxious odour

Dart frog: Poison

Velvet ant: Sting and noise



https://en.wikipedia.org/wiki/Aposematism#Defence_mechanism



He [honey badger](#)'s [reverse countershading](#) makes it conspicuous, [honestly signalling](#) its [ability to defend itself](#) through its aggressive temperament, and its sharp teeth and claws.

Aposematism



<http://youtube.com/watch?v=WZqXfFgu9MQ>

Aposematism



Examples:

Black widow spider: Neuro toxin

Honey badger: Aggressive



https://en.wikipedia.org/wiki/Aposematism#Defence_mechanism

Aposematism



<http://youtube.com/watch?v=v3WjnLn9p3U>

Aposematism



https://en.wikipedia.org/wiki/Aposematism#Defence_mechanism

Prevalence

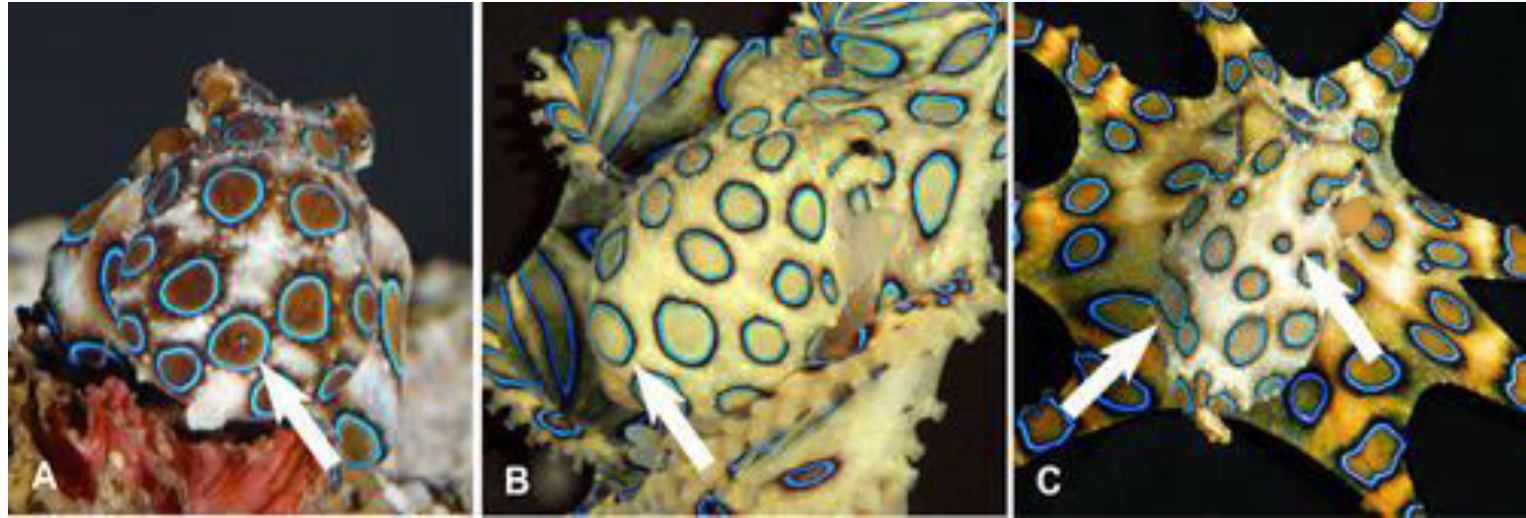
Both terrestrial and marine ecosystem

- Mostly insects

Marine animals

1. Nudibranch
2. Crown-of-thorns starfish

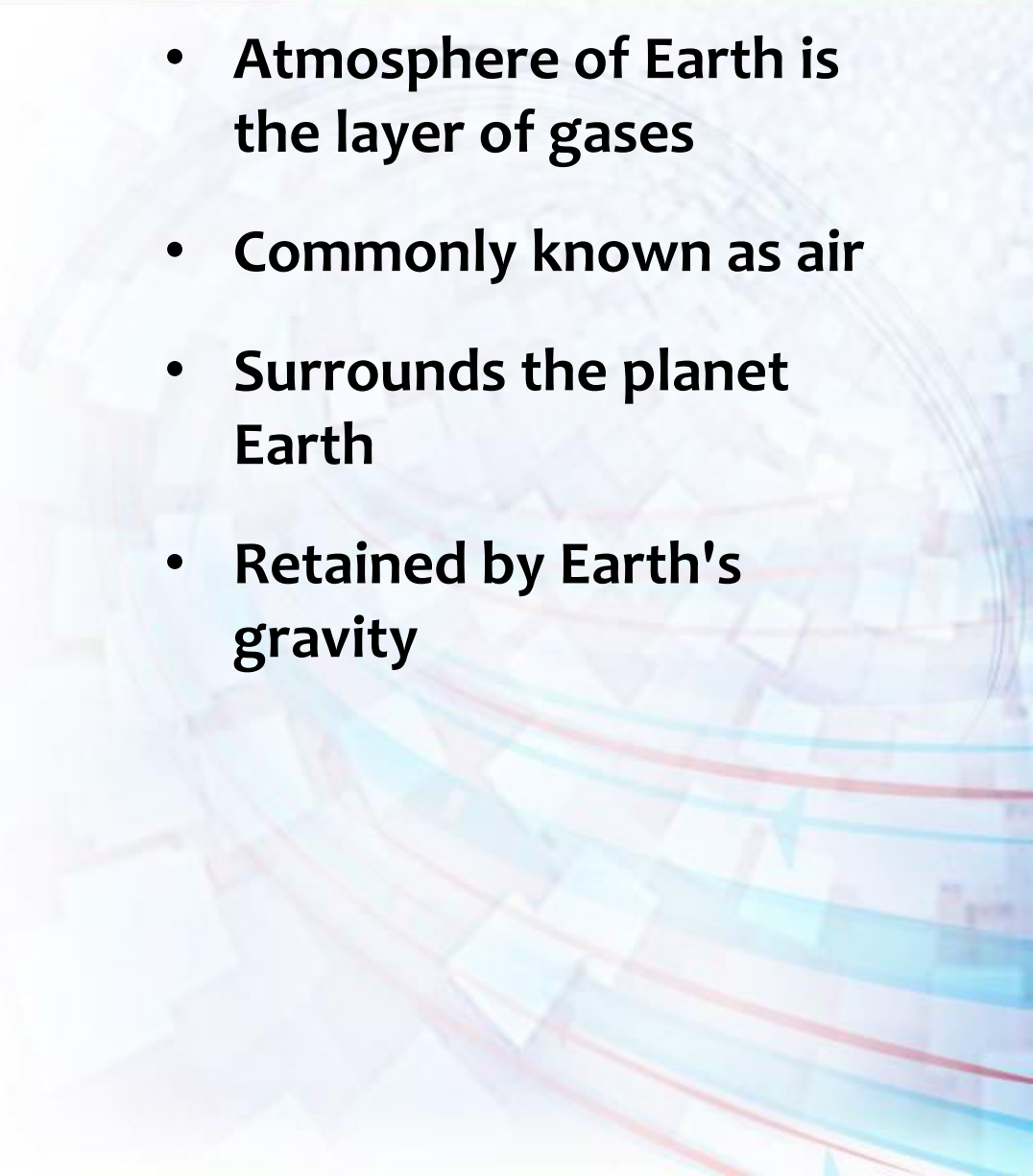
Aposematism



Iridescent blue rings on the mantles of the venomous octopus are considered by some to be aposematic

https://en.wikipedia.org/wiki/Aposematism#Defence_mechanism

Atmosphere and its Evolution

- **Atmosphere of Earth is the layer of gases**
 - **Commonly known as air**
 - **Surrounds the planet Earth**
 - **Retained by Earth's gravity**
- 

Atmosphere and its Evolution

Evolution

1. First Atmosphere

Composition of gases like solar nebula (disc of dirt and gases)

Aged 4 billion years ago

- Hydrogen
- Methane
- Helium
- Ammonia
- Water vapors

Atmosphere and its Evolution



https://en.wikipedia.org/wiki/Types_of_volcanic_eruptions

Evolution

2. Second Atmosphere

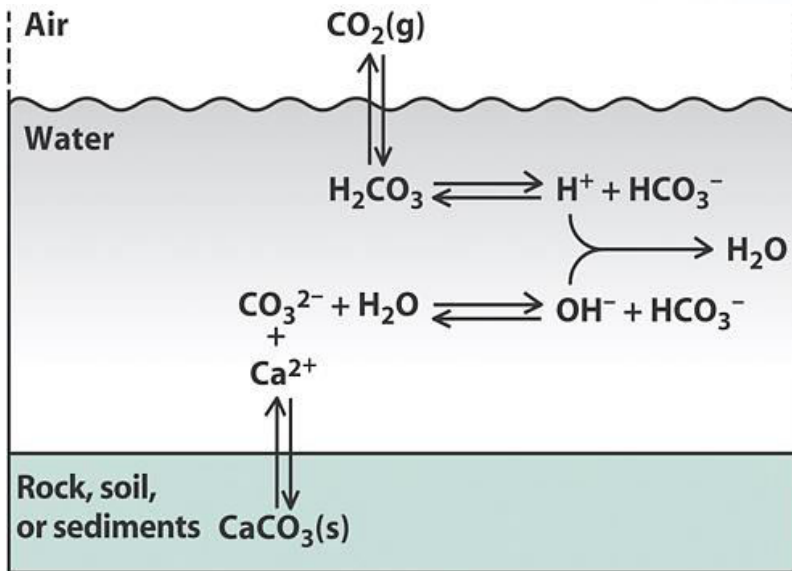
- Outgassing from volcanism
- Supplemented by gases produced during heavy bombardment of Earth by huge asteroids
- Added N_2 , CO_2 and inert gases

Atmosphere and its Evolution

Evolution

2. Second Atmosphere

- Carbon-dioxide emissions dissolved in water
- Reacted with metals such as calcium and magnesium
- Formed carbonates that were deposited as sediments.



<https://www.google.com/search?q=calcium+carbonate+formation+from+carbon+dioxide>

Atmosphere and its Evolution

Evolution

2. Second Atmosphere

- Water-related sediments have been found
- Dated from as early as 3.8 billion years ago
- 3.4 billion years ago, molecular nitrogen formed
- Earth temperature was warm.

Atmosphere and its Evolution

Evolution

2. Second Atmosphere

- 2.5 billion years ago, oxygen was developed
- Activity of photosynthesizing cyanobacteria
- Known as “Great Oxygenation Event”

Atmosphere and its Evolution

Evolution

2. Second Atmosphere

- Activity of photosynthesizing cyanobacteria



https://en.wikipedia.org/wiki/Great_Oxygenation_Event

Atmosphere and its Evolution

Evolution

2. Second Atmosphere

- Known as “Great Oxygenation Event”

Atmosphere and its Evolution

Evolution

3. Third Atmosphere

- Constant rearrangement of continental plates
- Inflow and outflow of CO_2 from large continental carbonate stores

Atmosphere and its Evolution

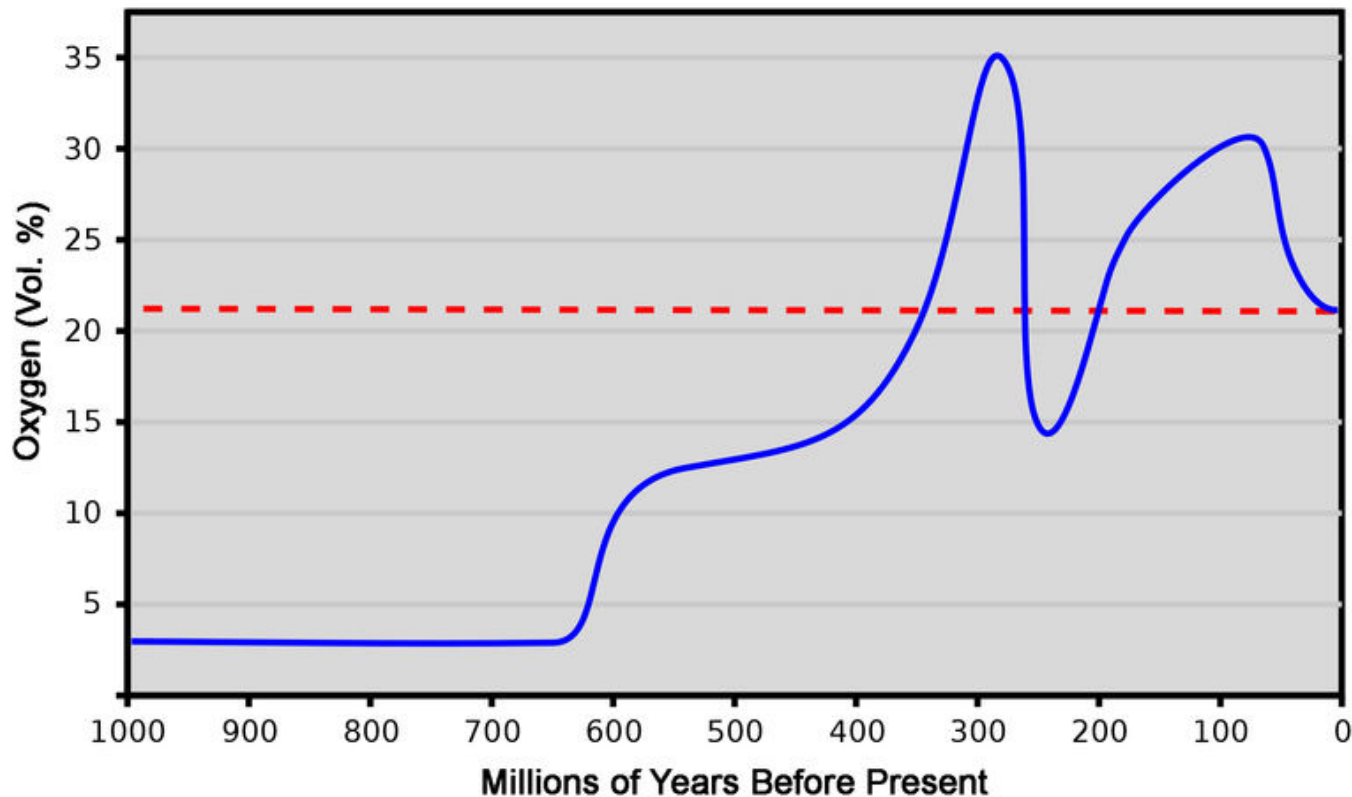
Evolution

3. Third Atmosphere

- O_2 synthesized in photosynthesis was being used in reducing materials
- Gradually, it was shifted to oxidizing atmosphere instead of reducing
- 541 million years ago, O_2 made stable 15% of atmosphere.

Atmosphere and its Evolution

Oxygen Content of Earth's Atmosphere
During the Course of the Last Billion Years



<https://en.wikipedia.org/wiki/File:Sauerstoffgehalt-1000mj2.png>

Atmosphere and its Evolution

Evolution

3. Third Atmosphere

Reasons of fluctuations in O_2

- Plants produced oxygen and consumed CO_2
- Volcanic eruptions release sulfur into the atmosphere

Atmosphere and its Evolution

Evolution

3. Third Atmosphere

Reasons of fluctuations in O_2

- Sulfur oxidizes itself
- Reduces the amount of oxygen in the atmosphere
- Volcanic eruption also released CO_2
- Exact causes of fluctuations are unknown.

Atmosphere and its Evolution



<http://youtube.com/watch?v=uCjmwckKeNaQ>

Composition of Atmosphere

Comprised of a mixture of:

- 1. Invisible permanent gases**
- 2. Variable gases**
- 3. Suspended microscopic particles**

Composition of Atmosphere

1. Permanent Gases

Form a constant proportion of the total atmospheric mass

2. Variable Gases

Distribution and concentration varies in space and time

3. Aerosols

Suspended particles and liquid droplets (excluding cloud droplets)

Composition of Atmosphere

1. Permanent Gases

78% Nitrogen (N_2)

21% Oxygen (O_2)

<1% Argon (Ar)

- **Relative percentages of the permanent gases remain constant up to 80 - 100km high.**
- **Turbulent mixing causes atmospheric composition to be fairly homogenous.**
- **Called as Homosphere**

Composition of Atmosphere

2. Variable Gases

- **Water vapor (H_2O):**
0 to 4%
- **Carbon Dioxide (CO_2):**
0.038%
- **Methane(CH_4):** 0.00017%
- **Ozone(O_3):** 0.000004%

Composition of Atmosphere

Variable Water vapors

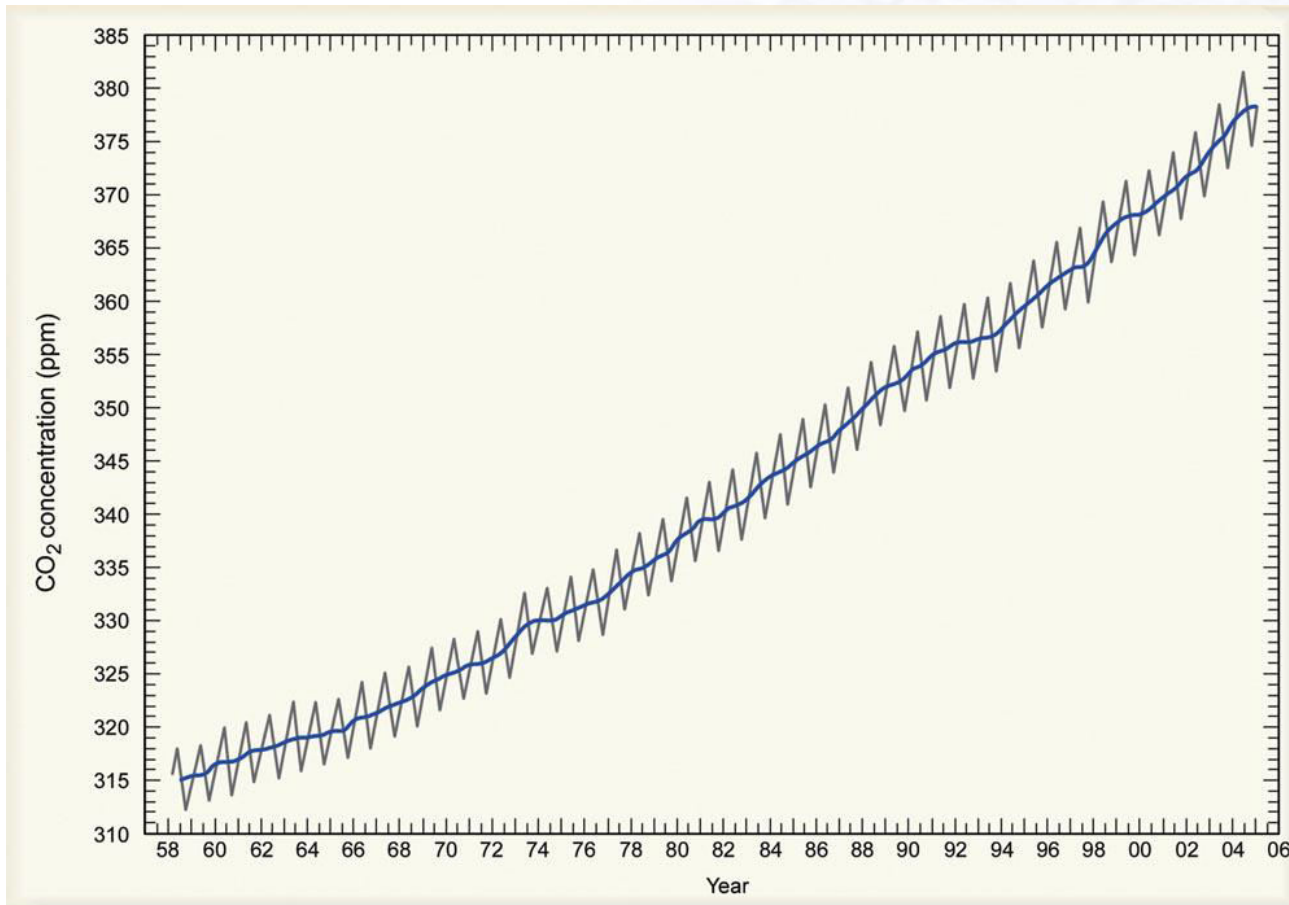
- Water vapor is invisible.
- Its not cloud droplets.
- 3rd major component
- Surface percentages vary between <1% in desserts to 4% in tropics.
- Typical mid-latitude value is about 1-2%.

Composition of Atmosphere

Variable Carbon Dioxide

- Small percentage of total atmosphere (380 ppm)
- But, very important green house gas

Composition of Atmosphere



© 2007 Thomson Higher Education

**Mauna Loa Observatory CO₂ trace
(annual variations embedded in the long-term record)**

Composition of Atmosphere

Variable Ozone

- Near the surface, ozone concentrations about 0.04-0.15 ppm
- In the upper atmosphere ozone concentration can reach ~15 ppm
- Upper atmospheric ozone is vital to blocking harmful radiation
- Ozone near the surface, however, harmful to life

Composition of Atmosphere

Variable Methane

- Concentrations of about 1.7 ppm
- Extremely potent green house gas - 21 times more powerful by weight than carbon dioxide
- Unexpected increase in last 400,000 years
- Increase attributed to agriculture, industries, bio-mass and fossil fuel burning

Composition of Atmosphere

3. Aerosols

- Small (or “tiny”) solid particles or liquid droplets (excluding clouds and rain)
- Aerosols can be man-made (anthropogenic) or naturally occurring (like ocean salt, dust, plant emissions)

Composition of Atmosphere

3. Aerosols

- Aerosols are not synonymous with pollution
- Some aerosols are very beneficial and, in fact, are required for precipitation processes to occur.

Composition of Atmosphere

Major constituents of dry air, by volume

Gas		Volume ^(A)	
Name	Formula	in ppmv ^(B)	in %
Nitrogen	N ₂	780,840	78.084
Oxygen	O ₂	209,460	20.946
Argon	Ar	9,340	0.9340
Carbon dioxide	CO ₂	400	0.04 ^[9]
Neon	Ne	18.18	0.001818
Helium	He	5.24	0.000524
Methane	CH ₄	1.79	0.000179
Not included in above dry atmosphere:			
Water vapor ^(C)	H ₂ O	10–50,000 ^(D)	0.001%–5% ^(D)

https://en.wikipedia.org/wiki/Atmosphere_of_Earth#Composition

Layers of Atmosphere

- Air pressure and density decrease with altitude
- Temperature shows different patterns.
- It increases with altitude in some regions.
- Temperature is useful tool to distinguish atmospheric layers.

Layers of Atmosphere

- Atmosphere can be divided into five layers.
- Highest to lowest

Layers of atmosphere

1. Exosphere: 700 to 10,000 km (440 to 6,200 miles)
2. Thermosphere: 80 to 700 km (50 to 440 miles)[12]
3. Mesosphere: 50 to 80 km (31 to 50 miles)

Layers of Atmosphere

**4. Stratosphere: 12 to 50 km
(7 to 31 miles)**

**5. Troposphere: 0 to 12 km
(0 to 7 miles)**

Layers of Atmosphere



http://youtube.com/watch?v=1Xtcqlv_EHs

Troposphere

- The lowest layer
- All weather conditions take place here.
- Conventional aviation takes place here.
- It contains:
- 75% of the atmosphere's mass
- 99% of the total mass of water vapor and aerosols

Troposphere

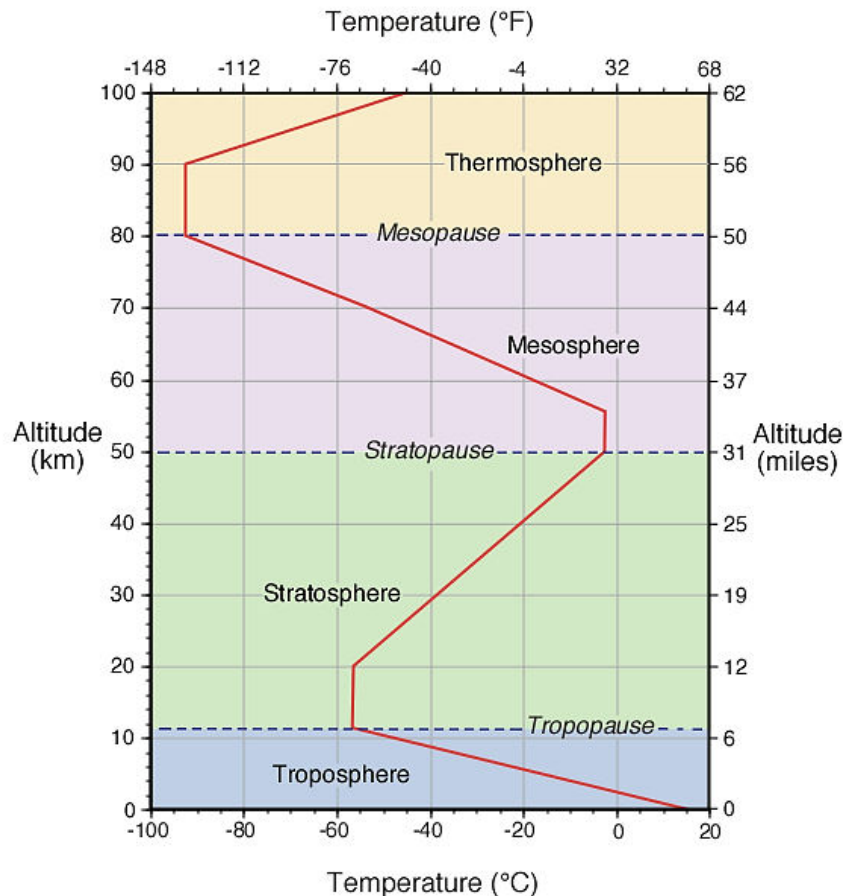
Height



[https://en.wikipedia.org/wiki/Troposphere#
Pressure_and_temperature_structure](https://en.wikipedia.org/wiki/Troposphere#Pressure_and_temperature_structure)

- 18 km (11 mi; 59,000 ft) in the tropics
- 17 km (11 mi; 56,000 ft) in the middle latitudes
- 6 km (3.7 mi; 20,000 ft) in the polar regions in winter.
- The total average height of the troposphere is 13 km.

Troposphere



<http://users.isp.com/retic/atmslayers.gif>

Properties

- Composition of air is homogenous except water vapors.
- Temperature of the troposphere decreases with altitude.
- Ground absorbs most of the sun's energy
- It makes the lower levels of the atmosphere warm.

Troposphere

Properties

- Radiating of heat away from earth at the top of the atmosphere
- It results in the cooling of that part of the atmosphere.

Troposphere

Properties

- Vapor pressure drops with decrease in temperature.
- Drop in vapor pressure is due to gravity.
- Pressure of the atmosphere is maximum at sea level and decreases with altitude.

Troposphere

Tropopause



The orange layer is the troposphere, the white layer is the stratosphere, and the blue layer is the mesosphere.

https://en.wikipedia.org/wiki/Troposphere#Pressure_and_temperature_structure

- Boundary region between the troposphere and the stratosphere
- Temperature decrease with increase in altitude.

Troposphere

Tropopause

Height

- 17 kilometres (11 mi) above equatorial regions
- 9 kilometres (5.6 mi) over the polar regions

END

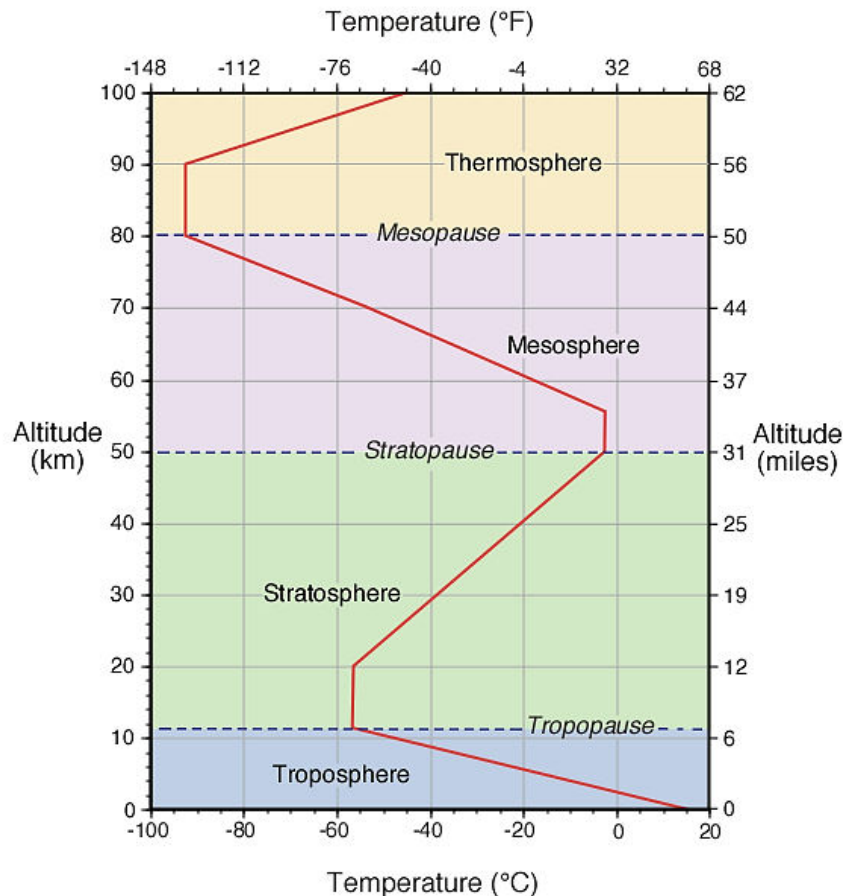
Stratosphere

- The second major layer of Earth's atmosphere
- Above the troposphere, and below the mesosphere

Height

- 20 km (66,000 ft; 12 mi) at equator
- 10 km (33,000 ft; 6.2 mi) at midlatitudes
- 7 km (23,000 ft; 4.3 mi) at the poles

Stratosphere

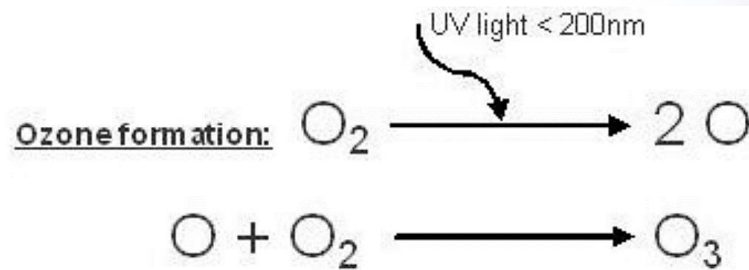


<http://users.isp.com/retic/atmslayers.gif>

Properties

- Low density/pressure of gases due to reduced gravity.
- Ozone is important gas.
- Temperature increases by increase in altitude.
- Temperatures range from -51°C near the tropopause
- To an average of -15°C near the mesosphere

Stratosphere



<http://www.itclamps.com/ozone.html>

- Increase in temperature is a result of the absorption of the Sun's ultraviolet radiation by the ozone layer.
- Speed of wind far exceed those in the troposphere, reaching near 220 km/h.
- Aircrafts fly near lower troposphere to take benefit of low air density and temperature optimizing fuel efficiency.

Stratosphere

- Bacterial life survives in the stratosphere, making it a part of the biosphere.
- Few bird species have been observed in upper troposphere.
- Rüppell's vulture
- Bar-headed geese

Stratosphere

Stratopause

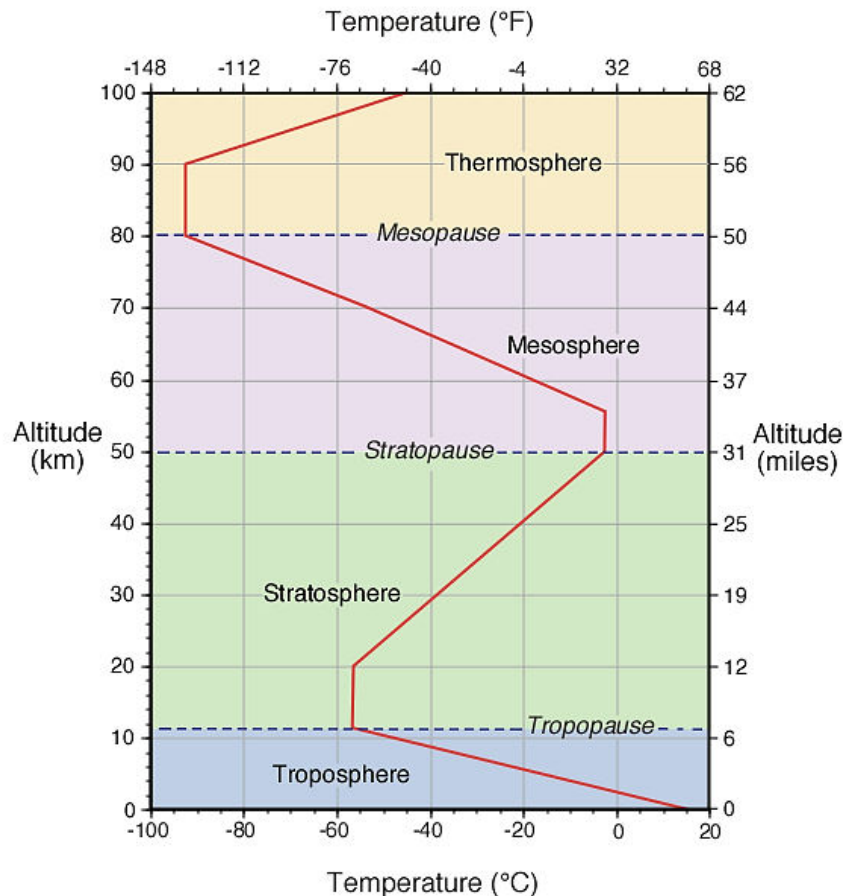
- **Boundary between stratosphere and the mesosphere**
- **Height: 50 to 55 kilometres high above the Earth's surface**
- **Pressure: 1/1000 of the pressure at sea level**
- **Temperature: -15 degrees Celsius**

END

Mesosphere

- Layer, directly above the stratosphere and directly below the thermosphere
- Lower boundary is 50 to 65 kilometres
- Upper boundary is around 85 to 100 kilometres
- It is called as “near space”.

Mesosphere



<http://users.isp.com/retic/atmslayers.gif>

Properties

- Temperature decreases with increasing height
- Due to decreasing absorption of solar radiation
- Rarefied, low dense atmosphere
- Increasing cooling by CO_2 radiative emission

Mesosphere



<https://en.wikipedia.org/wiki/Aurora>

Properties

- Temperatures in the upper mesosphere fall as low as -101°C varying according to latitude and season.
- Ionization starts here due to minimum gravity.
- Polar aurora is observed due to entry of solar winds.

Mesosphere

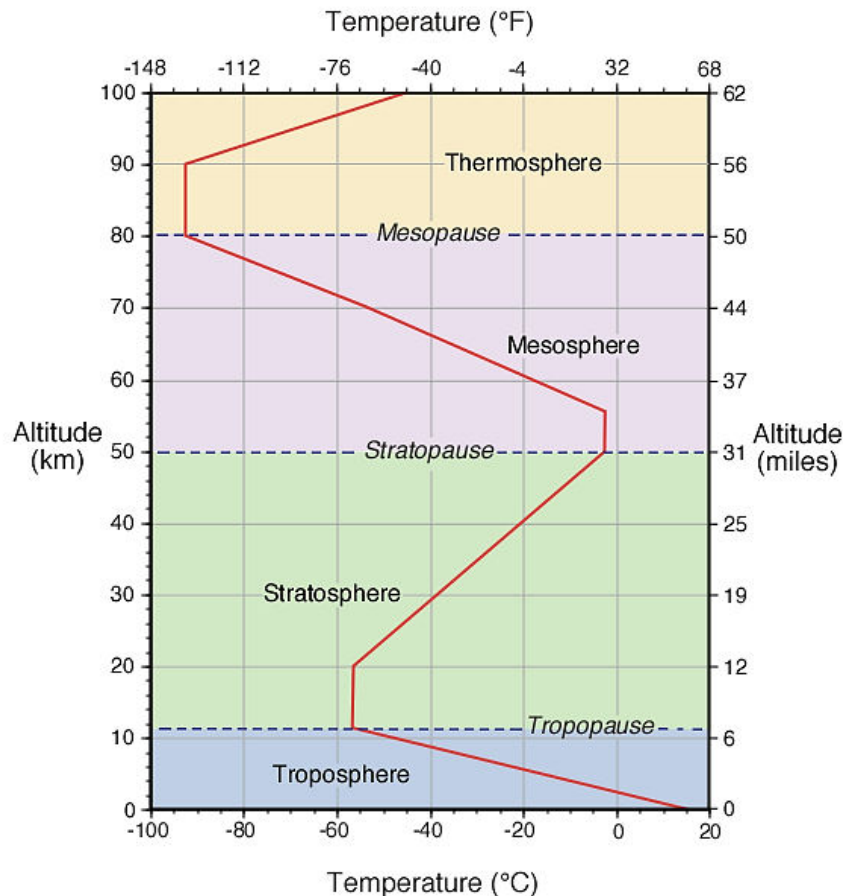
Mesopause

- Boundary between the mesosphere and the thermosphere
- Height: 85 – 100 km
- Temperature: 100 – 102 °C

Thermosphere

- Layer, directly above the mesosphere and below the exosphere
- Height: 80 km
- Extreme ultraviolet radiation causes photoionization/photodissociation of molecules, creating ions
- Known as ionosphere

Thermosphere



<http://users.isp.com/retic/atmslayers.gif>

Properties

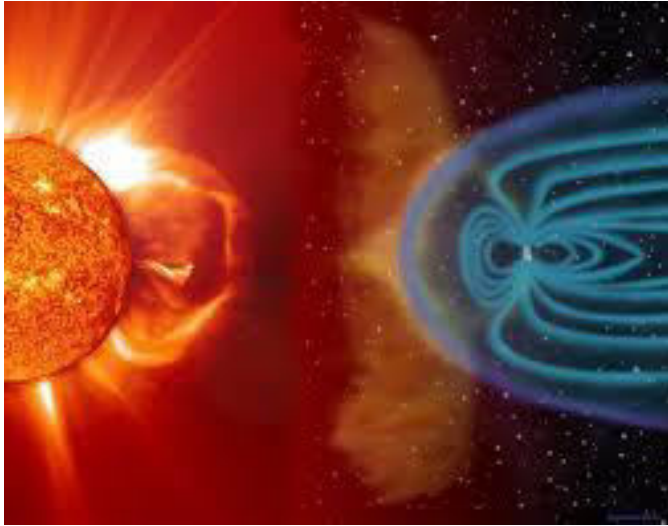
- Temperature increase with altitude due to absorption of highly energetic solar radiation.
- Temperature can raise 1700 °C or more.
- Highly diluted gas in this layer can reach 2,500 °C during the day.

Thermosphere

Properties

- Temperatures will significantly drop if solar activity is limited.
- Ions are far apart to transfer heat through conduction.
- Around 160 kilometres, the density is so low that molecular interactions are too infrequent to permit the transmission of sound.

Thermosphere



<https://www.google.com/search?q=solar+winds>

- Solar winds, extreme UV storms and atmospheric waves are observed here.
- Bombardment of particles is too intense to weaken the magnetic field on opposite end.
- Difficult to distinguish the boundary between thermosphere and exosphere.

Exosphere

- It is located directly above the thermosphere.
- Thin, atmosphere-like volume surrounding a planet
- Molecules are gravitationally bound to that body
- Density is too low for them to behave as a gas by colliding with each other.

Exosphere

- Atmosphere thins out and merges with interplanetary space.
- Very little is known about it due to lack of research.

Benefits of Atmosphere

- Atmosphere is responsible for sustaining life on earth.
- Without it, earth will be barren like moon and mars.
- Atmosphere is key element in survival of life.

Benefits of Atmosphere

Temperature

- Maintains the Earth's temperature
- Molecules of air gases absorb heat energy from sun.
- Conduction of heat energy by mixing of air
- CO₂, methane, nitrous oxide cause green house effect.

Benefits of Atmosphere



<https://www.shutterstock.com/image-vector/natural-human-enhanced-greenhouse-effect-diagram-656360482>

Benefits of Atmosphere

Protection against Radiations

- **Protective shield against radiation and cosmic rays**
- **Bombardment of UV rays and particles from sun**
- **Ozone filters most of the UV rays.**
- **Gases also absorb cosmic rays, gamma rays and x-rays**
- **Prevent mutations**

Benefits of Atmosphere

Physical Protection

- Solar system is full of debris, small particles leftover from planetary creation or collisions in the asteroid belt.
- 100 tons of space debris strikes Earth every single day.
- Friction with atmospheric gases destroys them long before they reach the ground.

Benefits of Atmosphere



<https://www.youtube.com/watch?v=jPu9DbstIDw>

Benefits of Atmosphere

Weather and Water

- Atmosphere holds around 12,900 cubic kilometers of water at any given time.
- Responsible for water cycle
- Transfer water vapors from oceans to dry areas.
- Responsible for precipitation.
- Winds control weather.

END

Carbon Cycle

- Carbon is main component of living organisms.
- 20% of human body is carbon by weight.
- CO₂ is major source of carbon.
- Atmosphere contains about 750 billion metric tons of carbon in form of CO₂.

Carbon Cycle

- Photosynthetic organisms take in CO_2 from the atmosphere
- Synthesize the carbon containing organic compounds
- This process is sometimes called carbon fixation.

Carbon Cycle

- **Animals eat plants and make their own organic compounds.**
- **Production of energy by breakdown of these organic compounds in respiration.**
- **Respiration releases CO_2 in air.**
- **Decaying organisms also produce CO_2 .**

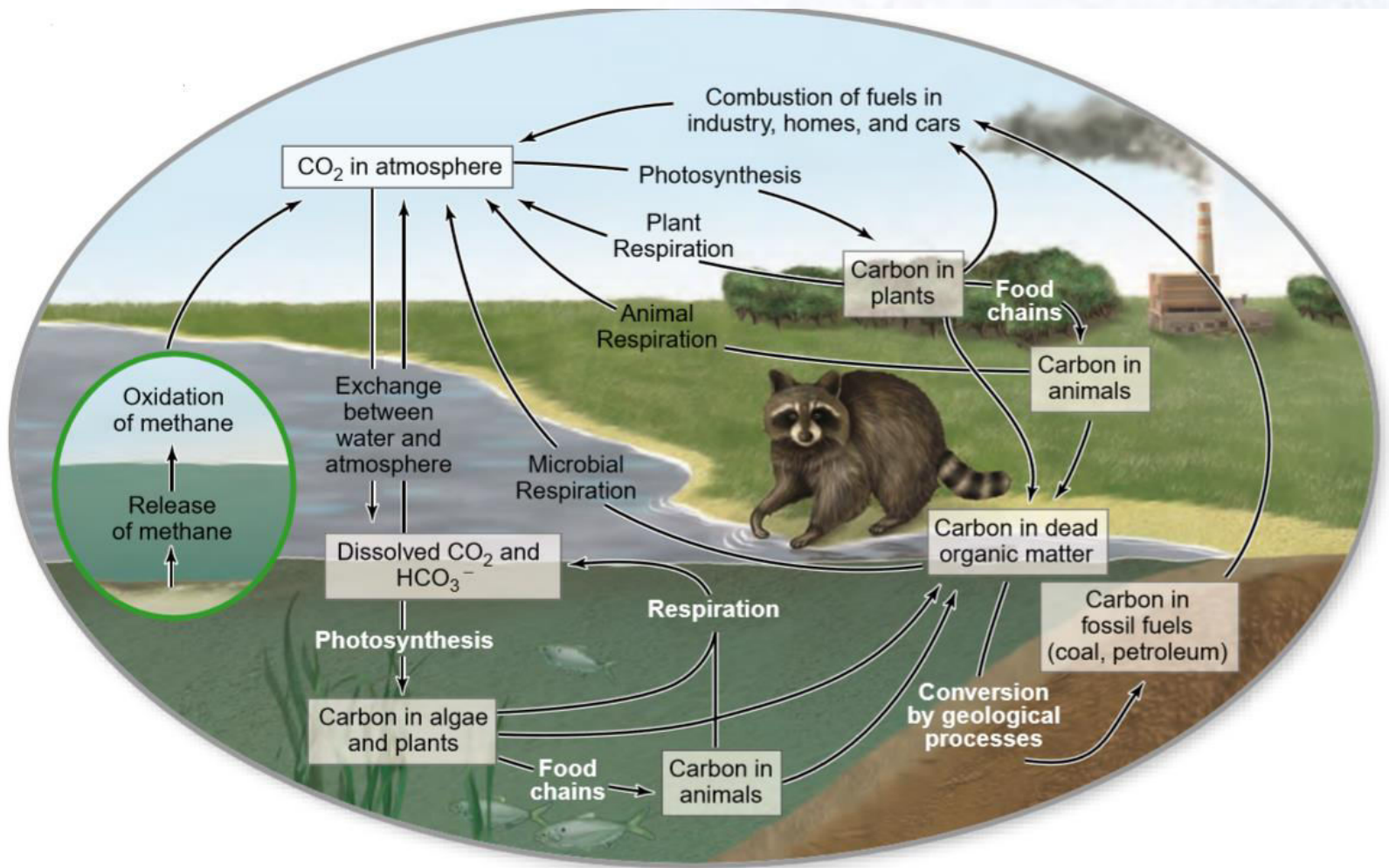
Carbon Cycle

- CO_2 released in air is again available for plants.
- In aquatic ecosystem, CO_2 reacts with the water to form bicarbonate ions.
- Dissolved CO_2 and bicarbonates are used by algae and aquatic plants in photosynthesis.

Carbon Cycle

- Methanogens, for example, are microbes that produce methane (CH_4) instead of CO_2 .
- Methane released in air is oxidized into CO_2 .
- Amount of CO_2 made by breakdown of organic compounds almost matches the amount of CO_2 used.
- Small mis-matches can have large consequences.

Carbon Cycle



Biology Mason, Losos and Singer, 9th ed.

Water Cycle

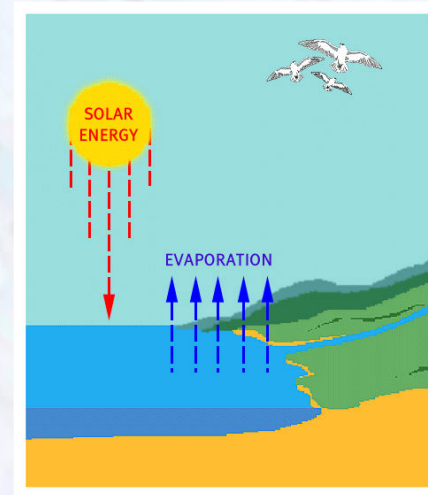
- Life depends on the presence of water.
- Adult human body contains 60% of water by weight.
- Water is compound which can be synthesized and broken.
- Synthesis in aerobic respiration
- Break down in photosynthesis

Water Cycle

- Rate of synthesis and breakdown is equal.

Evaporation

- In evaporation, water changes from its liquid state to a gaseous state.



https://en.wikipedia.org/wiki/Water_cycle

Water Cycle

Evaporation

- Some of the water in the oceans and freshwater bodies, such as lakes and rivers, is warmed by the sun and evaporates.



https://en.wikipedia.org/wiki/Water_cycle

Water Cycle

Evaporation

- During the process of evaporation, impurities in the water are left behind.
- As a result, the water that goes into the atmosphere is cleaner than it was on Earth.

Water Cycle

Condensation

- Opposite of evaporation
- Condensation occurs when a gas is changed into a liquid.



https://en.wikipedia.org/wiki/Water_cycle

Water Cycle

Condensation

- When the water droplets formed from condensation are very small.
- They remain suspended in the atmosphere.



https://en.wikipedia.org/wiki/Water_cycle

Water Cycle

Condensation

- These millions of droplets of suspended water form clouds in the sky or fog at ground level.



https://en.wikipedia.org/wiki/Water_cycle

Water Cycle



https://en.wikipedia.org/wiki/Water_cycle

Precipitation

- Under right temperature and atmospheric pressure conditions
- Small droplets of water in clouds form larger droplets and precipitation occurs.
- The raindrops fall to Earth.

Water Cycle

Surface Runoff

- Much of the water that returns to Earth as precipitation runs off the surface of the land
- Flows downhill into streams, rivers, ponds and lakes.



https://en.wikipedia.org/wiki/Water_cycle

Water Cycle

Surface Runoff

- Small streams flow into larger streams, then into rivers, and eventually the water flows into the ocean.



https://en.wikipedia.org/wiki/Water_cycle

Water Cycle

Surface Runoff

- Surface runoff is an important part of the water cycle.
- Much of the water returns again to the oceans, where a great deal of evaporation occurs.



END

https://en.wikipedia.org/wiki/Water_cycle

Water Cycle

Percolation (Infiltration)

- Percolation is an important process.
- Rain water soaks into (infiltrates) the ground, into the soil and underlying rock layers.

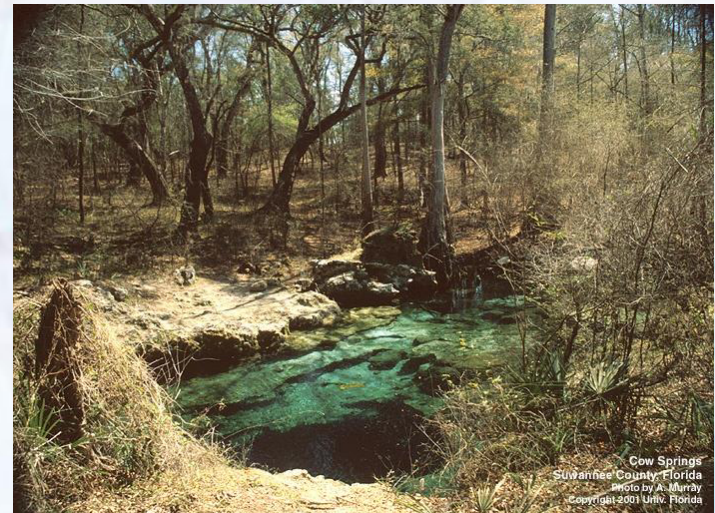


https://en.wikipedia.org/wiki/Water_cycle

Water Cycle

Percolation (Infiltration)

- Some of this water ultimately returns to the surface at springs or in low spots downhill.

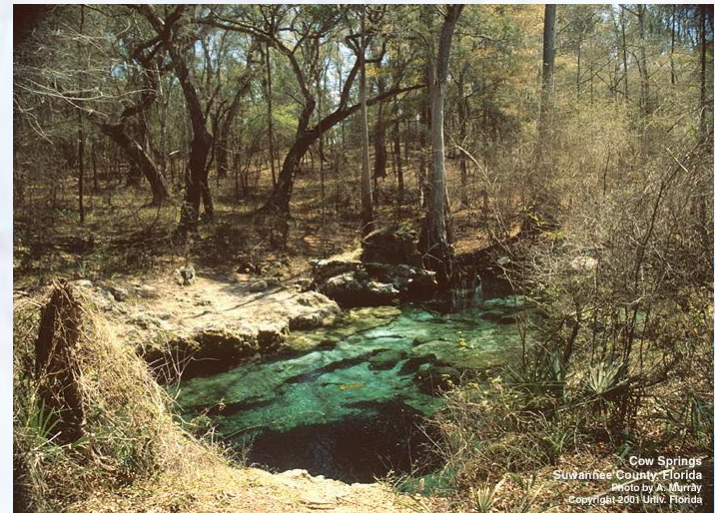


https://en.wikipedia.org/wiki/Water_cycle

Water Cycle

Percolation (Infiltration)

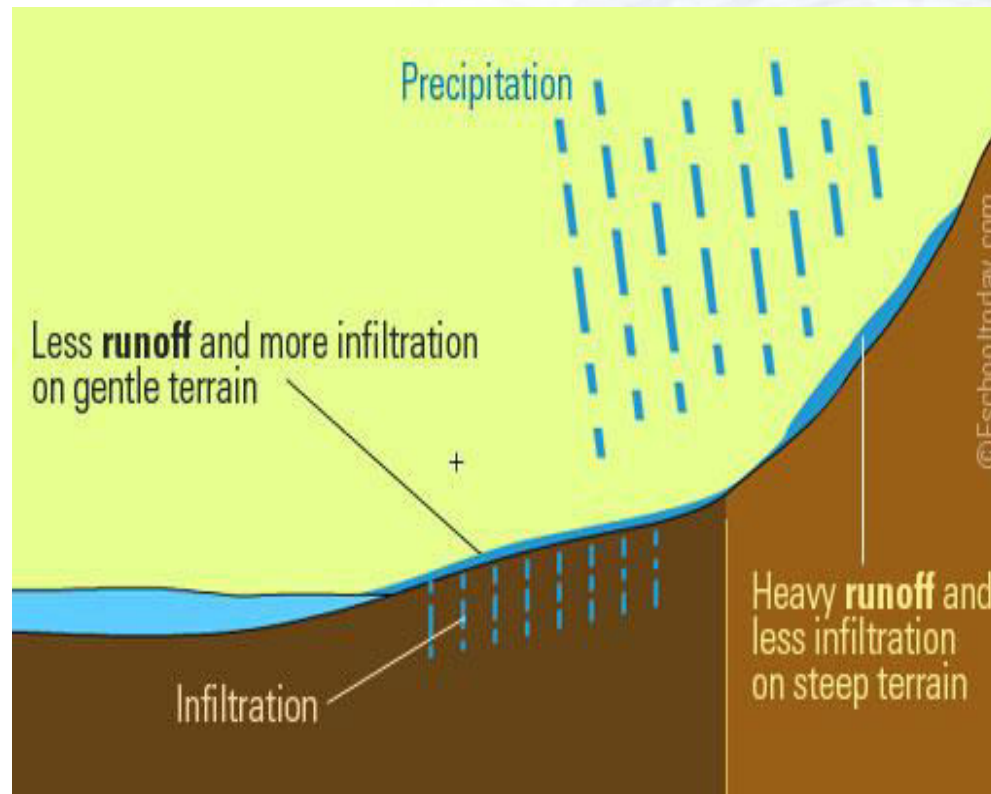
- Some of this water ultimately returns to the surface at springs or in low spots downhill.



https://en.wikipedia.org/wiki/Water_cycle

Water Cycle

Percolation (Infiltration) & Surface Runoff



https://en.wikipedia.org/wiki/Water_cycle

Water Cycle

Percolation (Infiltration)

- Some of the water percolates underground
- It is called groundwater.



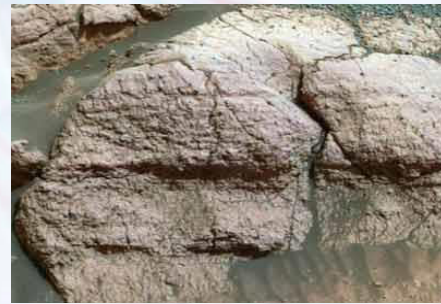
https://en.wikipedia.org/wiki/Water_cycle

Water Cycle

Groundwater

Water moves through the soil and rock layers.

- **Many of the impurities in the water are filtered out.**
- **This filtering process helps clean the water.**



https://en.wikipedia.org/wiki/Water_cycle

Water Cycle

Transpiration

- Plants absorb water from the soil.
- Water moves from the roots through the stems to the leaves.



https://en.wikipedia.org/wiki/Water_cycle

Water Cycle

Transpiration

- Water reaches the leaves.
- Some of it evaporates from the leaves, adding to the amount of water vapor in the air.



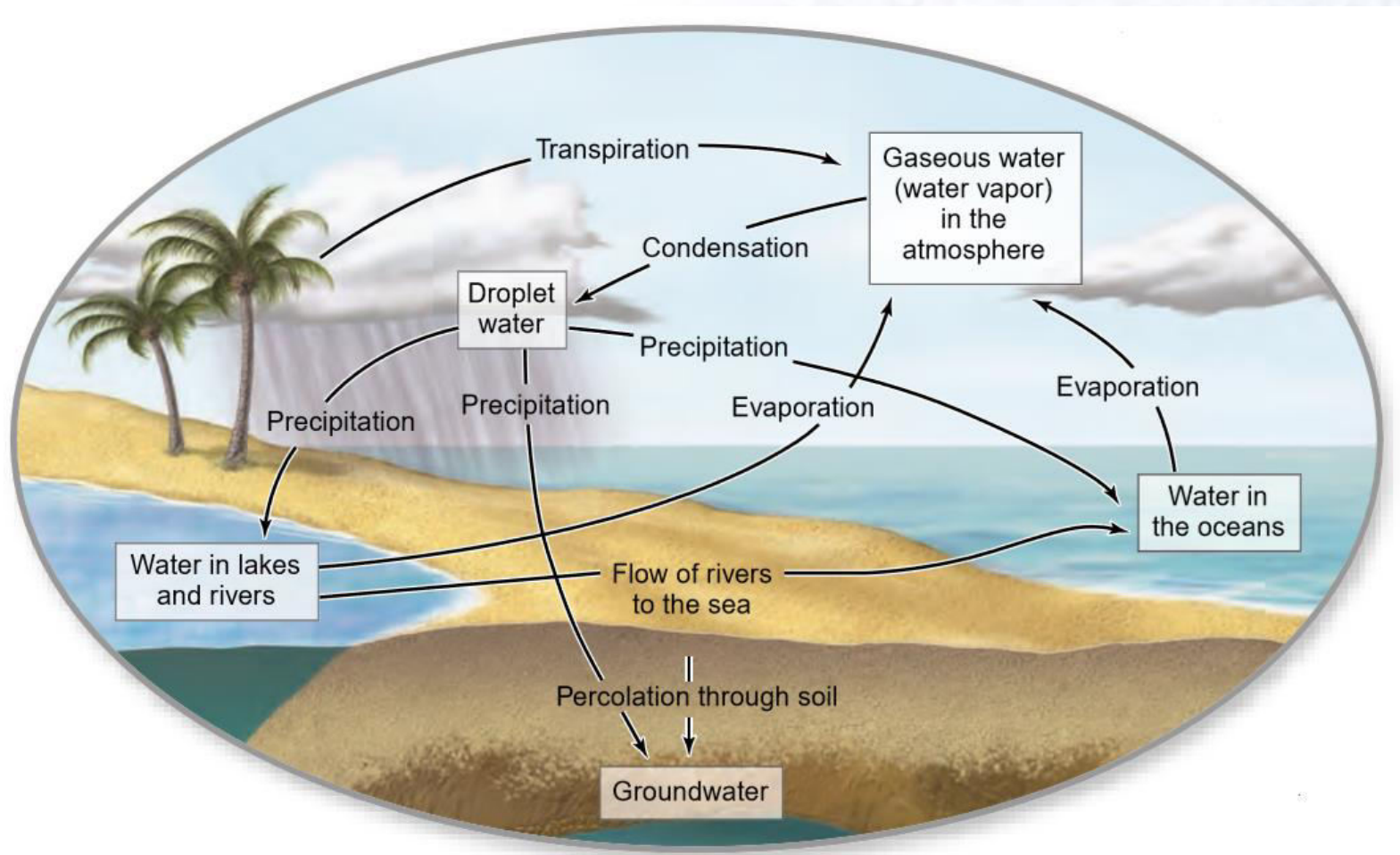
https://en.wikipedia.org/wiki/Water_cycle

Water Cycle

Transpiration

- This process of evaporation through plant leaves is called transpiration.
- In large forests, an enormous amount of water will transpire through leaves.

Water Cycle



Biology Mason, Losos and Singer, 9th ed.

Nitrogen Cycle - 1

- Nitrogen is a component of all proteins and nucleic acids.
- A living organism is 16% nitrogen by weight.
- Atmosphere is 78% nitrogen by volume.
- Atmospheric nitrogen is in its elemental form.
- It cannot be used in this form by living organism.

Nitrogen Cycle - 1

- Ultimate source of nitrogen is nitrogen containing organic compounds synthesized by plants or by algae or other microbes.
- Two commonly used nitrogen sources are ammonia (NH_3) and nitrate ions (NO_3^-).

Nitrogen Cycle - 1

- Prokaryotic microbes can synthesize ammonia and nitrate from N_2 in the atmosphere.
- It makes atmospheric nitrogen accessible to plants and algae.

Nitrogen Cycle - 1

- Other prokaryotes turn NH_3 and NO_3 into N_2 , making the nitrogen inaccessible.
- These activities balance the amount of accessible of nitrogen.
- All these processes form nitrogen cycle.

Nitrogen Cycle - 1

Sources of Nitrogen

- **Lightning**
- **Inorganic fertilizers**
- **Nitrogen Fixation**
- **Animal Residues**
- **Crop residues**
- **Organic fertilizers**

Nitrogen Cycle - 1

Forms of Nitrogen

- Urea $\rightarrow \text{CO}(\text{NH}_2)_2$
- Ammonia $\rightarrow \text{NH}_3$
(gaseous)
- Ammonium $\rightarrow \text{NH}_4$
- Nitrate $\rightarrow \text{NO}_3$
- Nitrite $\rightarrow \text{NO}_2$
- Atmospheric Dinitrogen
 $\rightarrow \text{N}_2$
- Organic N

Nitrogen Cycle - 1

Global Nitrogen Reservoirs

Nitrogen Reservoir	Metric tons nitrogen	Actively cycled
Atmosphere	3.9×10^{15}	No
Ocean → soluble salts	6.9×10^{11}	Yes
Biomass	5.2×10^8	Yes
Land → organic matter → Biota	1.1×10^{11} 2.5×10^{10}	Slow Yes

https://en.wikipedia.org/wiki/Nitrogen_cycle

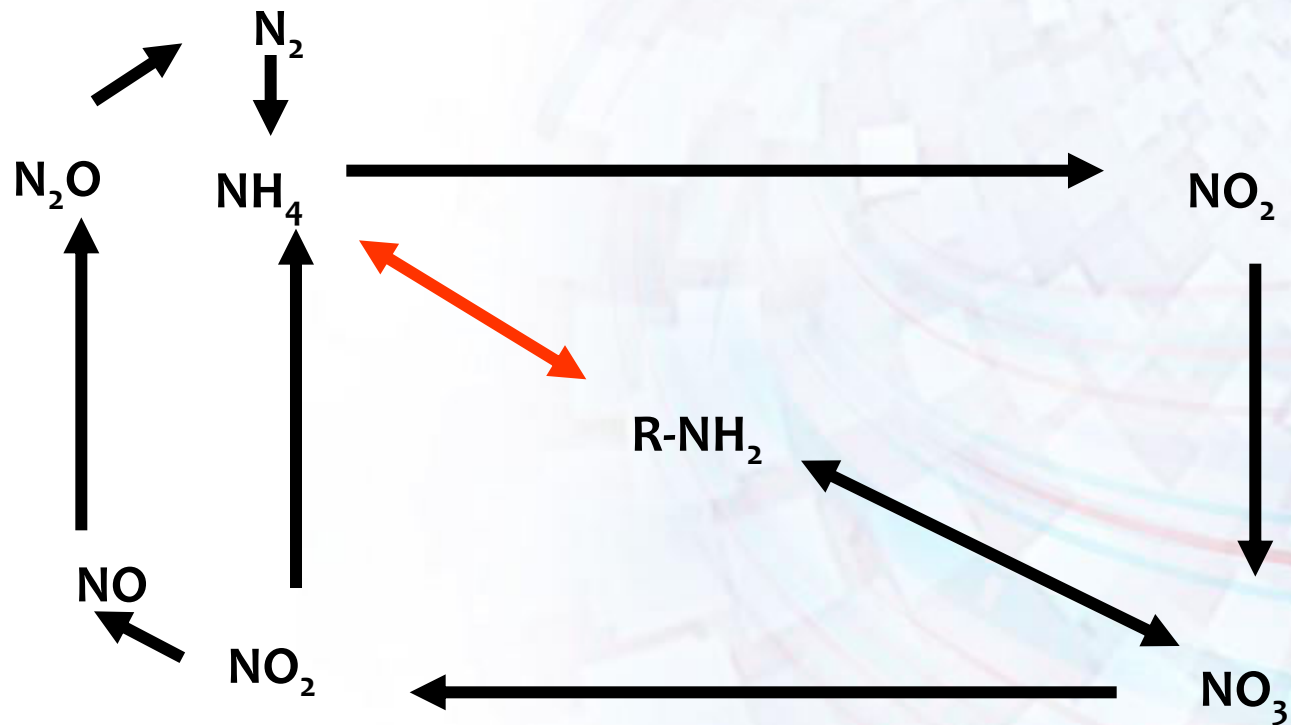
Nitrogen Cycle - 1

Phases of Cycle

1. **Ammonification/mineralization**
2. **Immobilization**
3. **Nitrogen Fixation**
4. **Nitrification**
5. **Denitrification**

Nitrogen Cycle - 1

Ammonification or Mineralization



https://en.wikipedia.org/wiki/Nitrogen_cycle

Nitrogen Cycle - 1

Ammonification

- **Decomposers:**
earthworms, termites, slugs, snails, bacteria, and fungi
- **Uses extracellular enzymes to initiate degradation of plant polymers.**

Nitrogen Cycle - 1

Ammonification

- Microorganisms use:

Proteases, lysozymes,
nucleases to degrade
nitrogen containing
molecules

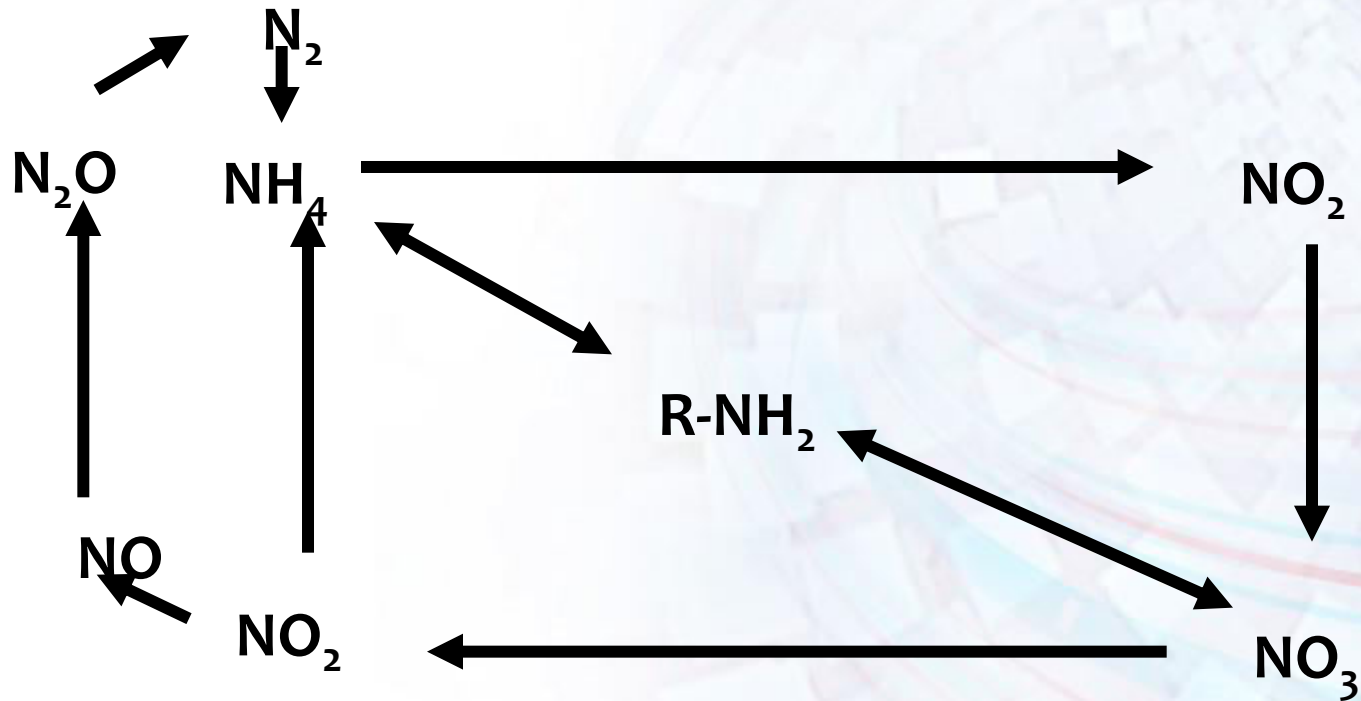
Nitrogen Cycle

Immobilization

- The opposite of mineralization
- Happens when nitrogen is limiting in the environment
- Nitrogen limitation is governed by C/N ratio
- C/N typical for soil microbial biomass is 20
- $C/N < 20 \rightarrow$ Mineralization
- $C/N > 20 \rightarrow$ Immobilization

Nitrogen Cycle

Nitrogen Fixation



Nitrogen Cycle

Nitrogen Fixation

- Energy intensive process :
- $$\text{N}_2 + 8\text{H}^+ + 8\text{e}^- + 16 \text{ ATP} = 2\text{NH}_3 + \text{H}_2 + 16\text{ADP} + 16 \text{ Pi}$$
- Performed only by selected bacteria and actinomycetes
- Performed in nitrogen fixing crops
(ex: soybeans, legumes)

Nitrogen Cycle

Microorganisms fixing Nitrogen

- *Azobacter*
- *Beijerinckia*
- *Azospirillum*
- *Clostridium*
- *Cyanobacteria*
- Require the enzyme nitrogenase
- Inhibited by oxygen
- Inhibited by ammonia (end product)

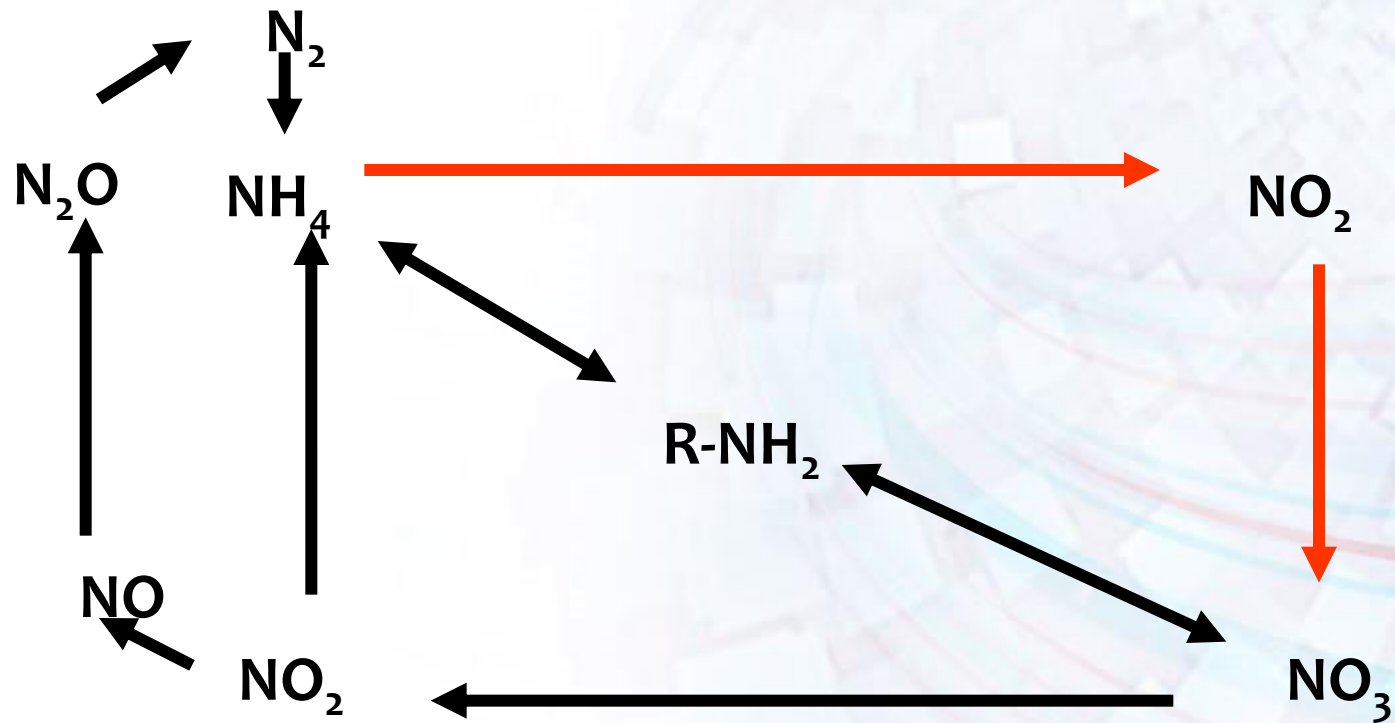
Nitrogen Cycle

Rates of Nitrogen Fixation

N₂ fixing system	Nitrogen Fixation (kg N/hect/year)
Rhizobium-legume	200-300
Cyanobacteria- moss	30-40
Rhizosphere associations	2-25
Free- living	1-2

Nitrogen Cycle

Nitrification

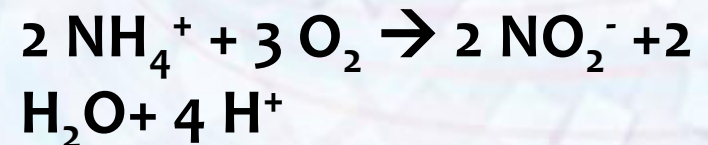


Nitrogen Cycle

Nitrification

Two step reactions that occur together :

- 1st step catalyzed by *Nitrosomonas*



- 2nd step catalyzed by *Nitrobacter*



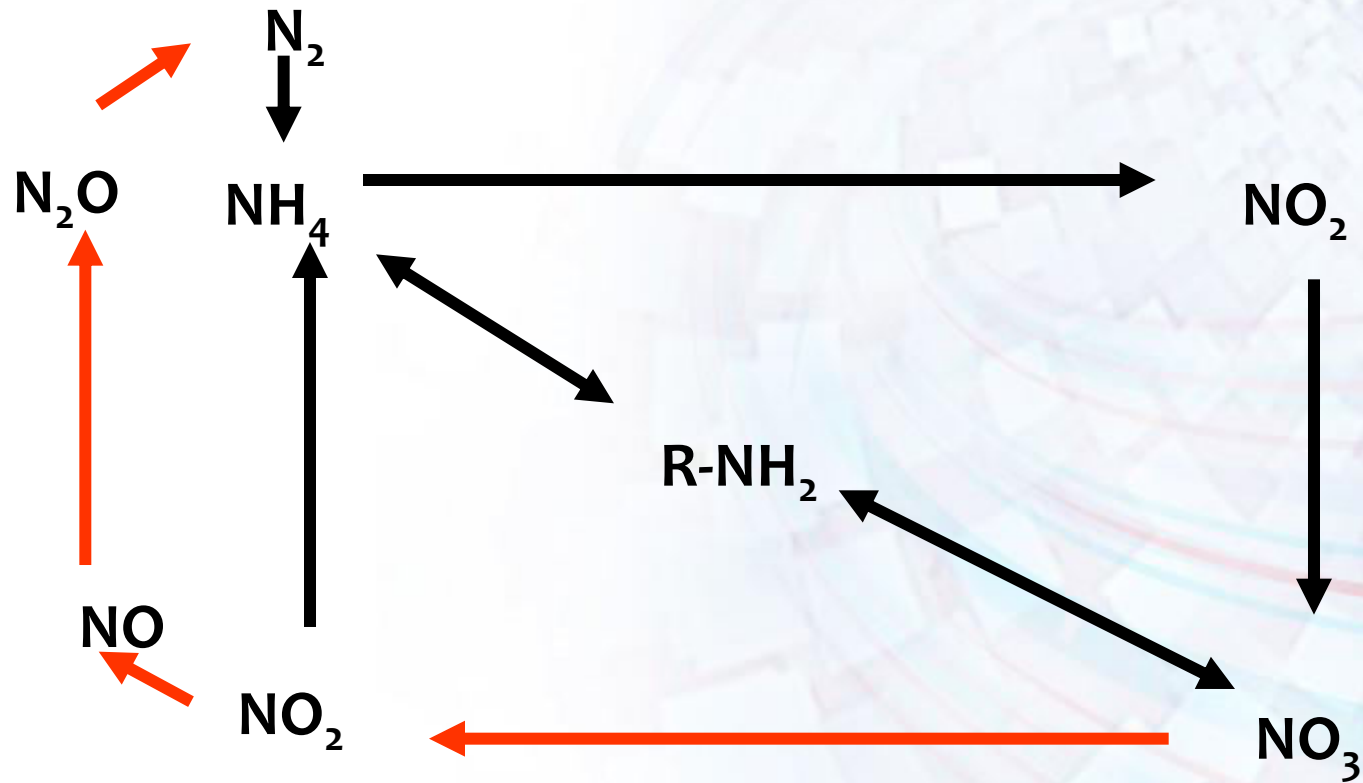
Nitrogen Cycle

Nitrification

- Optimal pH is between 6.6-8.0
- If $\text{pH} < 6.0 \rightarrow$ rate is slowed
- If $\text{pH} < 4.5 \rightarrow$ reaction is inhibited

Nitrogen Cycle

Denitrification



Nitrogen Cycle

Denitrification

- Removes a limiting nutrient from the environment
- Facultative anaerobic bacteria perform it.
- Oxidized nitrates are reduced.
- Glucose is electron donor.
- $4\text{NO}_3^- + \text{C}_6\text{H}_{12}\text{O}_6 \rightarrow 2\text{N}_2 + 6\text{H}_2\text{O}$
- Nitrate is the terminal electron acceptor

Nitrogen Cycle

Denitrification

- Inhibited by oxygen
- Not inhibited by ammonia

Nitrogen Cycle

www.7active.in

SAMPLE USE ONLY
NITROGEN CYCLE

<https://www.youtube.com/watch?v=LbBgPekjiyc>

Phosphorus Cycle

Phosphorus is important component of:

- 1. Nucleic acids**
 - 2. Membrane phospholipids**
 - 3. ATP**
- Phosphorus is mainly found in water, soil and rock in form of phosphates.**

Phosphorus Cycle

- The phosphorus cycle is the movement of phosphorus to and from between environment and organisms.
- Unlike to carbon and nitrogen, Phosphorus cannot be found in air in the gaseous state.

Phosphorus Cycle

- Phosphate salts are released from rocks through weathering.
- Dissolve in soil water and will be absorbed by plants.
- This is the slowest cycle.

Phosphorus Cycle

- **Animals absorb phosphates by eating plants or herbivores.**
- **When animals and plants die, phosphates will return to the soils or oceans again during decomposition.**

Phosphorus Cycle

- After that, phosphorus will end up in sediments or rock formations again, remaining there for millions of years.
- Eventually, phosphorus is released again through weathering and the cycle starts over.

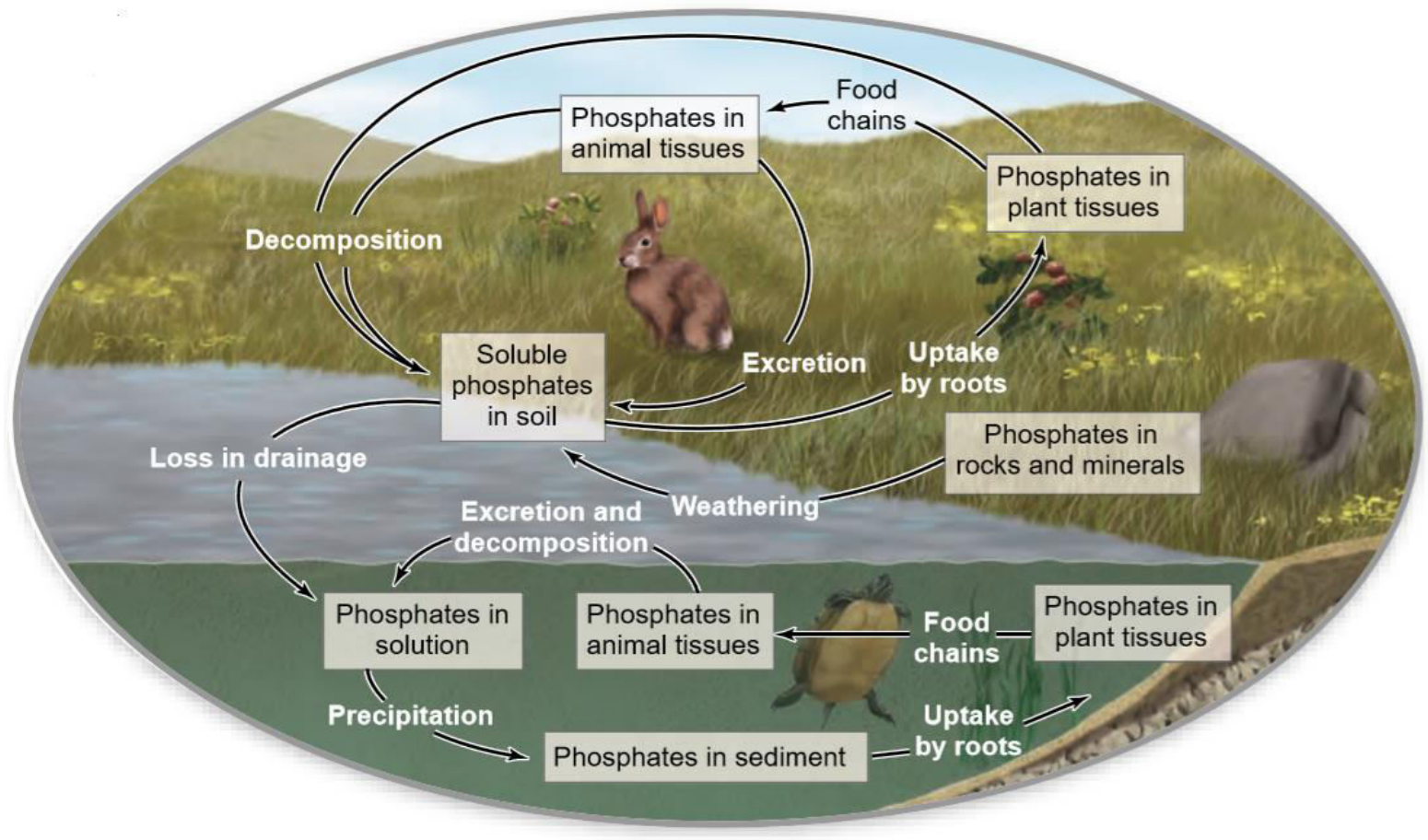
Phosphorus Cycle

- Like nitrogen, increased use of fertilizers increases phosphorus runoff into our waterways and contributes to eutrophication.



https://en.wikipedia.org/wiki/Phosphorus_cycle

Phosphorus Cycle



Biology Mason, Losos and Singer, 9th ed.

Extinction and Habitat Loss

Extinction

- **Termination of an organism or of a group of organisms (taxon), usually a species**
- **Moment of extinction is death of the last individual of the species.**
- **Capacity to breed and recover may have been lost before this point.**

Extinction and Habitat Loss

Extinction

- **Variety of causes, independently or in concert, are responsible for extinctions.**
- **Disruption of habitat**
- **Loss of ecosystem interactions e.g. disrupted food chain or web, symbiosis, predation**

Extinction and Habitat Loss

Extinction

- **Pollution**
- **Loss of genetic variation**
- **Catastrophic disturbances**
- **Natural or human-caused activities**

Extinction and Habitat Loss

Extinction

- **Disruption of ecosystem interactions is an important cause.**
- **One of the major disruptions is loss of habitat.**

Extinction and Habitat Loss

Percentage of Extinction of Species

Group	Habitat Loss
<i>E X T I N C T I O N S</i>	
Mammals	19
Birds	20
Reptiles	5
Fish	35

Extinction and Habitat Loss

- **Habitat loss is the most important cause of modern-day extinction.**
- **Human activities are negatively affecting all types of habitat, from rain forest to ocean floor.**
- **Natural habitats may be adversely affected by humans in three ways.**

Extinction and Habitat Loss

1. **Habitat destruction**
 2. **Pollution**
 3. **Habitat fragmentation**
- 

Extinction and Habitat Loss



<http://youtube.com/watch?v=TP-V4VLLHNE>

Habitat Destruction

- A process in which natural habitat becomes incapable of supporting its native species.
- Organisms inhabiting there are displaced or destroyed reducing biodiversity.
- Human activities are responsible for this.

Habitat Destruction

Causes

- Clearing habitats for agriculture and urban development
- Deforestation
- Mining
- Trawling
- Urbanization

Habitat Destruction



<https://www.hunker.com/13406919/effects-of-land-clearing>

Clearing habitats for agriculture and urban development

- Land clearing for agriculture is destructive.
- If extensive, it will bring irreversible changes.
- It will destroy whole ecosystem permanently.
- If 90% area is cleared, 50% of species will be lost.

Habitat Destruction

Clearing habitats for agriculture and urban development

- After clearing flora, fauna and topography, green house gas emissions, rise in soil salinity will damage it further.
- Land clearing should keep legislative regulation on its use.

Habitat Destruction

Deforestation

- **About 31% of Earth's land surface is covered by forests.**
- **Removal of a forest or stand of trees from land which is then converted to a non-forest use.**

Habitat Destruction

Deforestation

Reasons of deforestation

- Agriculture
- Fuel
- Timber for building
- Pastures & Plantation
- Urban development

Insufficient reforestation has resulted in habitat damage, biodiversity loss, and aridity.

Habitat Destruction



<http://youtube.com/watch?v=lc-J6hcSKa8>

Habitat Destruction

Impacts of Deforestation

- **Global warming**
- **Tropical deforestation is responsible for approximately 20% of world greenhouse gas emissions.**
- **Decay or burning of wood release carbon in atmosphere.**
- **Carbon store in soil is also released.**

Habitat Destruction



<https://en.wikipedia.org/wiki/Deforestation#Causes>

Impacts of Deforestation

- Water cycle
- Rate of transpiration by trees is reduced.
- Drier climate
- Low moisture in soil
- Less water available for plants.
- More erosion, flooding and land sliding

Habitat Destruction

Impacts of Deforestation

- Decline in biodiversity
- Cause of extinction
- Loss of genetic variations
- Affect interspecific interactions e.g. symbiosis

Habitat Destruction



<https://www.google.com/search?q=mining&rlz>

Impacts of Mining

- Mining needs removal of everything to get access to mine.
- It involves land clearing, deforestation and digging.
- Habitat is permanently damaged.
- It leaks pollutants.

Habitat Destruction

Impacts of Mining

- **Water bodies are exposed to heavy metals and toxic minerals**
- **Water is pumped out to allow miners to work.**
- **Water level shrinks.**

Habitat Destruction



<https://en.wikipedia.org/wiki/Trawling>

Impacts of Trawling

Trawling

- Pulling a fishing net through the water behind one or more boats
- Heavy catch of fish from wild.
- Using trawlers
- All wanted and unwanted animals are caught.

Habitat Destruction

Impacts of Trawling

- Fishing gear towing at the bottom of sea damages sea bed.
- Including coral shattering, damage to habitats and removal of seaweed

Habitat Destruction

Impacts of Urbanization

- Population shift from rural areas to urban areas
- Gradual increase in the proportion of people living in urban areas

Negative Impacts

- Deforestation
- Habitat destruction
- Habitat fragmentation
- Pollution

Habitat Fragmentation

- Habitat needs to be an intact body.
- Least interference is required to protect niches of organisms.
- Many species are too sensitive to tolerate changes in their niches.
- e.g. insects

Habitat Fragmentation

- **Breaking up of one patch of habitat into several smaller patches**
- **Reduction in the total area of the habitat**
- **Isolation of one habitat fragment from other areas of habitat**
- **Decrease in the average size of each patch of habitat**

Habitat Fragmentation



https://en.wikipedia.org/wiki/Habitat_fragmentation#Natural_causes

Causes

- Fragmentation by different obstructions
- Volcanism, fire, climate change, deforestation, colonies, roads, water drains, noise, mining
- Population is divided into unconnected patches
- It affects biodiversity and population size in that habitat.

Habitat Fragmentation

www.7active.in

**DESTRUCTION OF HABITATS
AND
FRAGMENTATION**

<http://youtube.com/watch?v=YMpMsSK3VPY>

Habitat Fragmentation

Edge Effect

- Proportion of the habitat that occurs on the boundary or edge, increases.
- Edge effects can significantly degrade a population's chances of survival.
- Changes in microclimate in edges are more dangerous than fragmentation itself.

Habitat Fragmentation



Rainforests of Manaus, Brazil, First monkeys needing large area disappeared., then insects followed by insectivorous birds.

Habitat Fragmentation

Edge Effect

- Population on edges are more exposed to sunlight.
- Parasites and predator effects can be more intense on edges.
- Plants are worst affected due to their immobility.

END

Effects of Habitat Fragmentation

Effects of Fragmentation

- **Loss of biodiversity**
- **Threatening or extinction of species**
- **Genetic diversity is also fragmented.**
- **Small populations having smaller pool of fitness maintaining alleles to survive**

Effects of Habitat Fragmentation

Effects of Fragmentation

- Gene flow and migration are reduced.
- Species do not migrate to fragments.
- Reproductive isolation and inbreeding
- It causes inbreeding depression.

Effects of Habitat Fragmentation

Effects of Fragmentation

- Inbreeding increases the level of homozygosity.
- Facilitating the expression of deleterious alleles that reduce the fitness
- It may result in local extinction.

Effects of Habitat Fragmentation

Effects of Fragmentation

- Genetic drift is increased.
- Random changes to the genetic make up of populations
- It leads to reductions in genetic diversity.

Effects of Habitat Fragmentation

Effects of Fragmentation

- Natural selection prefers large population size.
- Small population are weak in adaptations
- May be eliminated by the process of natural selection.

Effects of Habitat Fragmentation

Effects of Fragmentation

- Greatly affect the predator-prey dynamics of many species.
- By changing species diversity and population size
- Reduction in area of safe zones for prey making them vulnerable to predators
- It may increase their area of refuge on opposite

END

Invasive Species

- **Colonization, a natural process by which a species expands its geographic range, occurs in many ways.**

Examples:

- **Seed dispersal by birds**
- **Lowering of sea levels join two isolated land masses**
- **Flooding**

Invasive Species

- **Migratory animals carrying other animals as symbiotic or parasitic partners**
- **Human activities are the major reason of invasion of species.**

Invasive Species

- **Exotic Species** are species that occur outside of their natural ranges because of human activity.
- An exotic species becomes **invasive** when it becomes established in its new environment and out competes native species.

Invasive Species

- **Types of invasive species**
 1. **Animals**
 2. **Plants**
 3. **Pathogens**
- **They cause an imbalance in the natural environment.**
- **They outcompete the native species.**

Invasive Species



https://en.wikipedia.org/wiki/Invasive_species

Zebra Mussel

- Originally from the Caspian and Black Sea.
- Accidentally introduced to the Great Lakes in 1988.
- Can harm other fresh water mussels
- Can block intake pipes for power plants
- Can cause problems for recreational activities.

Invasive Species



Kudzu

- native to southern Japan and south east China.
- It occurs as an invasive species in the U.S.
- it is considered a noxious weed that climbs over trees or shrubs and grows so rapidly.
- It kills them by heavy shading.

https://en.wikipedia.org/wiki/Invasive_species

Invasive Species



https://en.wikipedia.org/wiki/Invasive_species

Asian Tiger Mosquito

- Originally from Asia.
- Introduced to Hawaii in the late 1800's.
- Continental United States in 1985.
- Can be a vector for disease.
- Ex. West Nile Virus

Invasive Species



https://en.wikipedia.org/wiki/Invasive_species

Lionfish

- Native to the Pacific Ocean
- Introduced to the water around the United States in 1992 through the aquarium trade.
- Preys on native species.
- Has poisonous spines.

Invasive Species

END

- **Strategies and legislations about introduction of exotic species**
- **Protection of native species is of utmost priority.**
- **For purpose of conservation of native biodiversity**

Overexploitation

- Over harvesting
- Harvesting a renewable resource to the point of diminishing returns
- Risk of extinction appears even when the species is initially very abundant.

Examples:

1. Passenger pigeons
 2. Bison
- in N. America

Overexploitation

Cascade Effect

- Removal of apex predator in environment
- It leads to dramatic increase in their prey species.
- Unchecked prey can then overexploit their own food resources
- Shortage in food resources disturbs whole food web.

Overexploitation

Cascade Effect

- Cascade of overexploitation at different trophic levels
- Population numbers dwindle, possibly to the point of extinction.

Example:

- Overhunting of sperm whale; an apex predator

Overexploitation



<http://youtube.com/watch?v=GlcoMDw6mRo>

Overexploitation

Commercial Motivations

- International trade
- Furs
- Skin
- Bones
- Other body parts
- Trophy & Game hunting
- Hunting for food
- Medicines

Overexploitation

Threatened species

- Chinchilla, vicuña, otter, and many cat species for fur
- Elephants for tusk
- Cod, bluefin tuna and swordfish fishing in N. America
- Mahogany trees in West Indies
- Cedar forests of Lebanon

END

Land Pollution - 1

Pollution

- Introduction of contaminants into the natural environment that cause adverse change.
- Undesirable change in physical, chemical, or biological characteristics of any component of the environment.

Land Pollution - 1

Pollution

Classified as:

- 1. Point source**
- 2. Non point source**

1. Point source

Single identifiable source of pollution

Example:

- Land fill**
- Factory drain**
- Sewages pipes**

Land Pollution - 1

Pollution

Non Point source

Resulting from many diffuse sources

Example:

- Sediment
- Nutrients
- Toxic contaminants and chemicals
- Pathogens

Land Pollution - 1

Types of Pollution

1. Land
2. Air
3. Water
4. Light
5. Noise

Land Pollution - 1

- Followed by industrial Revolution, natural habitats were destroyed and the environment was polluted.
- Common problem worldwide
- Direct result of the activities that humans engage in

Land Pollution - 1

- **Pollution of the Earth's natural land surface by industrial, commercial, domestic, and agricultural activities**
- **Includes visible litter and waste along with the soil**
- **The deposition of solid or liquid waste materials on land or underground can contaminate the soil and groundwater**

Land Pollution - 1



http://youtube.com/watch?v=wt_siDmpfBo

Land Pollution - 1

- Land pollution is comprised of:
 1. Solid waste
 2. Soil pollution

Land Pollution - 1

Solid Waste

- **Semisolid or solid matter that are created by human or animal activities**
- **Disposed because they are hazardous or useless**

Land Pollution - 1

Solid Waste

Sources of solid waste

- Wastes from agriculture
- Wastes from mining
- Wastes from industries
- Solids from sewage treatment
- Ashes
- Garbage

Land Pollution - 1

Solid Pollution

- Caused by chemicals in pesticides
- Results from unhealthy methods of soil management.
- Harmful practices of irrigation methods.
- Causes

END

Land Pollution - 2

Sources

- Littering
- Herbicides, insecticides, and pesticides
- Construction debris and waste
- Mining waste
- Overcrowded landfills
- Deforestation
- Chemical and nuclear plants

Land Pollution - 2

Sources

- Industrial factories
- Oil refineries
- Human sewage
- Oil and antifreeze leaking from motor vehicles

Land Pollution - 2

Causes

- Increase in urbanization
- Increase in agricultural land
- Domestic waste
- Agricultural activities
- Industrial activities

Land Pollution - 2

Effects

- **Exterminates wild life**
- **Vegetation that provides food and shelter is destroyed**
- **It can seriously disrupt the balance of nature, and, in extreme cases, can cause human fatalities**
- **Most pesticides kill or damage life forms other than those intended.**

Land Pollution - 2

Effects

- Pesticides can damage crops; kill vegetation; and poison birds, animals, and fish
- Skin problems are often diagnosed due to land pollution.
- One of the leading causes for birth defects.

Land Pollution - 2

Effects

- Land pollution indirectly affects the respiratory system of human beings
- Can incur breathing problems and a number of diseases, which may affect the health of babies as well.

Land Pollution - 2

Effects

- **Impact it has on the surrounding environments**
- **Every form of land pollution kills off the habitats of land animals as well as aquatic life**
- **The more we deplete these habitats the more the ecosystem is thrown out of balance.**

Land Pollution - 2

Preventions

- People should be educated and made aware about the harmful effects of littering.
- Items used for domestic purposes ought to be reused or recycled.
- Personal litter should be disposed properly.

Land Pollution

Preventions

- **Organic waste matter should be disposed in areas that are far away from residential places**
- **Use of pesticides should be reduced.**

Water Pollution - 1

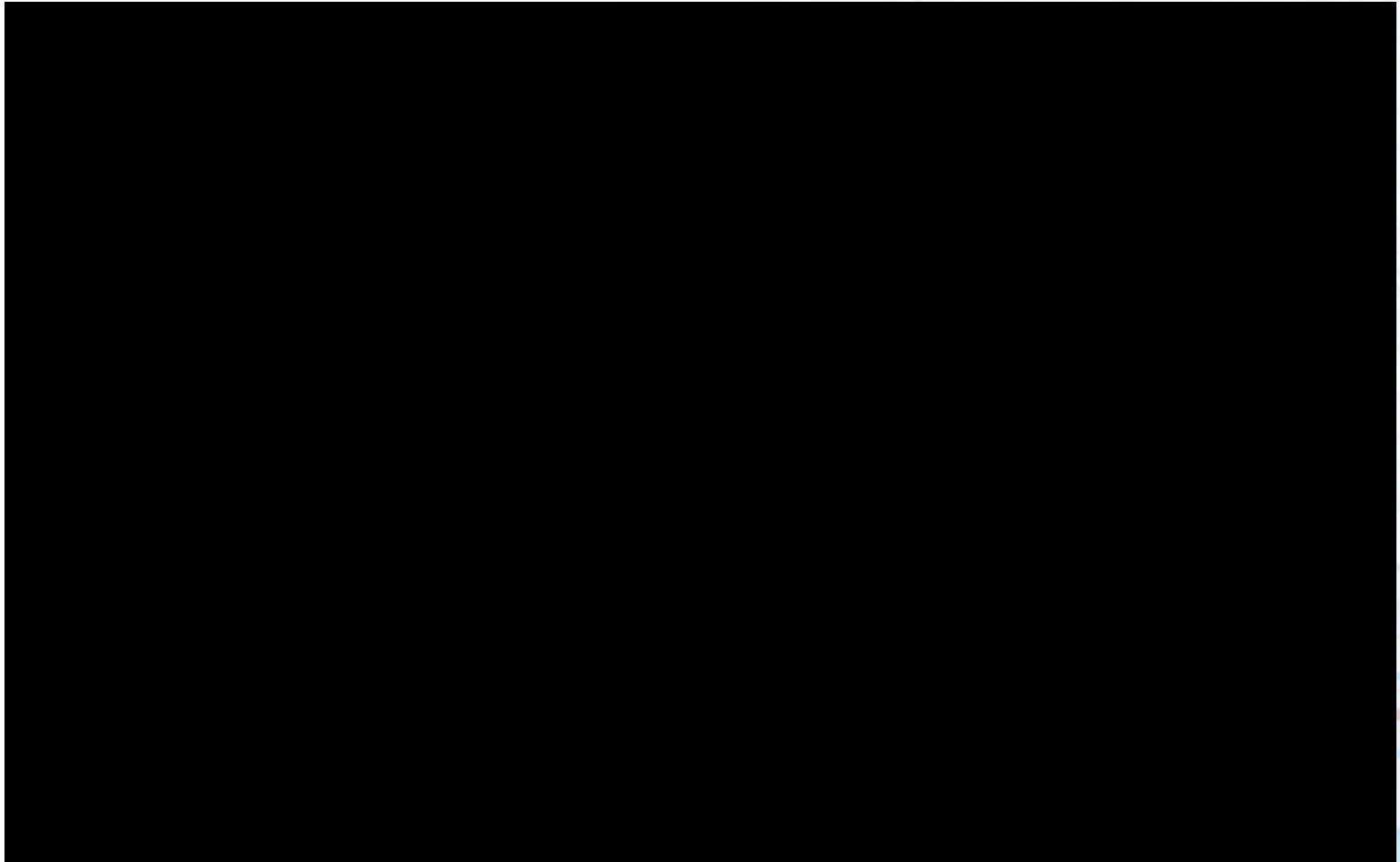
- Any physical or chemical change in water that adversely affects the health of humans and other organisms
- Varies in magnitude by location
- Global issue
- Lack of disease-free water anywhere in the world

Water Pollution - 1

Categories

1. Sewage
2. Disease-causing agents
3. Sediment pollution
4. Inorganic plant and algal nutrients
5. Organic compounds
6. Inorganic chemicals
7. Radioactive substances
8. Thermal pollution

Water Pollution - 1



<http://youtube.com/watch?v=Sp572udnPVg>

Water Pollution - 1

Sewage

- The release of wastewater from drains or sewers
- Includes human wastes, soaps, and detergents

Causes serious environmental problems:

- Enrichment of nutrients

Water Pollution - 1

Sewage

- **Fertilization of a body of water by high levels of plant and algal nutrients (nitrogen and phosphorus)**
- **Increase in Biological Oxygen Demand (BOD)**

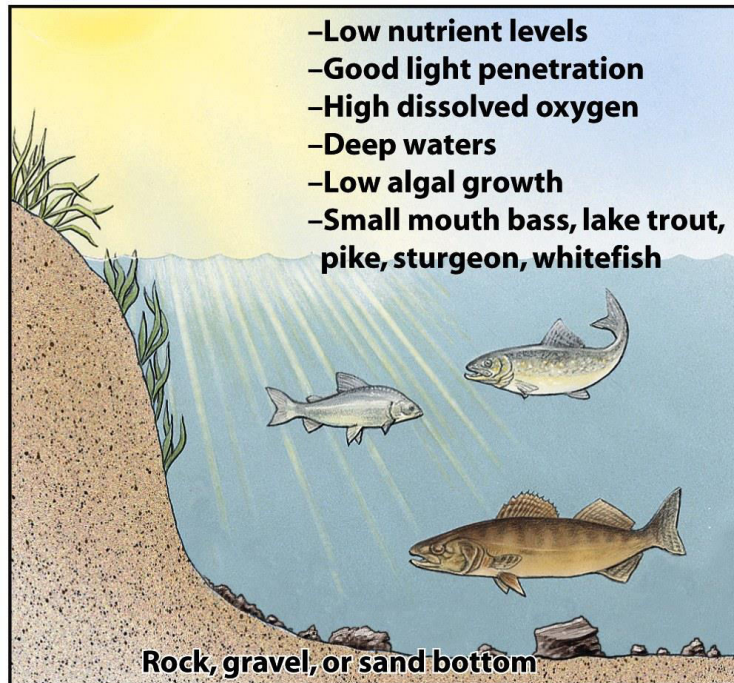
Water Pollution - 1

Sewage

- Amount of oxygen needed by microorganisms to decompose biological wastes
- As BOD increases Dissolve Oxygen (DO) decreases

Water Pollution - 1

Oligotrophic lake



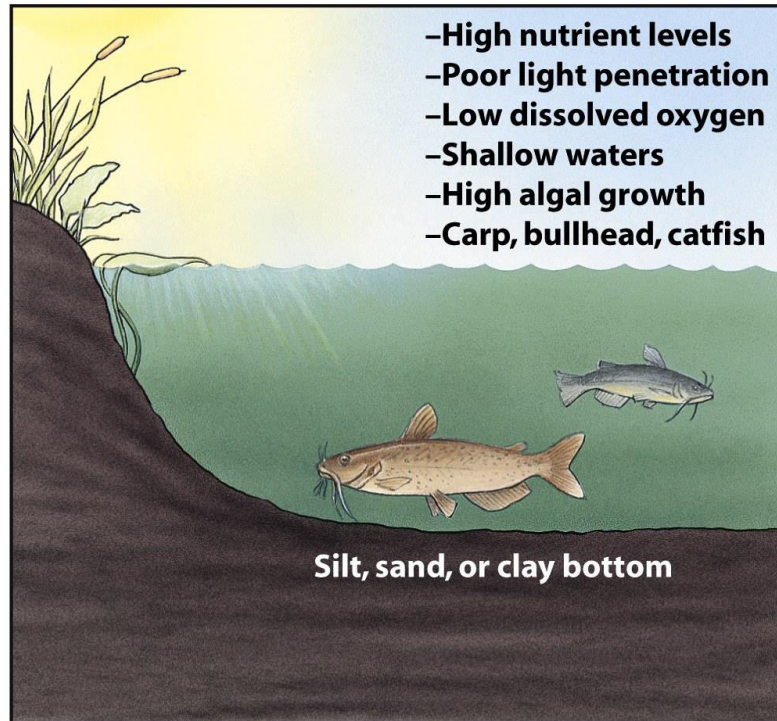
https://en.wikipedia.org/wiki/Water_pollution

Sewage

- Sewage- Eutrophication
- Oligotrophic
 - Unenriched, clear water that supports small populations of aquatic organisms

Water Pollution - 1

Eutrophic lake



https://en.wikipedia.org/wiki/Water_pollution

Sewage

- Sewage- Eutrophication
- Eutrophic-
 - Slow-flowing stream, lake or estuary enriched by inorganic plant and algal nutrients such as phosphorus
 - Often due to fertilizer or sewage runoff

Water Pollution - 1

Sewage

- **Disease-causing Agents**
- **Infectious organisms that cause diseases**
 - **Originate in the wastes of infected individuals**
- **Common bacterial or viral diseases:**
 - **Typhoid, cholera, bacterial dysentery, polio, and infectious hepatitis**

Water Pollution - 1

Sewage

- **Sediment Pollution**
- **Excessive amounts of suspended soil particles**

– **Originates from**

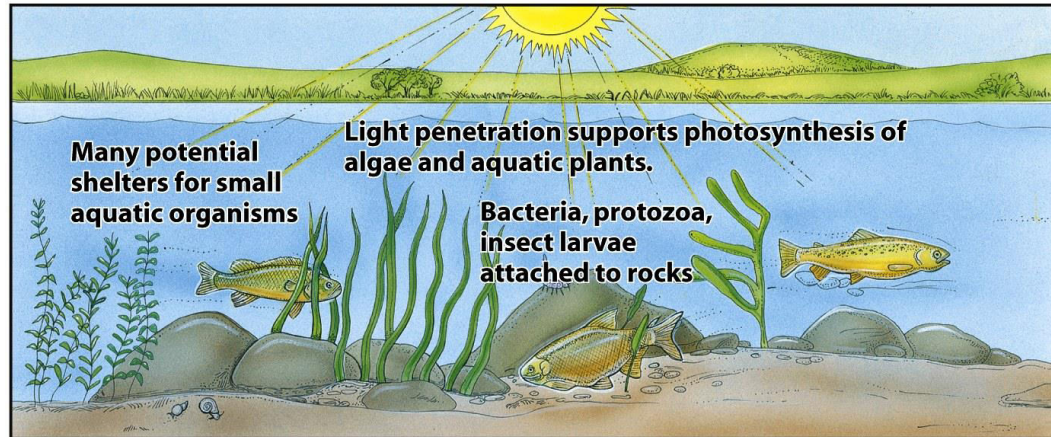
Erosion of agricultural lands, forest soils exposed by logging, degraded stream banks, overgrazed rangelands, strip mines, and construction

Water Pollution - 1

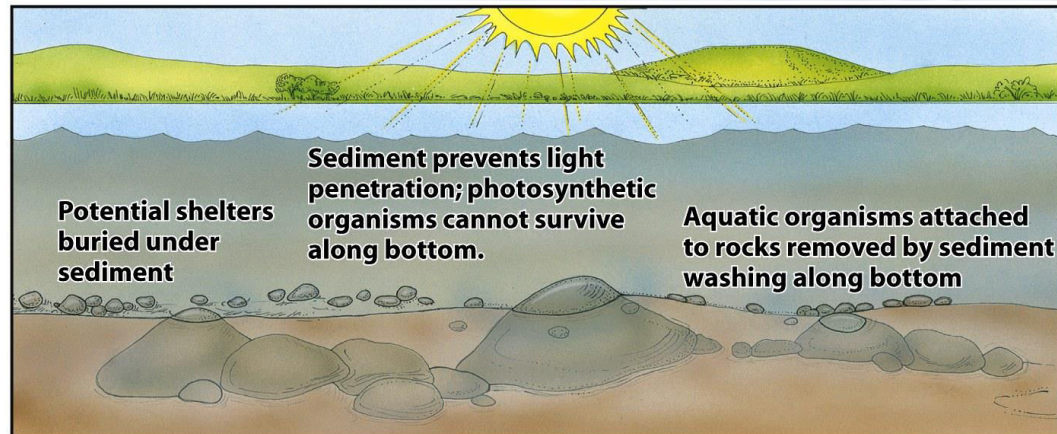
Sewage

- **Sediment Pollution**
- **Problems**
 - **Limits light penetration**
 - **Covers aquatic animals and plants**
 - **Brings insoluble toxins into waterways**

Water Pollution - 1



Stream ecosystem with low level of sediment



Same stream with high level of sediment

https://en.wikipedia.org/wiki/Water_pollution

Water Pollution - 2

Inorganic Plant and Algal Nutrients

- Chemicals such as nitrogen and phosphorus that stimulate the growth of plants and algae
 - Harmful in large concentrations

Water Pollution - 2

Inorganic Plant and Algal Nutrients

- **Sources:**
 - Human and animal wastes, plant residues, atmospheric deposition, and fertilizer runoff
- **Causes:**
 - Enrichment, bad odors, and a high BOD

Water Pollution - 2

Organic Compounds

- Chemicals that contain carbon atoms
 - Natural examples: sugars, amino acids, and oils
 - Human-made examples: pesticides, solvents, industrial chemicals, and **PLASTICS**

Water Pollution - 2

Dead Zone; The Great Pacific Patch of Garbage



<http://youtube.com/watch?v=QHK2Zg5OibI>

Water Pollution - 2

Inorganic Chemicals

- Contaminants that contain elements other than carbon
 - Examples: acids, salts, and heavy metals
- Do not degrade easily
- Lead
 - Found in old paint, industrial pollutants, leaded gasoline

Water Pollution - 2

Inorganic Chemicals

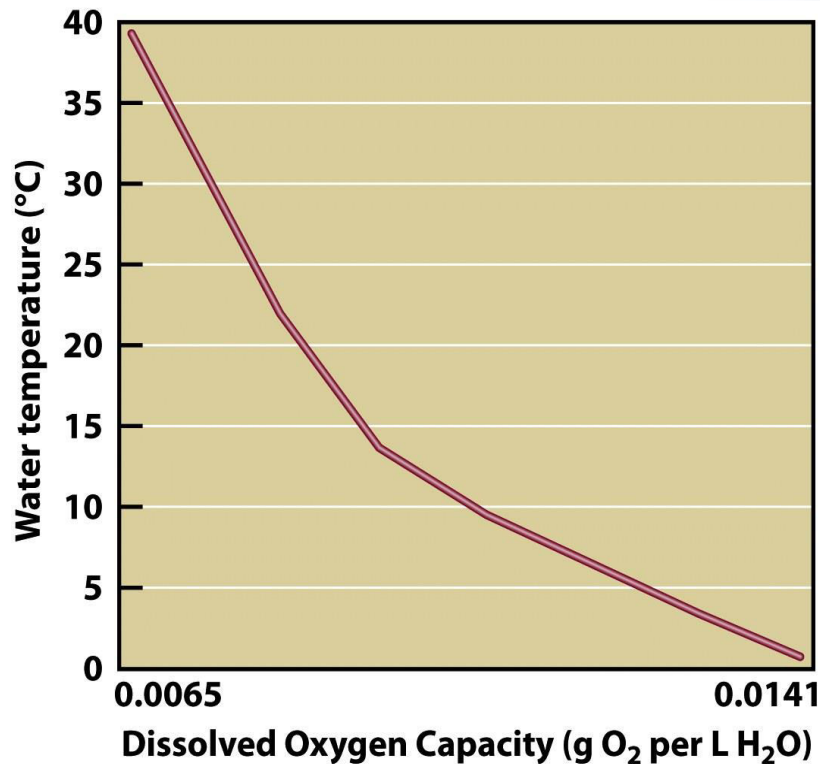
- **Mercury**
 - **Mercury bioaccumulates in the muscles of top predators of the open ocean**

Water Pollution - 2

Radioactive Substances

- **Contain atoms of unstable isotopes that spontaneously emit radiation**
- **Sources**
 - **Mining**
 - **Processing radioactive materials**
 - **Nuclear power plants**
 - **Natural sources**

Water Pollution - 2



https://en.wikipedia.org/wiki/Water_pollution

Thermal Pollution

- Occurs when heated water produced during industrial processes is released into waterways
- Organisms affected
 - Temperature affects reproductive cycles, digestion rates, and respiration rates
 - Warm water holds less DO than cold water

Water Pollution - 2

Water Pollution from Agriculture

- **Agriculture is leading source of water pollution**
 - **Animal wastes and plants residues have high BOD**
 - **Chemical pesticides can leach into groundwater**
- **Almost all streams and rivers are polluted with agricultural pesticides.**

Water Pollution - 2

Industrial & Municipal Wastes in Water

- Different industries generate different pollutants
 - Food processing plants- high BOD
 - Paper mills- High BOD and toxic compounds
- Few industries recover toxins before they go into the waste stream.

END

Air Pollution - 1

- Harmful or excessive quantities of substances including gases, particles, and biological molecules are introduced into Earth's atmosphere.
- It may cause diseases, allergies and even death of organisms.
- Both human and natural activities contribute.

Air Pollution - 1

Air Pollutants

- Chemicals added to the atmosphere by natural events or human activities in high enough concentrations to be harmful
- Two categories of air pollutants
 1. Primary Air Pollutant
 2. Secondary Air Pollutants

Air Pollution - 1

Primary Air Pollutant

- Harmful substance that is emitted directly into the atmosphere

Secondary Air Pollutant

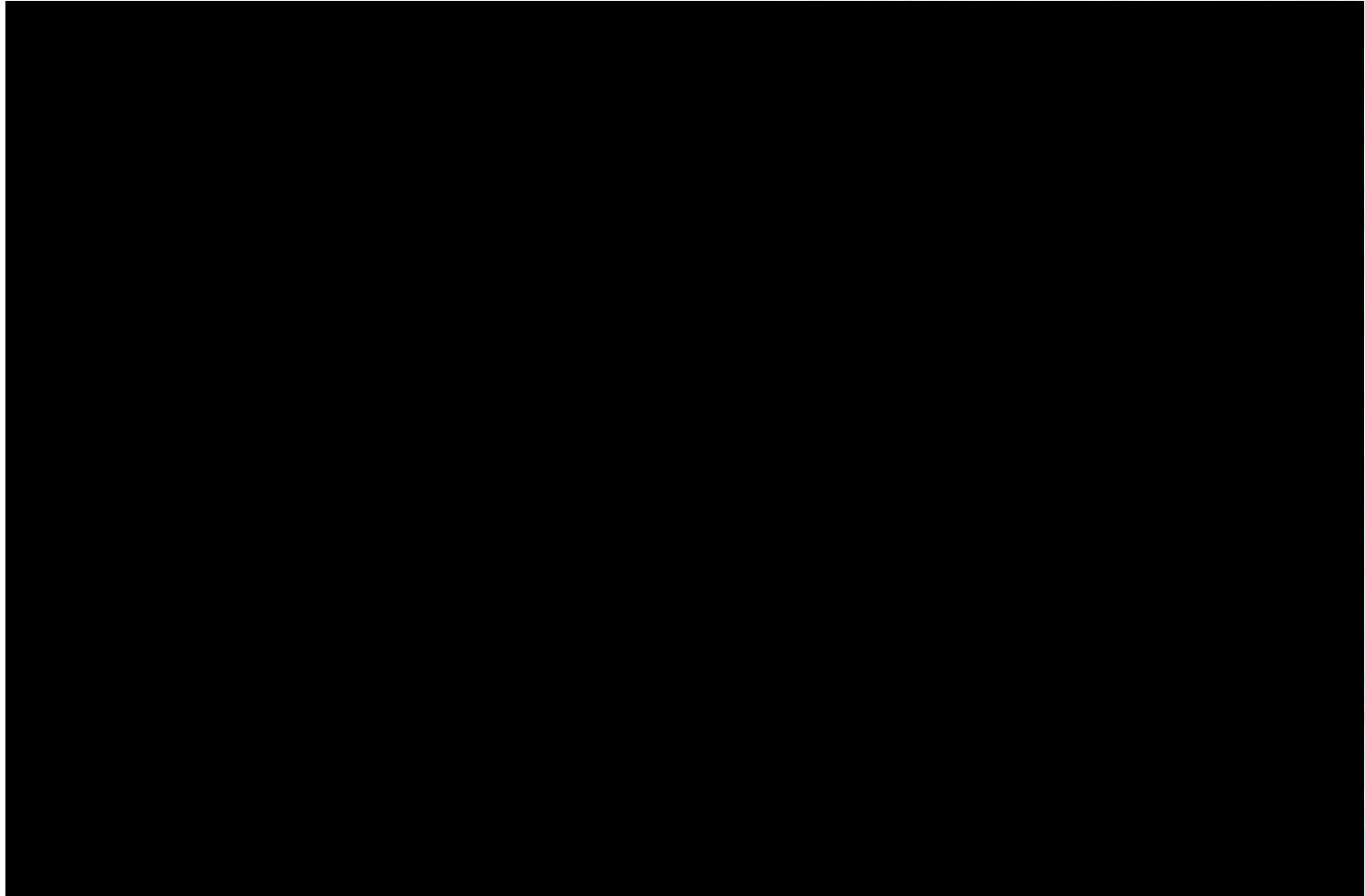
- Harmful substance formed in the atmosphere
- Primary air pollutant reacts with substances normally found in the atmosphere or with other air pollutants.

Air Pollution - 1

Major Classes of Air Pollutants

- **Particulate Material**
- **Nitrogen Oxides**
- **Sulfur Oxides**
- **Carbon Oxides**
- **Hydrocarbons**
- **Ozone**

Air Pollution - 1



<http://youtube.com/watch?v=jSmmmYfO7pU>

Air Pollution - 1

Particulate Material

- Thousands of different solid or liquid particles suspended in air
- Includes: soil particles, soot, lead, asbestos, sea salt, and sulfuric acid droplets

Air Pollution - 1

Particulate Material

Dangerous for two reasons

- 1. May contain materials with toxic or carcinogenic effects**
- 2. Extremely small particles can become lodged in lungs**

Air Pollution - 1

Nitrogen and Sulfur Oxides

- **Nitrogen Oxides**
 - **Gases produced by the chemical interactions between atmospheric nitrogen and oxygen at high temperature**
 - **Problems**
 - **Greenhouse gases**
 - **Cause difficulty breathing**

Air Pollution - 1

Nitrogen and Sulfur Oxides

- **Sulfur Oxides**
 - **Gases produced by the chemical interactions between sulfur and oxygen**
 - **Causes acid precipitation**

END

Air Pollution - 2

Carbon Oxides and Hydrocarbons

Carbon Oxides

- Gases carbon monoxide (CO) and carbon dioxide (CO₂)
- Greenhouse gases

Air Pollution - 2

Carbon Oxides and Hydrocarbons

- **Hydrocarbons**
 - Diverse group of organic compounds that contain only hydrogen and carbon (ex: CH₄ - methane)
 - Some are related to photochemical smog and greenhouse gases

Air Pollution - 2

Ozone

- **Tropospheric Ozone**
 - **Man- made pollutant in the lower atmosphere**
 - **Secondary air pollutant**
 - **Component of photochemical smog**

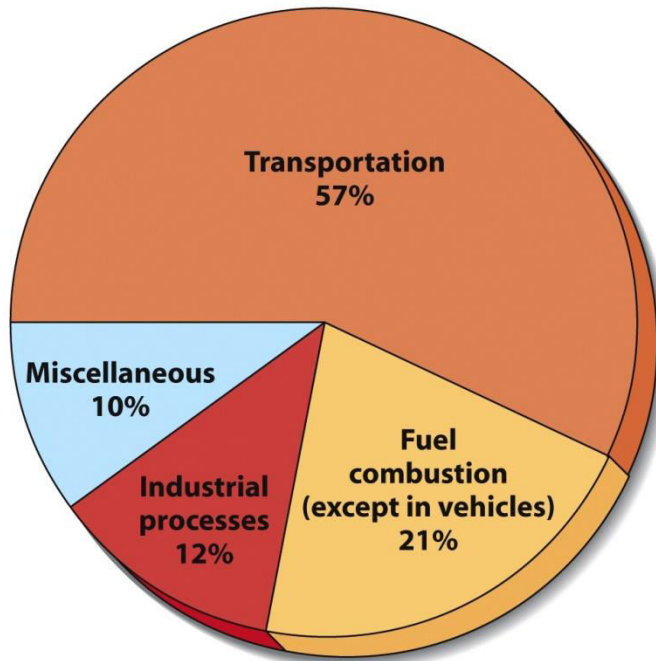
Air Pollution - 2

Ozone

- **Stratospheric Ozone**
 - **Essential component that screens out UV radiation in the upper atmosphere**
 - **Man- made pollutants (ex: CFCs) can destroy it**

Air Pollution - 2

Sources of Outdoor Air Pollution



- Two main sources
 - Transportation
 - Industry
- Intentional forest fires is also high

https://en.wikipedia.org/wiki/Air_pollution

Air Pollution - 2



https://en.wikipedia.org/wiki/Air_pollution

Urban Air Pollution

- **Photochemical Smog**
 - **Brownish-orange haze formed by chemical reactions involving sunlight, particulate susoension, nitrogen oxide, and hydrocarbons**

Air Pollution - 2



<http://youtube.com/watch?v=s2sHlyvNoaE>

Air Pollution - 2

Effects of Air Pollution

- **Low level exposure**
 - **Irritates eyes**
 - **Causes inflammation of respiratory tract**
- **Can develop into chronic respiratory diseases**

END

Light Pollution

- Light pollution, also known as photo pollution
- Excessive, misdirected or obtrusive artificial light
- Major side-effect of urbanization
- It is blamed for compromising health, disrupting ecosystems and spoiling aesthetic environments.

Light Pollution

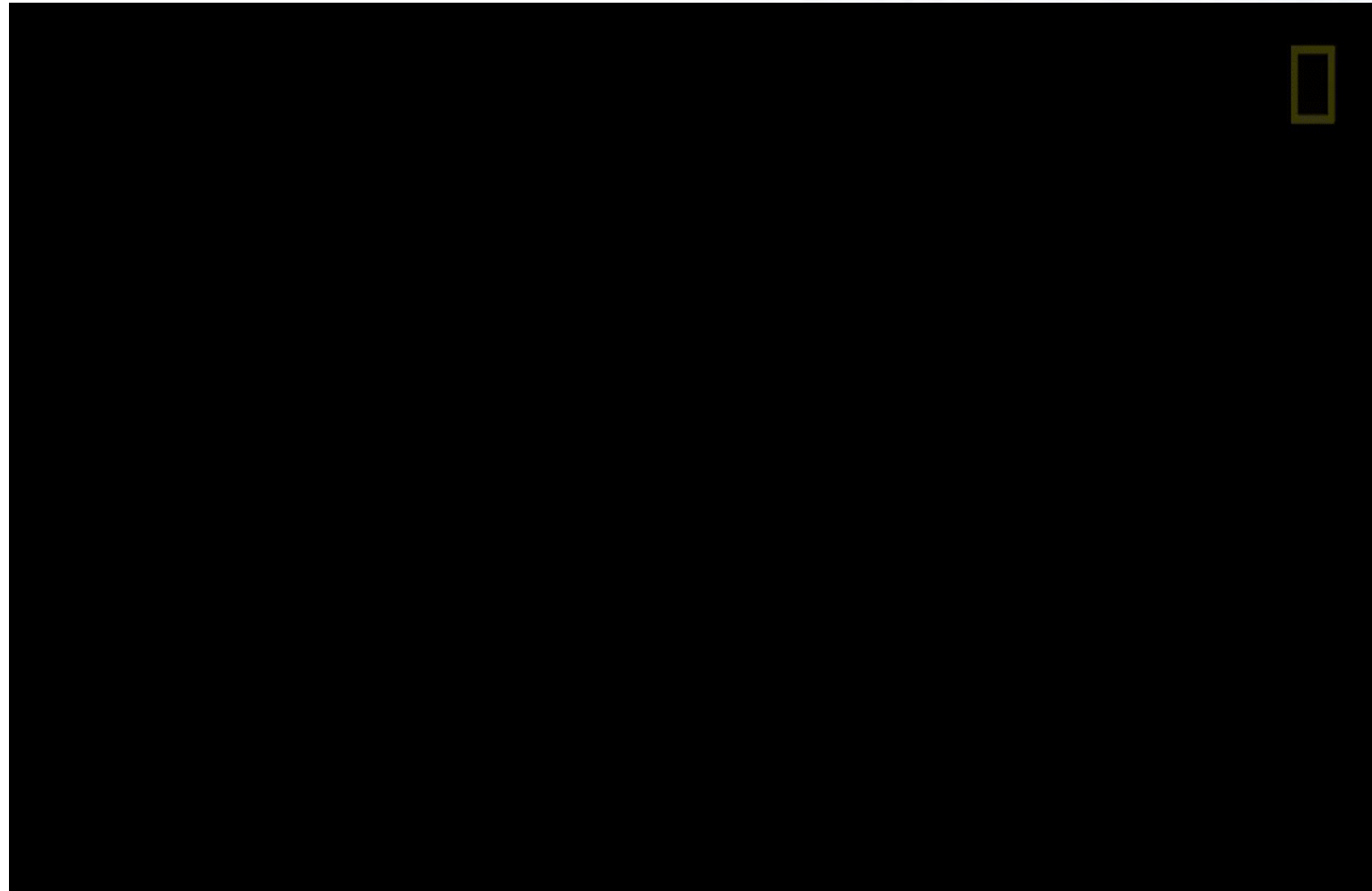


Nanthn Brandt Astrophotography, 2014



BBC, 2017

Light Pollution



https://www.youtube.com/watch?v=V_A78zDBwYE

Light Pollution

Impacts

- The degradation of photic habitat by artificial light.
- The alteration of natural light levels in the outdoor environment.
- The alteration of light levels in the outdoor environment due to man-made sources of light.

Light Pollution

Impacts

- Indoor light pollution is such alteration of light levels in the indoor environment
- Due to sources of light, which compromises human health.
- The introduction by humans, directly or indirectly, of artificial light into the environment.

Light Pollution

Effects

Humans

- Stress
- Headache
- Seasonal Affective Disorder

Ecosystem

- Ecosystem cycles of natural light

Light Pollution

Effects

Animals

- Negative impacts on plant and animal physiology
- Sleep
- Foraging behavior
- Predation
- Reproduction
- Animal navigation
- Alter competitive interactions
- Change predator-prey relations

Light Pollution



https://en.wikipedia.org/wiki/Light_pollution

Example

- Reptiles are strongly affected by light pollution.
- Sea turtles like to build their nests in remote and very dark beaches.
- Coastal lights interfere with their ability to find safe nesting areas for their eggs.

Light Pollution

Effects

- **Artificial lights can confuse the baby turtles and take them to crawl away from the ocean into the roads or communities.**

END

Noise Pollution

- Sound that is unwanted or disrupts one's quality of life is called as noise.
- When there is lot of noise in the environment, it is termed as noise pollution.
- Sound becomes undesirable when it disturbs the normal activities such as working, sleeping, and during conversations.

Noise Pollution

- It is an underrated environmental problem because of the fact that we can't see, smell, or taste it.
- World Health Organization stated that “Noise must be recognized as a major threat to human well-being”

Noise Pollution

Effects

- Noise pollution can damage physiological and psychological health.
- High blood pressure, stress related illness, sleep disruption, hearing loss, and productivity loss
- It can also cause memory loss, severe depression, and panic attacks.

Noise Pollution

Sources

- **Transportation systems are the main source of noise pollution in urban areas.**
- **Construction of buildings, highways, and streets cause a lot of noise, due to the usage of air compressors, bulldozers, loaders, dump trucks, and pavement breakers.**

Noise Pollution

Sources

- **Industrial noise also adds to the already unfavorable state of noise pollution.**
- **Loud speakers, plumbing, boilers, generators, air conditioners, fans, and vacuum cleaners add to the existing noise pollution.**

Noise Pollution

www.sciencetuts.com

NOISE POLLUTION

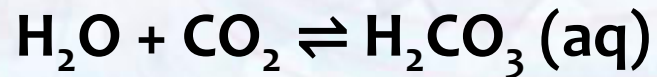
<http://youtube.com/watch?v=DdlmG-jphkM>

Acid Rain

- Form of precipitation that is unusually acidic.
- It has elevated levels of hydrogen ions (low pH).
- Deposition of a mixture
 1. Wet (rain, snow, sleet, fog, cloud water and dew)
 2. Dry (acidifying particles, dust and gases) acidic components

Acid Rain

- Unpolluted rain has an acidic pH, but usually no lower than 5.7.
- This low pH is due to carbonic acid



- Nitric acid is produced as result of lightening.

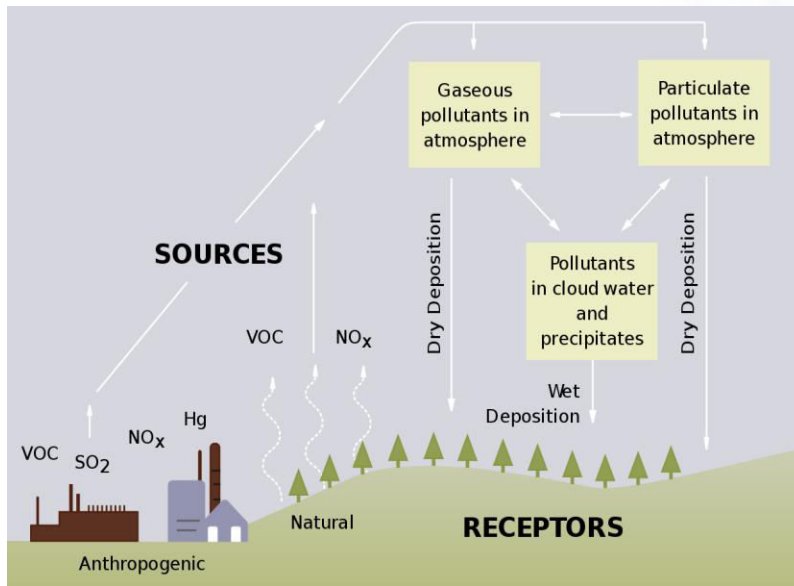
Acid Rain

Causes

- **Natural**
- Volcanic eruption releases SO_2 in air.
- Nitrogen fixation adds NO_2 in air.
- Electrical activity in the atmosphere such as lightning produces NO_2 .
- Acid deposits in glacial ice.

Acid Rain

Causes



https://en.wikipedia.org/wiki/Acid_rain

- **Human Activities**
- **Sulfur and nitrogen compounds from human sources**
- **Electricity generation, factories, and motor vehicles**
- **Greatest contribution by burning coal in power plants**

Acid Rain

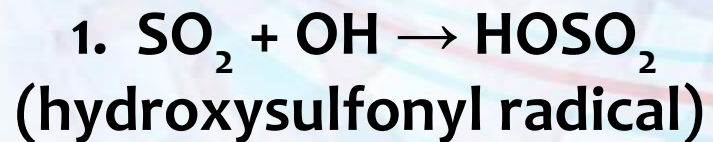
Causes

- **Human Activities**
- **Gases can be carried hundreds of kilometers in the atmosphere before they are converted to acids and deposited.**
- **Taller factories funnels disperse smoke even farther.**

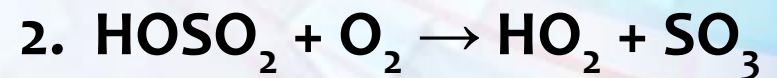
Acid Rain

Chemical Processes

- Gas phase chemistry
- In the gas phase sulfur dioxide is oxidized by reaction with the hydroxyl radical via an intermolecular reaction



which is followed by:



Acid Rain

Chemical Processes

- **Chemistry in cloud droplets**
- **In the presence of water, sulfur trioxide (SO₃) is converted rapidly to sulfuric acid**
$$\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4 (\text{aq})$$
- **Nitrogen dioxide reacts with OH to form nitric acid**
- $\text{NO}_2 + \text{OH} \rightarrow \text{HNO}_3$

Acid Rain



<https://www.youtube.com/watch?v=1PDjVDlrFec>

Acid Rain

Effects

- Not many things can grow in acidic conditions
- Low pH and high aluminum concentrations can damage or kill fish and aquatic populations
- At pH lower than 5 most fish eggs will not hatch and lower pH can kill adult fish.

Acid Rain

Effects

- Soils can be damaged by the hydronium ion, which mobilizes aluminum and encourages leaching of minerals such as magnesium essential for plant life.
- Forests suffer from soil damage, however most food crops are unharmed because the nutrients lost are replaced in fertilizer.

Acid Rain

Effects

- **Monuments made of Calcium Carbonate (limestone and marble) will react with acid rain to form Gypsum**
- **Increases the oxidation rate of metals such as copper and bronze**

Acid Rain

END



https://en.wikipedia.org/wiki/Acid_rain

Disruption of Water Cycle & Coastal Ecosystems

Stages of Water cycle

1. **Evaporation**
2. **Transpiration**
3. **Condensation**
4. **Precipitation**
5. **Surface Run off**
6. **Infiltration**

Disruption of Water Cycle & Coastal Ecosystems

Deforestation

- Directly affects rate of transpiration
- Lesser volume of moisture in air by plants
- Increased rate of erosion due to weak support from root system.
- Making water in seas more muddy.

Disruption of Water Cycle & Coastal Ecosystems

Pollution (Smog & Acid rain)

- Precipitation brings all the toxic chemicals in air back into surface runoff.
- Releasing them into water bodies.
- Infiltration of these chemicals adds toxicity in underground water.
- Toxic chemicals are recycled in water cycle through evaporation.

Disruption of Water Cycle & Coastal Ecosystems

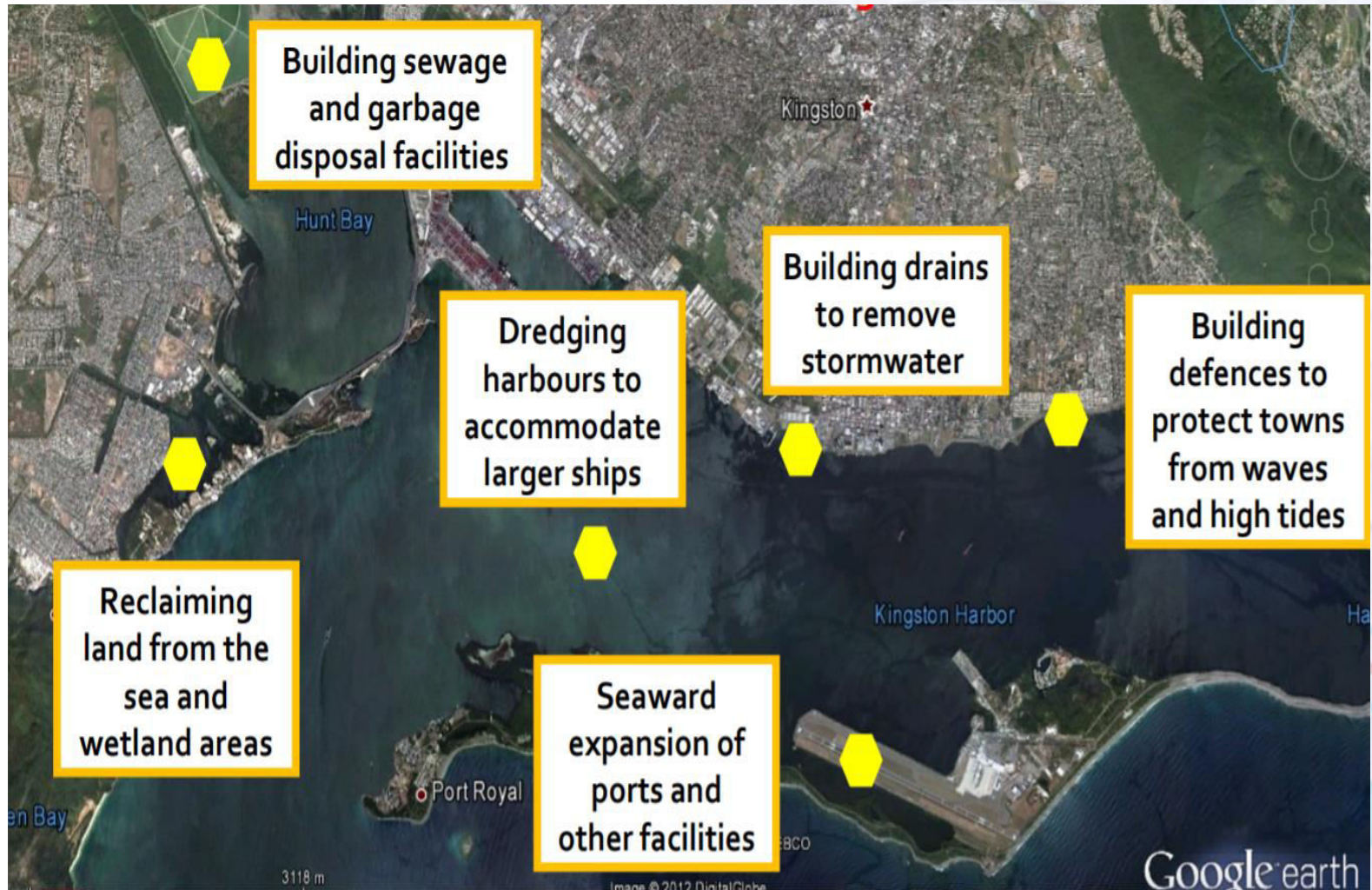
Coastal Ecosystems

- 3 billion people now live in coastal areas
- 1 thousand migrate to coastal areas in China each day
- 14 of the world's 17 largest cities are located along the coast

Disruption of Water Cycle & Coastal Ecosystems



Disruption of Water Cycle & Coastal Ecosystems



Disruption of Water Cycle & Coastal Ecosystems

Coastal Ecosystems

- To reclaim lands, mangroves and other wetland areas are destroyed.
- Coastal defences (dams and dykes) disrupt the natural function of waves and tides.
- Dredging of harbors damage reefs and produce sediments which also smother reefs

Disruption of Water Cycle & Coastal Ecosystems

Coastal Ecosystems

- Drains may deliver excess freshwater, solid waste and pollutants to coastal areas.
- Untreated sewage may be released in the sea causing pollution

Disruption of Water Cycle & Coastal Ecosystems



<https://en.wikipedia.org/wiki/Eutrophication>

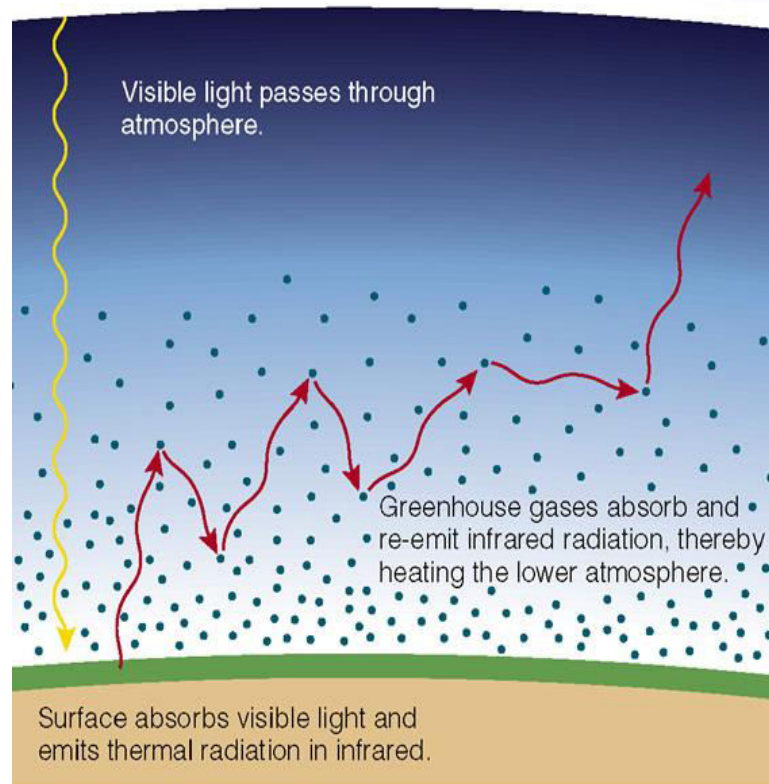
Coastal Ecosystems

- Over enrichment of nutrients in water causes eutrophication.
- Especially in estuaries
- Excessive growth of algae
- Depletion of oxygen in water disturbing flora and fauna

Greenhouse Effect

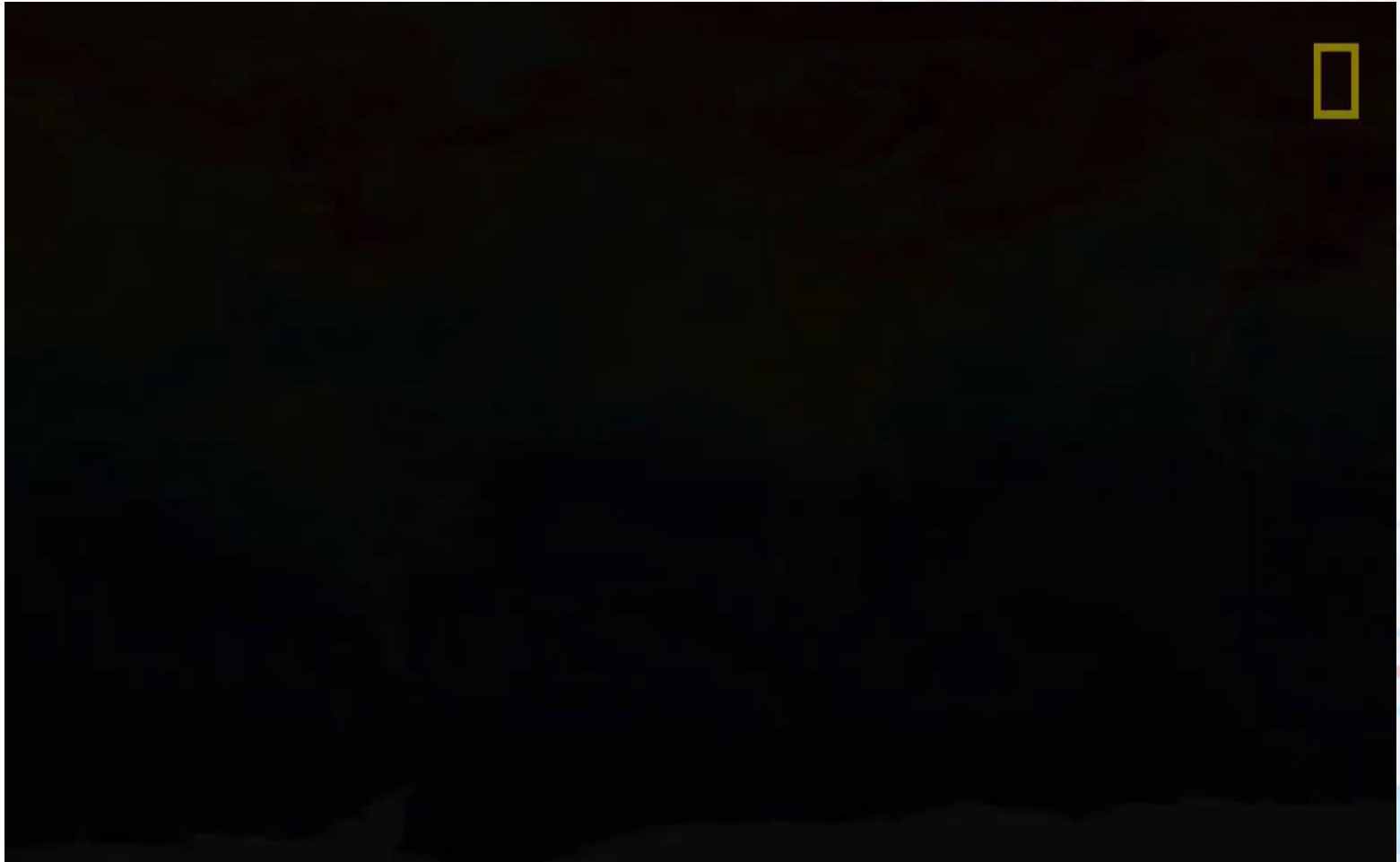
- The ground is heated by visible and (some) infrared light from the Sun.
- The heated surface emits infrared light.
- The majority of Earth's atmosphere (N_2 and O_2) are not good greenhouse gas.
- Carbon dioxide, methane, nitrous oxide and water vapors act as greenhouse gases.

Greenhouse Effect



- The small amount of greenhouse gases (H_2O , CO_2 , methane) traps (absorb and re-emit) the infrared radiation.
- Increasing the temperature of the atmosphere

Greenhouse Effect



https://www.youtube.com/watch?v=G4H1N_yXBIA

Greenhouse Effect

Emission of Gases

– CO₂

- 40% of carbon dioxide emissions stem from the burning of fossil fuels for the purpose of electricity generation.
- 20% and 13 % of carbon dioxide emissions comes from the burning of gasoline in internal-combustion engines of cars and trucks.

Greenhouse Effect

Emission of Gases

– CO₂

- Aviation causes 3.5% of global warming, and the figure could rise to 15% by 2050.
- Buildings structure account for about 12% of carbon dioxide emissions.

Greenhouse Effect

Emission of Gases

- CO_2
 - 25% of carbon emissions by the burning and cutting of about 34 million acres of trees each year

Greenhouse Effect

Emission of Gases

— Water Vapors

- Causes about 36-70% of the greenhouse effect on Earth (not including clouds)

Greenhouse Effect

Emission of Gases

— Methane

- Methane is more than 20 times as effective as CO₂ at trapping heat in the atmosphere.
- 145% increase in the last 100 years
- From sources such as rice paddies, bovine flatulence, bacteria in bogs and fossil fuel production

Greenhouse Effect

Emission of Gases

— Methane

- In flooded fields, anaerobic conditions develop and the organic matter in the soil decomposes.

Greenhouse Effect

Emission of Gases

– Nitrous oxide (N_2O)

- Naturally produced by oceans and rainforests

END

Man-made sources:

- Nylon and nitric acid production, the use of fertilizers in agriculture, cars with catalytic converters and the burning of organic matter

Ozone Depletion

- O_3
- Gas composed of three atoms of oxygen
- Bluish gas that is harmful to breathe.
- Nearly 90% of the Earth's ozone is in the stratosphere.
- Ozone absorbs a band of ultraviolet radiation called UVB.

Ozone Depletion



EGEE 102 - Pisupati

Ozone Depleting Substances

- Halocarbon refrigerants
- Solvents
- Propellants in aerosols
- Chlorofluorocarbons
- Hydro
chlorofluorocarbons
- Methyl bromide
- Carbon tetrachloride
- Methyl chloroform

Ozone Depletion



<http://youtube.com/watch?v=aU6pxSNDPhs>

Ozone Depletion

Effects

- Increased UV penetration
- Skin cancer in humans, basal and squamous cell carcinomas, have been strongly linked to UVB exposure.
- Malignant melanoma, lethal skin cancer related to UVA and UVB
- Cortical cataracts and UVB exposure

Ozone Depletion

Effects

- Increased tropospheric ozone
- Increased production of vitamin D linked with UVB
- Cyanobacteria are sensitive to UV, indirectly affecting crops

- Ban on Use of CFCs since 1996

- Problem will be resolved by half of 21st century

END

Global Warming

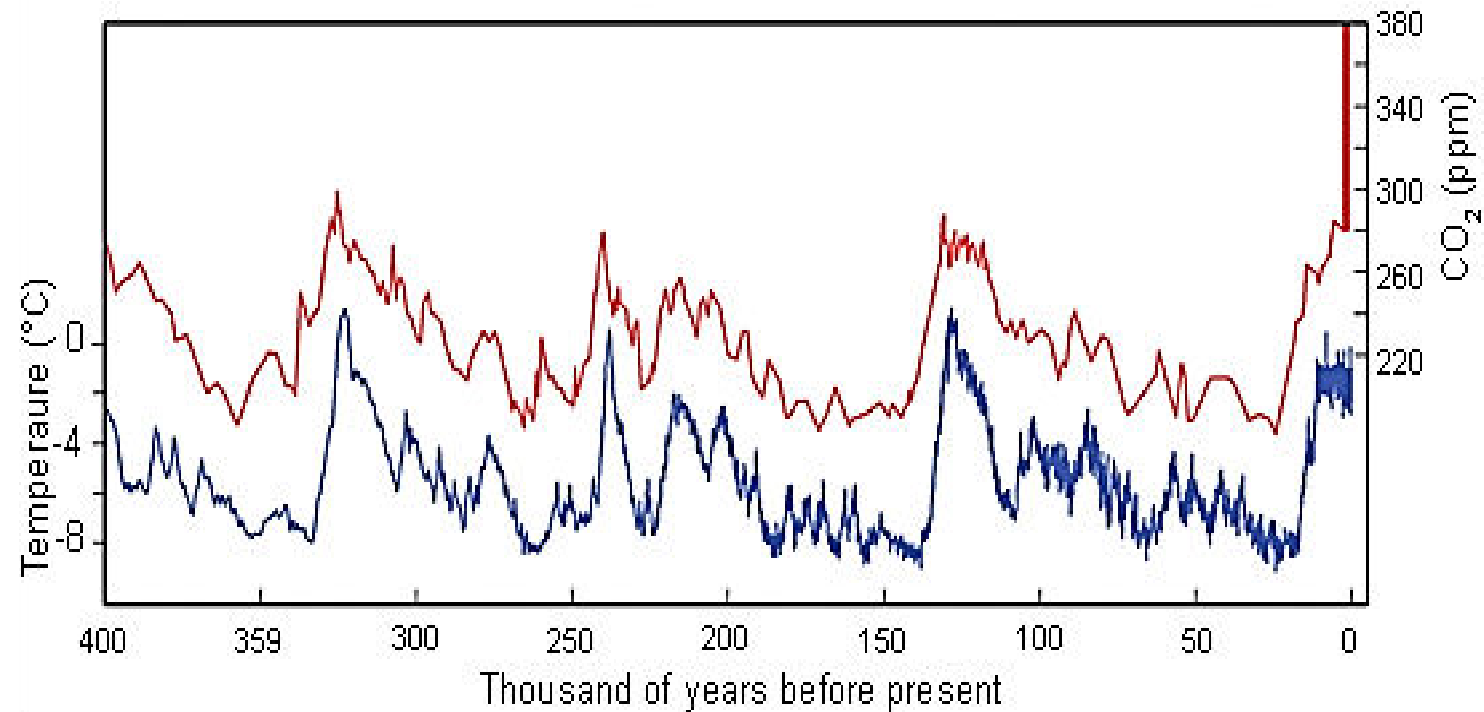
- There is a gradual increase in the average temperature of the Earth's atmosphere in the last 100 years.
- It has risen about 1°C since 1900.
- Sources: Greenhouse effect, pollution, deforestation

Global Warming

- The atmospheric concentration of CO₂ measured from Antarctic ice core data implies that Earth's climate has been pretty stable over the past 400,000 years.
- It also shows a rapid increase of about 30% in the past few centuries.
- From 270 ppm (parts per million) to 370 ppm

Global Warming

Figure 1: Changes in Carbon Dioxide and Temperature in the last 400,000 years

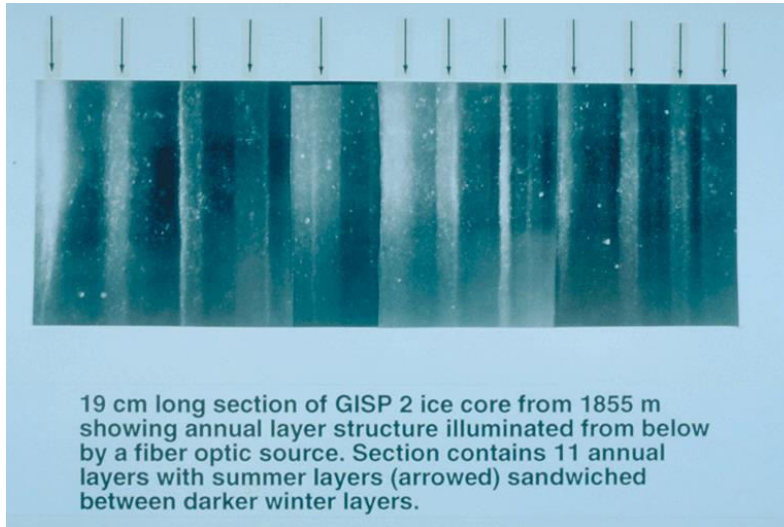


A. V. Fedorov et al. Science 312, 1485 (2006)17. 18.
Reproduced from EPA Climate Change Website

Global Warming

- Precise measurements of atmospheric CO₂ concentration is available only in the last few decade
- Information about atmospheric CO₂ concentration and temperatures in the past can be inferred by several different methods:
 1. Tree-ring
 2. Deep ocean sediment
 3. Ice core records
 4. Corals

Global Warming



NOAA Paleoclimatology Website

- Located high in mountains and in polar ice caps
- Ice has accumulated from snowfall over many millenia.
- Scientists drill through the deep ice to collect ice cores.
- These cores contain dust, air bubbles, or isotopes of oxygen, that can be used to interpret the past climate of that area.

Global Warming

- An increase in atmospheric temperature will lead to the increase in the water vapor content of the troposphere.
- Because water vapor is a strong greenhouse gas, the increase in H₂O vapor in turn causes enhanced greenhouse effect.
- Raising the temperature more

Global Warming

- Higher atmospheric temperature will cause more evaporation of water.
- Which leads to even higher temperature.
- Clouds reflect sunlight
- They decrease the solar energy input into Earth's atmosphere during the day
- But they trap IR radiation from the Earth in night

Global Warming

- CO_2 cycle is thermostat of earth.
- If Earth warms up a bit, then:
- Carbonate minerals formation in the oceans at a higher rate.
- The rate at which the oceans dissolve CO_2 gas increases
- Pulling CO_2 out of the atmosphere

Global Warming

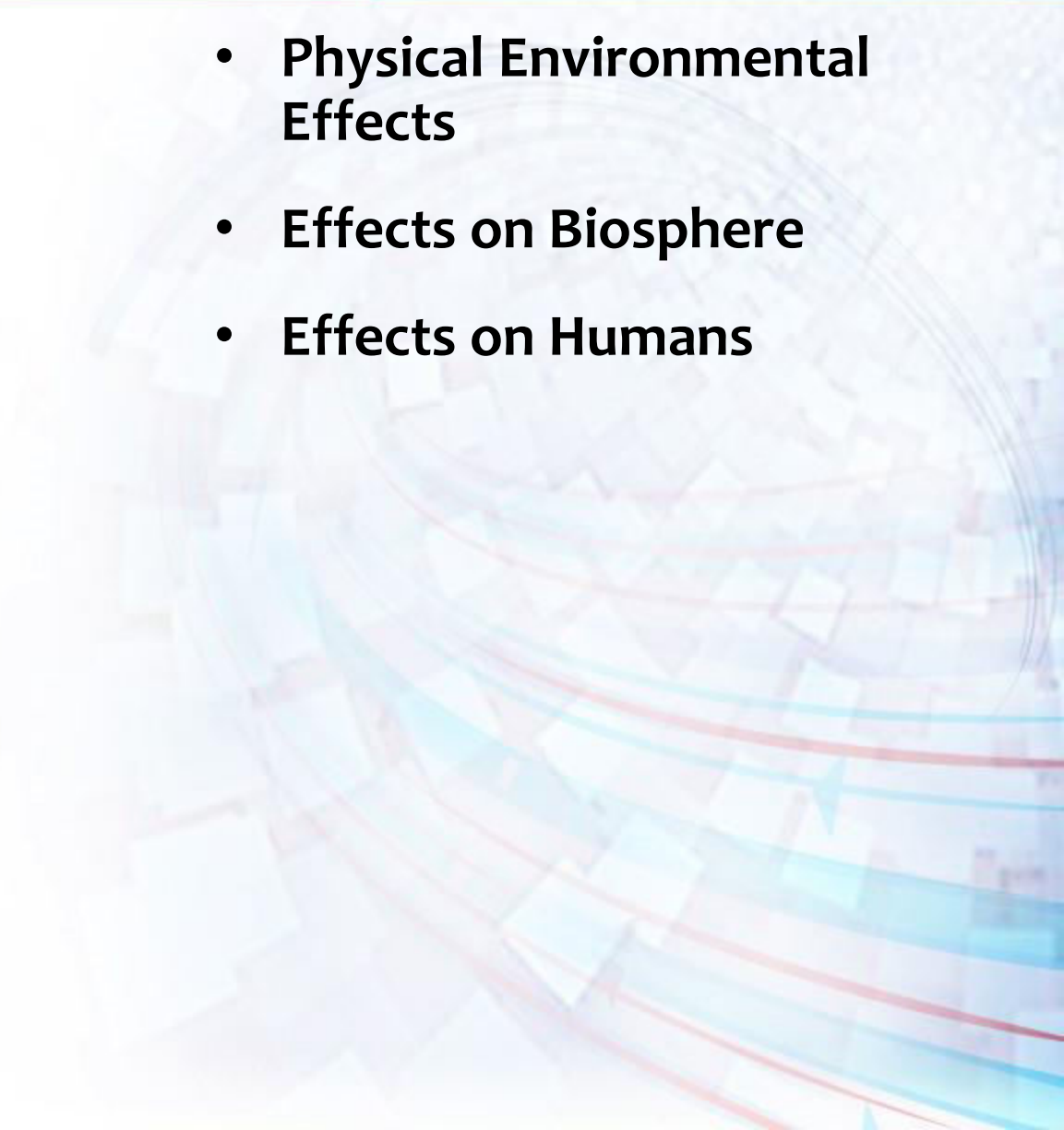
- The reduced atmospheric CO₂ concentration leads to a weakened greenhouse effect
- That counteracts the initial warming and cools the planet back down.

Global Warming

END

- If Earth cools a bit
- Carbonate minerals formation slows down in the oceans.
- The rate at which the oceans dissolve CO_2 gas decreases, allowing the CO_2 released by volcanism to build back up in the atmosphere.
- The increased CO_2 concentration strengthens the greenhouse effect and warms the planet back up

Effects of Global Warming - 1

- **Physical Environmental Effects**
 - **Effects on Biosphere**
 - **Effects on Humans**
- 

Effects of Global Warming - 1



https://www.youtube.com/watch?v=G4H1N_yXBIA

Effects of Global Warming - 1

- Arctic sea ice shrinking

- Decrease in extent – 2.7% per decade since 1978
- Max. area of seasonally frozen ground decreased by 7% in Northern Hemisphere since 1900

Effects of Global Warming - 1

Figure 2.20 Seasonal melting of the Greenland Ice Sheet



Note: The areas in orange/red are the areas where there is seasonal melting at the surface of the ice sheet.

Source: Steffen and Huff 2005

Effects of Global Warming - 1

- Global sea level rise

- Average rate of 1.8 mm per year (1961-2003)
- Likely sea level rise during the 21st century - 5 mm per year
- Low lying areas, small island states may disappear, salt water intrusion into aquifers, floods
- Low-lying coastal cities will be at the forefront of impacts

Effects of Global Warming - 1

- Non-polar glaciers

- Widespread retreat during the 20th century



Mt. Kilimanjaro-- February 17, 1993

Mt. Kilimanjaro-- February 21, 2000

https://en.wikipedia.org/wiki/Global_warming
#Physical_environmental

Effects of Global Warming - 1

- Continental Precipitation

- Significant increase in North & South America, northern Europe, north & central Asia
- Drying in Mediterranean, southern Africa, parts of South Asia

END

Effects of Global Warming - 2

- Extreme Weather

- Frequency of drought, heat waves, typhoons, hurricanes and cyclones increased from 1960 until 2010.
- Extremely wet or dry events within the monsoon period have increased since 1980.

Effects of Global Warming - 2

- **Changes in ocean properties**
 - Increase in atmospheric CO₂ concentrations have led to an increase in dissolved CO₂
 - Increase in ocean acidity
 - Oxygen levels decrease because oxygen is less soluble in warmer water
 - An effect known as ocean deoxygenation

Effects of Global Warming - 2

- **Changes in Ecosystem**
 - Earlier timing of spring events
 - Poleward and upward shifts in plant and animal ranges
 - Expansion of deserts
 - Ocean acidification threatens damage to coral reefs, fisheries, protected species, and other natural resources of value to society.

Effects of Global Warming - 2

- Impacts on Humans

- Changed precipitation patterns
- Warming of low latitudes
- Agricultural production
- Length of growing season will change.
- Droughts, extreme events will decrease yields
- Global economic inequality

Effects of Global Warming - 2

- Impacts on Humans

- Biodiversity shifts – changes in crop varieties
- New pests and diseases
- Food security
- Thermal stress – heat waves, cold spells
- Extreme events, weather disasters (personal injuries, damage & disruption to infrastructure)

Effects of Global Warming - 2

- **Impacts on Humans**
 - **Infectious diseases (malaria, dengue)**
 - **Air pollution - impact of some air pollutants**

Effects of Global Warming - 2

- **Recent Climatic Trends in Pakistan (Department of Meteorology)**
 - **Rise in mean temperature of 0.6-1.0°C in arid coastal areas, arid mountains and hyper arid plains.**
 - **0.5 to 0.7% Increase in solar radiation over southern half of country.**
 - **3-5% decrease in cloud cover in central Pakistan with increase in sunshine hours.**

Effects of Global Warming - 2

- **Recent Climatic Trends in Pakistan**
 - **5% increase in net irrigation water requirement with no change in rainfall.**

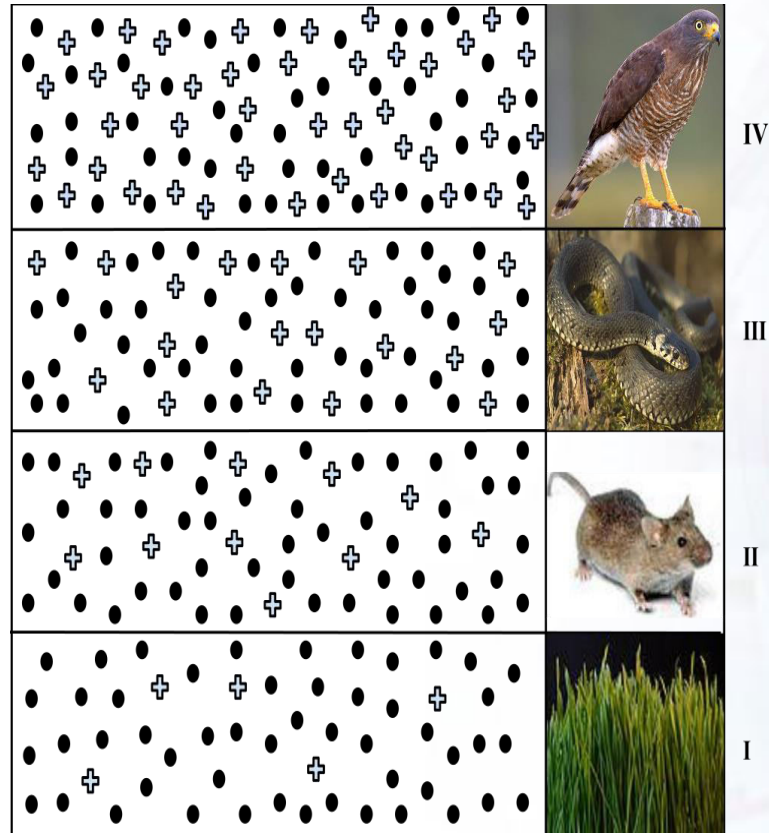
END

Biomagnification

- Increasing concentration of toxic substance
- In the tissues of tolerant organisms at successively higher levels in a food chain
- It occurs as result of three activities.

Biomagnification

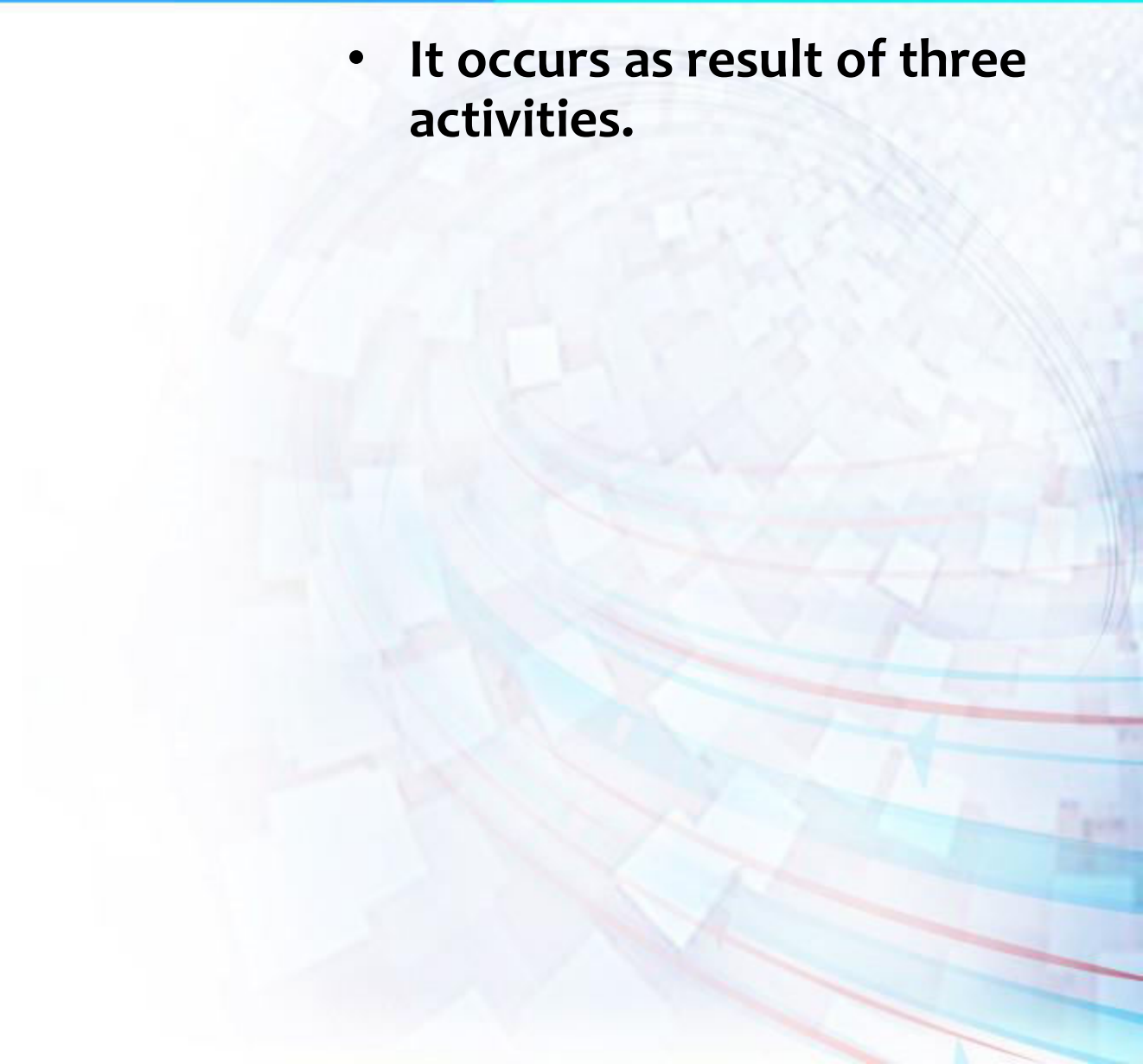
- Concentration of the persistent toxins increases higher up the food chain.



<https://en.wikipedia.org/wiki/Biomagnification>

Biomagnification

- It occurs as result of three activities.



Biomagnification

1. Persistence

- Substance cannot be broken down by environmental processes

2. Food chain energetics

- Substance's concentration increases progressively as it moves up a food chain

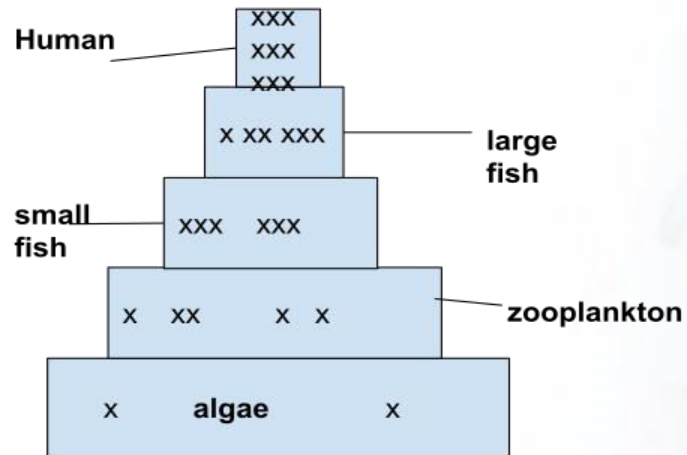
Biomagnification

3. Low or non-existent rate of internal degradation or excretion of the substance

- **often due to water insolubility**

Biomagnification

Higher amount of toxins as trophic level increases in the food chain.



Legend

 = amount of toxins (parts per million)

X = ddt concentration



0.04 ppm



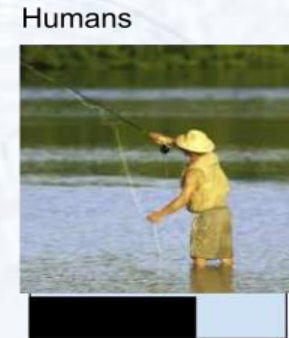
0.23 ppm



2.07 ppm



7.36 ppm



13.55 ppm

<https://en.wikipedia.org/wiki/Biomagnification>

Biomagnification

- **Biomagnification**
- Occurs across trophic levels (food chain).
- **Bioaccumulation**
- Occurs within a trophic level
- Increase in the concentration of a substance in certain tissues of organisms' bodies due to absorption from food and the environment.

Biomagnification

- **Bioconcentration**
- Occurs when uptake from the water is greater than excretion

Biomagnification

Pollutants that undergo biomagnification

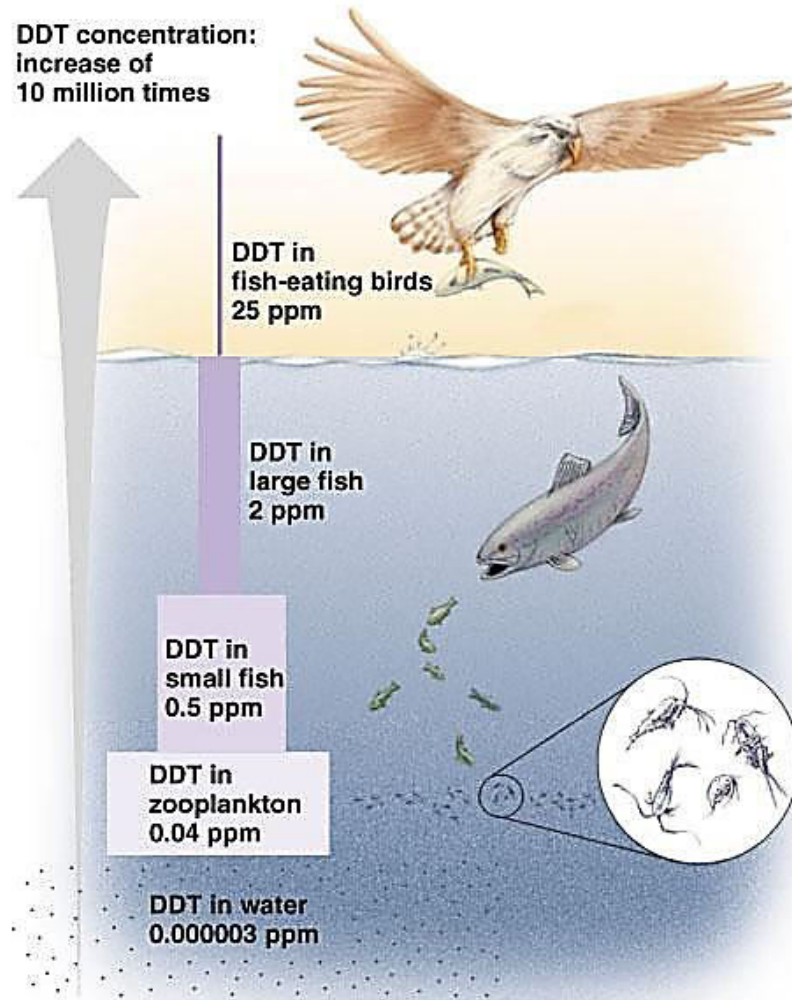
- **DDT**
(dichlorodiphenyltrichloroethane)
- **Hexachlorobenzene (HCB)**
- **PCBs (polychlorinated biphenyls)**
- **Toxaphene**
- **Monomethylmercury**

Biomagnification

DDT (Case Study)

- Insecticide used to control malaria and typhus by killing mosquitoes and lice.
- Commonly used after WWII
- Inventor received Noble Prize
- Overused on crops as a pesticide
- Concentration of DDT increased 10 million times.

Biomagnification



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Biomagnification

DDT (Problems)

- Egg shell thinning
- DDT interferes with metabolism of calcium
- Thin shells in predator birds such as osprey, bald eagles, brown pelicans
- Birds unable to incubate their eggs without breaking them.
- Acts as a hormone disrupter, mimics estrogen

Biomagnification

DDT (Problems)

- Has impacted sex ratio in some birds
- It has a half life of 15 year
- It takes 15yrs for its quantity to be $\frac{1}{2}$ its original
- If we start with 100 kg, we will still have ~ 1 kg after 100 years.

Biomagnification

DDT (Problems)

- Banned in US in 1972
- Still used overseas to prevent malaria especially in Africa
- It saves millions of lives annually in Africa.

END

Type of Wastewater

Wastewater

- Humans capture and divert a lot of the water for agriculture, industrial and municipal use.
- After the water is used, it is returned to the environment in a different condition, usually as dirty water (wastewater).

Type of Wastewater

- **Wastewater is any water that has been affected by human use.**
- **Wastewater is "used water" from these sources:**
- **Domestic, industrial, commercial or agricultural activities**
- **Surface runoff**
- **Storm water**
- **Sewer inflow or sewer infiltration**

Type of Wastewater

Components

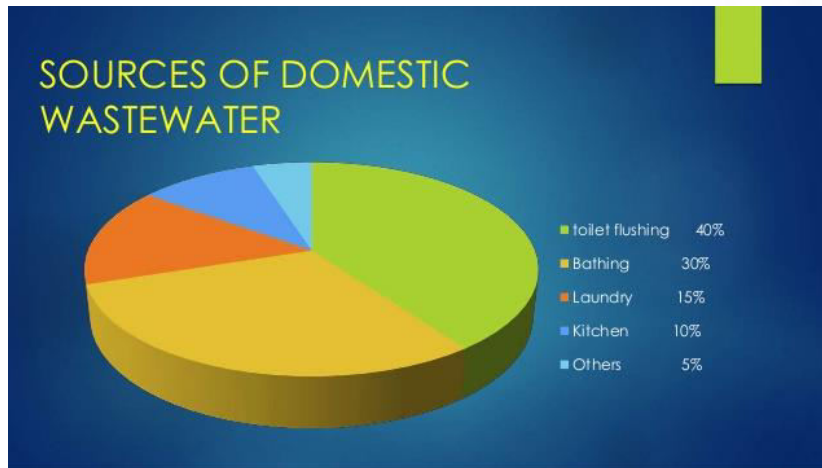
- Pathogens
- Salts
- Acids
- Bases
- Metals
- Toxic inorganic compounds
- Organic matter
- Suspended solids

Type of Wastewater

- **Types**

1. **Domestic wastewater from households**
2. **Municipal wastewater from communities (also called sewage)**
3. **Industrial wastewater from industrial activities**
4. **Agricultural wastewater from agricultural activities**
5. **Storm water**

Type of Wastewater



[https://www.google.com/search?q=domestic waste+water&rlz](https://www.google.com/search?q=domestic+waste+water&rlz)

Domestic wastewater

- Flush toilets
- Sinks
- Dishwashers
- Washing machines
- Bath tubs
- Showers

Type of Wastewater

Municipal wastewater

- Residential
- Shops
- Houses
- Offices
- Schools
- Hospitals
- Motor service stations
- Others

Type of Wastewater



<https://en.wikipedia.org/wiki/Wastewater>

Industrial wastewater

- Huge water use in industrial activities
- Every industry uses water for chemical or physical processes.
- Quantity and quality depends on the type of industry.



Industrial waste is the **waste** produced by **industrial** activity which includes any material that is rendered useless during a manufacturing process such as that of factories, **industries**, mills, and mining operations.

Type of Wastewater



<https://en.wikipedia.org/wiki/Wastewater>

Agricultural wastewater

- Irrigation
- Livestock watering and cleaning
- Aquaculture
- 69% to 90% of global fresh water use in agriculture
- Bulk of it is returned to the soil, waterways or discharged with added nutrients and contaminants

Type of Wastewater

Storm wastewater



<https://en.wikipedia.org/wiki/Wastewater>

- Rainfall that flows over the ground surface
- It is created when rain falls on roads, driveways, parking lots, rooftops and other paved surfaces that do not allow water to soak into the ground.
- Stormwater runoff is the number one cause of stream impairment in urban areas.

Type of Wastewater

Classification on basis of contamination

- Gray water
- Black water

Black Water: This is wastewater that originates from toilet fixtures, dishwashers, and food preparation sinks. It is made up of all the things that you can imagine going down the toilets, bath and sink drains. They include poop, urine, toilet paper and wipes; body cleaning liquids, anal cleansing water and so on. They are known to be highly contaminated with dissolved chemicals, particulate matter and is very pathogenic

Gray Water: This is wastewater that originates from non-toilet and food fixtures such as bathroom sinks, laundry machines, spas, bathtubs and so on. Technically it is sewage that does not contain poop or urine. Gray water is treated very differently from Black water and is usually suitable for re-use.

Yellow Water: This is basically urine collected with specific channels and not contaminated with either black water or gray water.

Type of Wastewater

Gray water

- Generated in households or office buildings from streams other than toilets
- Contains fewer pathogens
- Easier to treat and reuse onsite for:
 - Toilet flushing
 - Landscape or crop irrigation
 - Other non-potable uses

Type of Wastewater

Black water

- **Generated in toilets**
- **Contains pathogens**
- **Contain feces, urine, water and toilet paper from flush toilets**
- **Difficult to treat and recycle**

END

Impacts of Wastewater on Environment

- All types of wastewater act as pollutant.
- Negative impacts on environment
- Industrial effluents may contain acids, alkalis, salts, poisons, oils and in some cases harmful bacteria.
- Mines, especially gold and coal mines, are responsible for large quantities of acid water.

Impacts of Wastewater on Environment

- **Agricultural pesticides, fertilizers and herbicides may wash into rivers and stagnant water bodies.**
- **Sewage as well as domestic and farm wastes were often allowed to pollute rivers and dams.**

Impacts of Wastewater on Environment



<http://youtube.com/watch?v=fzRQkPPkuLU>

Impacts of Wastewater on Environment

Non-persistent (degradable) Pollutants

- Domestic sewage
- Fertilizers
- Some industrial wastes
- Disease causing microbes
- These can be broken down by chemical reactions or by natural bacteria.

Impacts of Wastewater on Environment

Non-persistent (degradable) Pollutants

- Product of break down will be non-polluting substances such as carbon dioxide and nitrogen.
- If the pollution load is high, this process can lead to low oxygen levels and eutrophication.
- This damage is reversible.

Impacts of Wastewater on Environment

Persistent Water pollutants

- Some pesticides (DDT, dieldrin)
- Some leachate components from landfill sites (municipal, industrial)
- Petroleum and petroleum products (oil pollution)
- Heavy metals such as lead, mercury, cadmium

Impacts of Wastewater on Environment

Persistent Water pollutants

- **Radioactive materials such as strontium-90, cesium-137, radium-226, and uranium**
- **High risk to the health of organisms**
- **Disturbance of food chain by affecting certain trophic levels**

Impacts of Wastewater on Environment

Persistent Water pollutants

- This includes substances that degrade very slowly or cannot be broken down at all.
- They may remain in the aquatic environment for years or longer periods of time.
- The damage they cause is either irreversible or repairable only over decades or centuries.

END

Treatment of Wastewater - 1

Septic Tanks

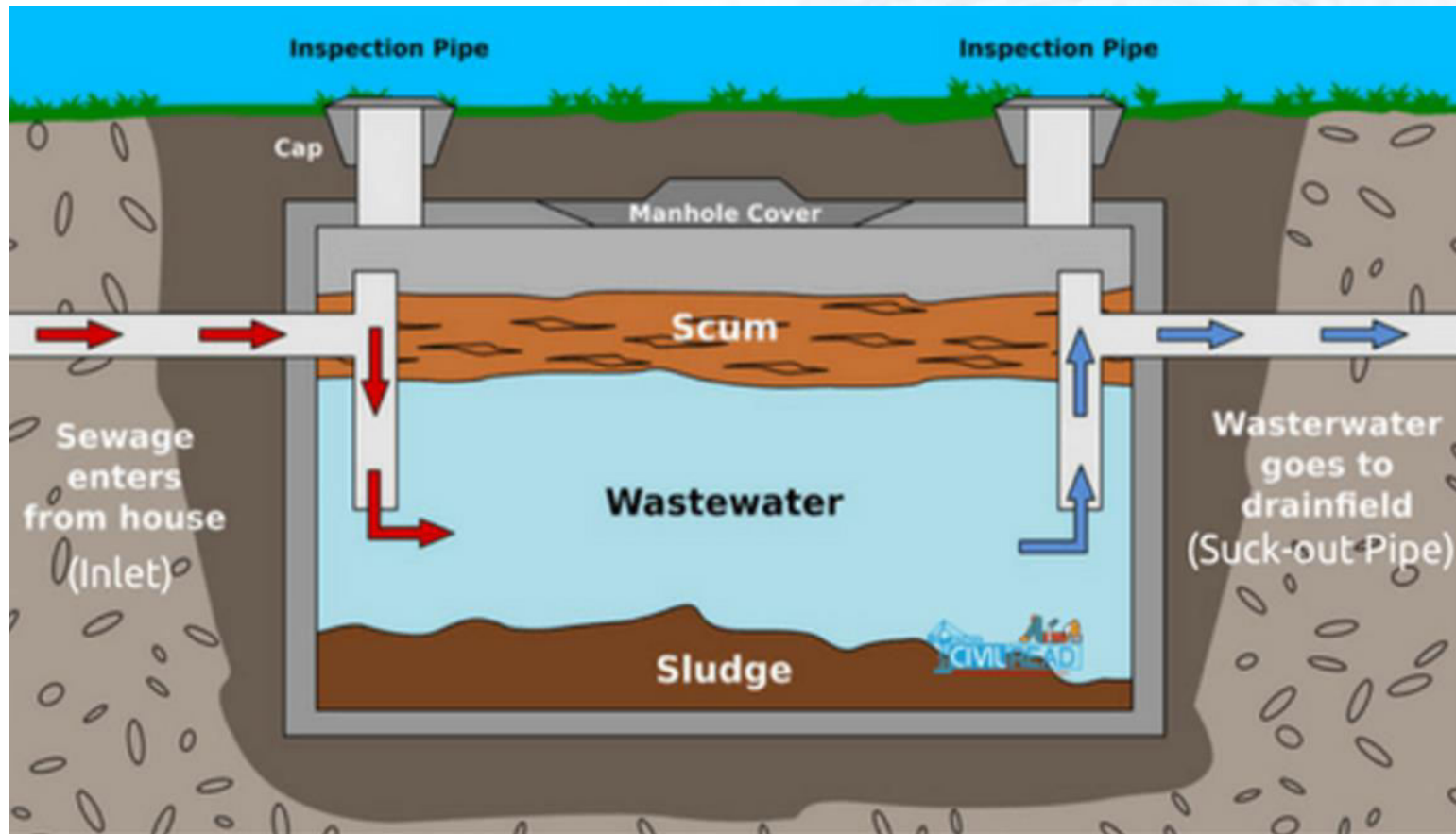
- Bacteria in sewage degrade organic matter
- Tank buried in ground to treat sewage from an individual home
- Wastewater flows into tank
- Solid material (“sludge”) settles to bottom of tank

Treatment of Wastewater - 1

Septic Tanks

- **Clean water flows out of tank into ground through subsurface drains**

Treatment of Wastewater - 1



<http://www.watsantraining.com/blog/106-components-and-design-of-a-septic-tank>

Treatment of Wastewater - 1

Lagoons (Stabilization Ponds)

- 3-5' deep
- Sunlight, algae, oxygen interact to clean wastewater

Conventional Sewage Treatment

- Processes
 1. Collection
 2. Treatment

Treatment of Wastewater - 1

Conventional Sewage Treatment

Collection

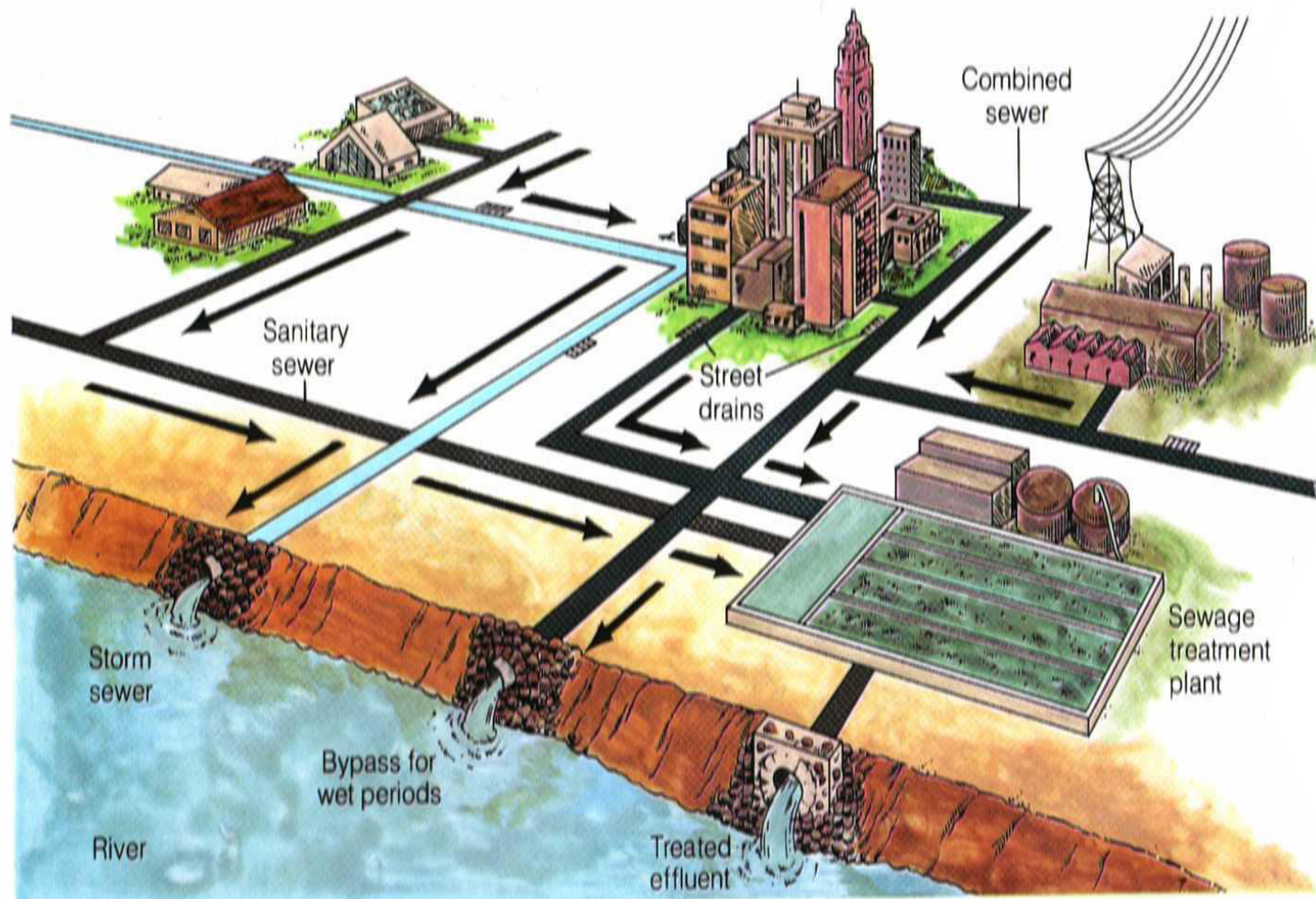
- **Combined Sewers**

Carry stormwater and wastewater

- **Separated Sewers**

1. **Sanitary Sewers**
2. **Storm Sewers**

Treatment of Wastewater - 1



Treatment of Wastewater - 1

Conventional Sewage Treatment

Processes

1. **Primary**
2. **Secondary**
3. **Advanced**

Treatment of Wastewater - 1

Primary Treatment

- Mechanical process
- Screening
- Grinding
- Grit Chamber (heavy particles - “grit”)
- Sedimentation Tank (suspended solids - “sludge”)
- Chlorination of effluent

Treatment of Wastewater - 1

Secondary Treatment

- Biological process
- Uses bacteria to remove organic matter dissolved in water (ODW) and lower BOD

Types

- Trickling Filter
- Activated Sludge

Treatment of Wastewater - 1



<http://youtube.com/watch?v=QoBLswO6xhk>

Treatment of Wastewater - 1

Secondary Treatment

- **Secondary Treatment With Trickling Filters**
- **Effluent leaves sedimentation tank and flows through trickling filters**
- **Bed of stones 3-10' deep through which sewage passes**

Treatment of Wastewater - 1

Secondary Treatment

- Bacteria gather on stones and multiply, consuming ODW
- Cleaner water trickles through pipes at bottom of filter for additional treatment

END

Treatment of Wastewater

Secondary Treatment

- **Secondary Treatment With Activated Sludge**
- **Effluent leaves sedimentation tank and is pumped to an aeration tank**
- **Effluent is mixed with air and sludge loaded with bacteria (“activated sludge”)**

Treatment of Wastewater

Secondary Treatment

- **Sludge contacts with raw sewage, and bacteria in the sludge then decompose the raw sewage, lowering ODW and BOD**

Treatment of Wastewater

Secondary Treatment

- Activated sludge-sewage mixture is called mixed liquor
- Mixed liquor leaves aeration tank
- Flows to another sedimentation tank where suspended solids settle out for reuse as activated sludge

Treatment of Wastewater

Secondary Treatment

- Effluent is chlorinated
- Activated sludge in sedimentation tank is reused

Treatment of Wastewater

Advanced Tertiary Treatment

- Physical and chemical processes that specific pollutants left in wastewater after primary and secondary treatment
- Extremely costly
- It cost twice as much to build a secondary treatment plant

Treatment of Wastewater

Advanced Tertiary Treatment

- Bleaching to remove coloration
- Disinfection to kill pathogens
- Coagulation-sedimentation with alum
- Adsorption using activated charcoal
- Electrodialysis for salt removal

Treatment of Wastewater



<http://youtube.com/watch?v=2oXk2XfDhuY>

Conservation of Endangered Species

- Preserving populations and species in danger of decline or extinction
- There should be something left to be conserved.

Conservation of Endangered Species

Pristine Restoration

- In ecosystems where all species have been effectively removed
- Conservationists might attempt to restore the plants and animals that are the natural inhabitants of the area
- If these species can be identified.

Conservation of Endangered Species

Pristine Restoration

- Focus is on reconstructing the processes that operated in the natural habitat.

Conservation of Endangered Species



a.



b.

The University of Wisconsin–Madison Arboretum has pioneered restoration ecology.

- a. The restoration of the prairie was at an early stage in November 1935.
- b. The prairie as it looks today. This picture was taken at approximately the same location as the 1935 photograph

Conservation of Endangered Species

Removing introduced species

- Sometimes the habitat has been destroyed by a single introduced species.
- In such a case, habitat restoration involves removing the introduced species.
- It is important to act quickly if an introduced species is to be removed.

Conservation of Endangered Species

Cleanup and Rehabilitation

- Habitats are seriously degraded by chemical pollution.
- They cannot be restored until the pollution is cleaned up.
- Example: Nashua River in New England
- Cleaned up after being heavily polluted by dye plants.

Conservation of Endangered Species

Captive Breeding Programs

- Recovery programs
- Particularly focused to on one or a few species
- Sometimes involve direct intervention in natural populations to avoid an immediate threat of extinction.
- It might be successful or failed effort.

Conservation of Endangered Species



State Symbols USA

Captive Breeding Programs

- The Hawaiian goose was practically extinct in the wild
- 12 birds were taken into captivity
- A population of 9000 was released back into the wild
- The rats eat the eggs and the nestlings of the geese.

Conservation of Endangered Species



America Zoo

Captive Breeding Programs

- Pere David's deer was a native species of China
- In 1865, 18 were taken into zoological collections
- Meanwhile it became extinct in the wild
- By 1981 there were 994 individuals scattered through zoological collections.

Conservation of Endangered Species

National Parks

- First the area that is suitable for the creation of a reserve has to be identified and delimited.
- This requires surveys to collect data on key species.
- Property may have to be expropriated.
- A legal framework may need to be set up to control human activities in the area.

Conservation of Endangered Species

National Parks

- If part of the area has been degraded due to bad land use it may need restoring.
- Alien species that have penetrated the area may need excluding or eliminating.

Conservation of Endangered Species

National Parks

- **Constant management will be needed to maintain the habitat of the species being conserved.**
- **This may mean arresting natural succession.**
- **Examples in Pakistan**
 1. **Hingol national park**
 2. **Kirthar national park**
 3. **Chitral national park**

Conservation of Endangered Species

Hingol National Park



Hingol National Park Quick Facts



- The Largest NP of Pakistan
- Area : 619043 ha
- Location: Lasbella (34%), Awaran 58% and Gwadar (8%)

This video is Copyright of Balochistan Forest and Wildlife department.
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Permission to be taken prior using any part of it.

<http://youtube.com/watch?v=56SXB2sScAE>

Evolution

- **Change in the heritable characteristics of biological populations over successive generation**
- **These characteristics are the expressions of genes.**
- **Which are passed on from parent to offspring during reproduction.**
- **Variations come from evolution and causes biodiversity.**

Evolution

History of Evolution

1. **Classical**
2. **Medieval**
3. **Pre-Darwinian**
4. **Darwinian Revolution**
5. **Modern Synthesis**

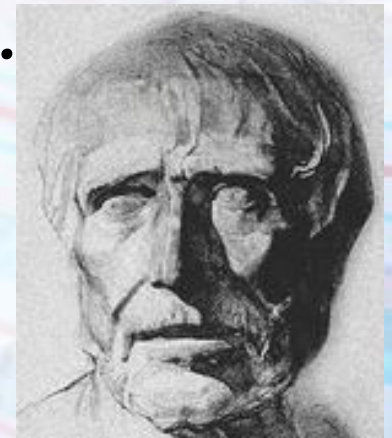
Evolution

History of Evolution

1. Classical

- All living things were related and that they had changed over time.
- One type of organism could descend from another type.

Lucretius



Evolution

History of Evolution

2. Medieval

- Fixed natural possibilities
- Life forms are fixed
- Aristotle added that new types of living things could come to be.

Aristotle



Evolution

History of Evolution

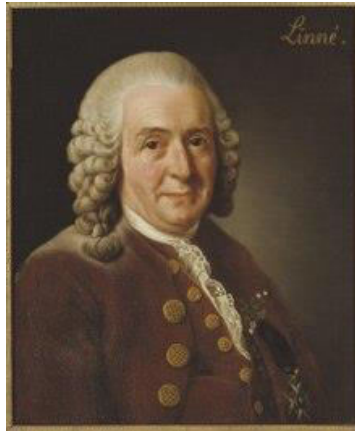
3. Pre-Darwinian Period

- Began in 17th century
- Explanation of natural phenomena in terms of physical laws
- John Ray introduced the term “Species”
- Linnaeus in 1735 explicitly recognized the hierarchical nature of species relationships.

Evolution



Buffon



Linnaeus



Lamarck

https://en.wikipedia.org/wiki/Evolution#History_of_evolutionary_thought

History of Evolution

3. Pre-Darwinian Period

- Introduction of five kingdom classification
- Buffon suggested that species could degenerate into different organisms.
- Erasmus Darwin proposed that all warm-blooded animals could have descended from a single microorganism.

Evolution

History of Evolution

3. Pre-Darwinian Period

- First evolutionary scheme was proposed by Jean-Baptiste Lamarck in 1809.
- He proposed the theory of changes/mutations over the time.
- It was a break from the concept of fixed creation but faced strong resistance.

Evolution

Malthus



Darwin



https://en.wikipedia.org/wiki/Evolution#History_of_evolutionary_thought

History of Evolution

4. Darwinian Revolution

- Malthus (1798) proposed that population growth creates a struggle for resource.
- Darwin influenced by Malthus's ideas, proposed the ground breaking “Theory of natural selection” in 1859.

Evolution

History of Evolution

4. Darwinian Revolution

- Huxley applied Darwin's ideas to humans, using paleontology and comparative anatomy.
- Gregor Mendel proposed the laws of genetics in 1865.
- Weismann made the important distinction between germ cells and somatic cells.

Evolution

History of Evolution

4. Darwinian Revolution

- He demonstrated that heredity passes through the germ line only.
- De Vries proposed that heritable material is in nucleus and transferred to cytoplasm.

Evolution

History of Evolution

5. Modern Synthesis

- It connected natural selection and population genetics, based on Mendelian inheritance, into a unified theory that applied generally to any branch of biology.
- Watson & Crick model of DNA in 1953.

Evolution

- **Evidences of evolution are based upon:**
 1. **Biogeography**
 2. **Paleontology**
 3. **Comparative Anatomy**
 4. **Molecular Biology**

END

Evolution

Biogeography

- Study of the geographic distribution of plants and animals
- Study the life-forms in different parts of the world have distinctive evolutionary histories
- Geographical regions have characteristic biotas.

Evolution

Biogeography

- Closely related taxa tend to be closer together than more distantly related groups.
- Similar environments are found in different areas but the same species may not be found in all places where they could be

Evolution

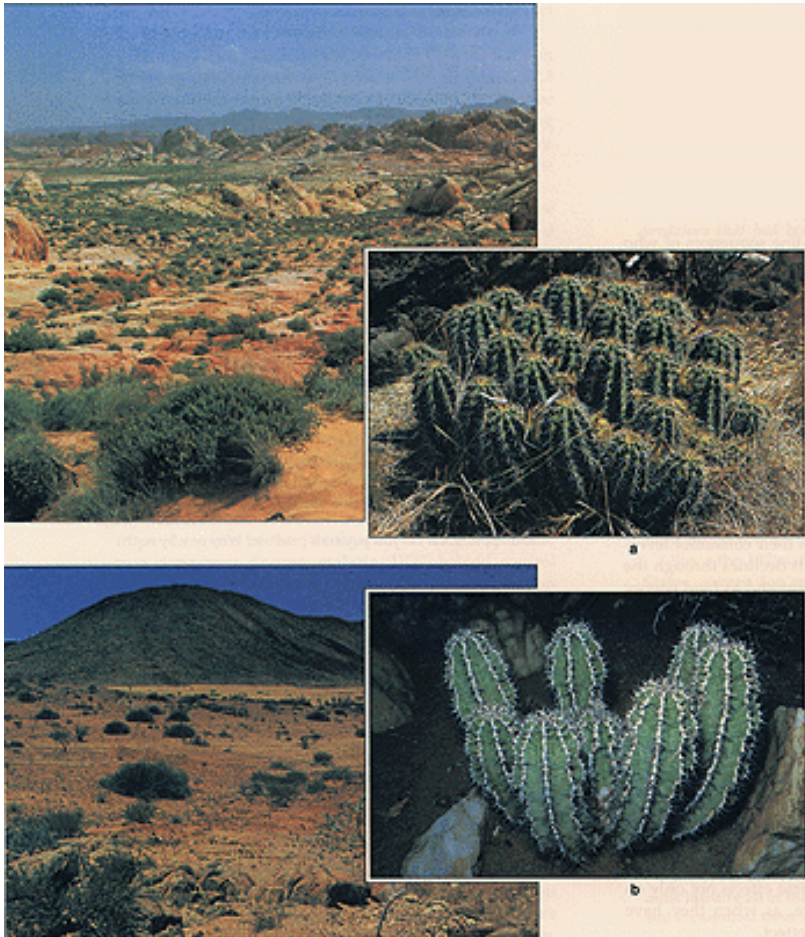
Biogeography

- Not closely related species in similar environments may appear similar due to convergence.

Evidences

1. Regions with identical climate have different floras and faunas (Buffon's Law).

Evolution

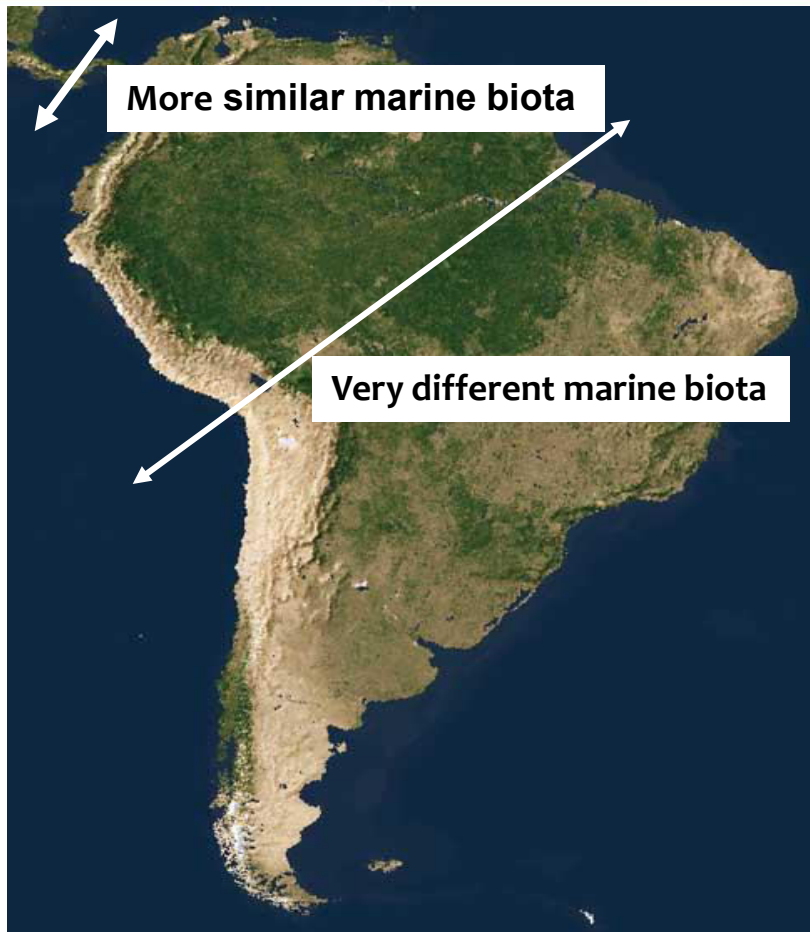


Courtesy of K.J. Sytsma

**Cactaceae in North
American deserts**

**Euphorbiaceae in southern
African deserts**

Evolution



Biogeography

2. Geographic barriers closely associated with breaks between taxonomic groups.

<https://en.wikipedia.org/wiki/Biogeography>

Evolution

Biogeography

3. Within a region, organisms are often closely related

- Across environmental gradients and lower taxonomic groups often show narrower distributions than higher.



https://en.wikipedia.org/wiki/Wallace_Line

Evolution

Biogeography

Disjunct Distribution

- A taxon with a disjunct distribution is one that has two or more groups that are related but considerably separated from each other geographically.
- Barriers

Evolution

Biogeography

Disjunct Distribution

- Physical barriers
- Climatic barriers
- Biological barriers

Evolution

Biogeography

Plate Tectonics

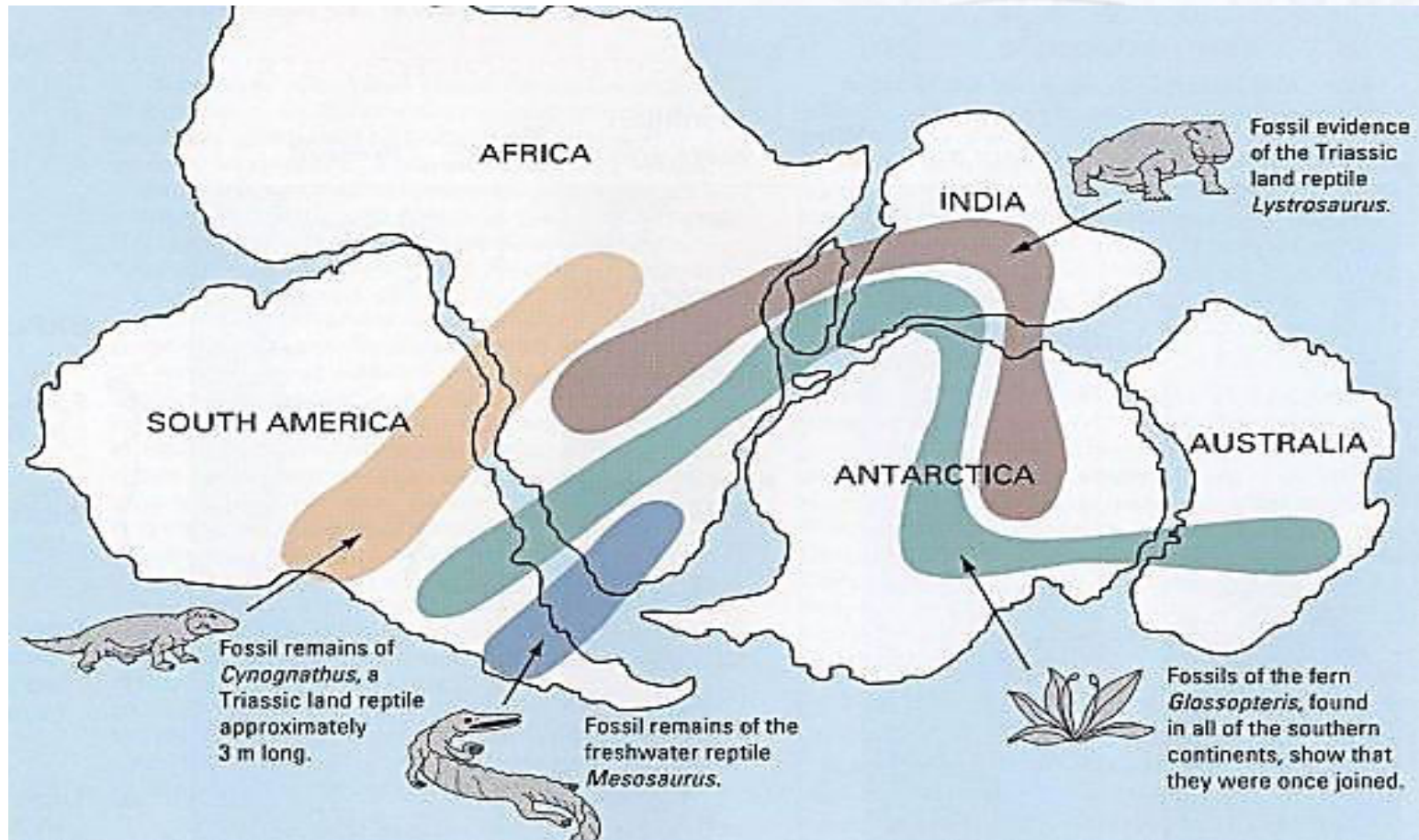
**Glossopteris Permian –
“fern”**

**Mesosaurus – Freshwater
Permian Reptile**

**Cynognathus – Triassic
land reptile**

**Lystrosaurus – Triassic
land reptile**

Evolution



<https://www.google.com/search?q=plate+tectonics+and+dinosaurs+distribution>

Evolution & Paleontology

- Paleontology is the study of fossils.
- It provides some of the most direct evidence for evolution.
- Fossils (to dig) are evidence of plants and animals that existed in the past and have become incorporated into the earth's crust.

Evolution & Paleontology

- **Nicolas Steno (1638-86)**
Father of geology and stratigraphy



https://en.wikipedia.org/wiki/Nicolas_Steno

Evolution & Paleontology

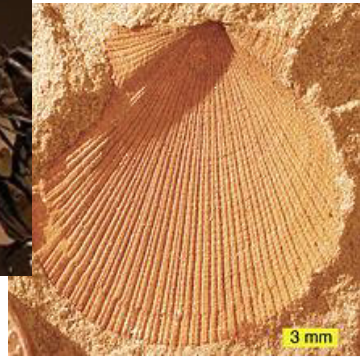
James Hutton

- Evolution of earth must be old .
- Observable processes produce small changes that accumulate over time

William Smith

- Different rock layers contain distinct fossils

Evolution & Paleontology



<https://en.wikipedia.org/wiki/Fossil>

Fossil

- Any preserved remains, impression, or trace of any once-living thing from a past geological age.

Examples: Bones, shells, exoskeletons, stone imprints of animals or microbes, objects preserved in amber, hair, petrified wood, oil, coal, and DNA remnants

Evolution & Paleontology

- Emerging field of geology lead to a new concept of the age of the Earth.
- The history of the earth extends back through vast time periods.
- The processes at work today are the same as those that have been operating throughout the entire history of Earth.

Evolution & Paleontology

These concepts became known as:

Uniformitarianism or Actualism

- Georges Cuvier (1769-1832) proposed that:
- Fossils resemble but are not exactly the same as modern species
- Many past species are extinct

Evolution & Paleontology

- **Mary Anning (1799 – 1847) discovered several marine fossils of Jurassic period.**
- **Darwin proposed his theory of natural selection based upon evidences from fossil record.**
- **Fossils provide the missing links between ancestor and descendants.**

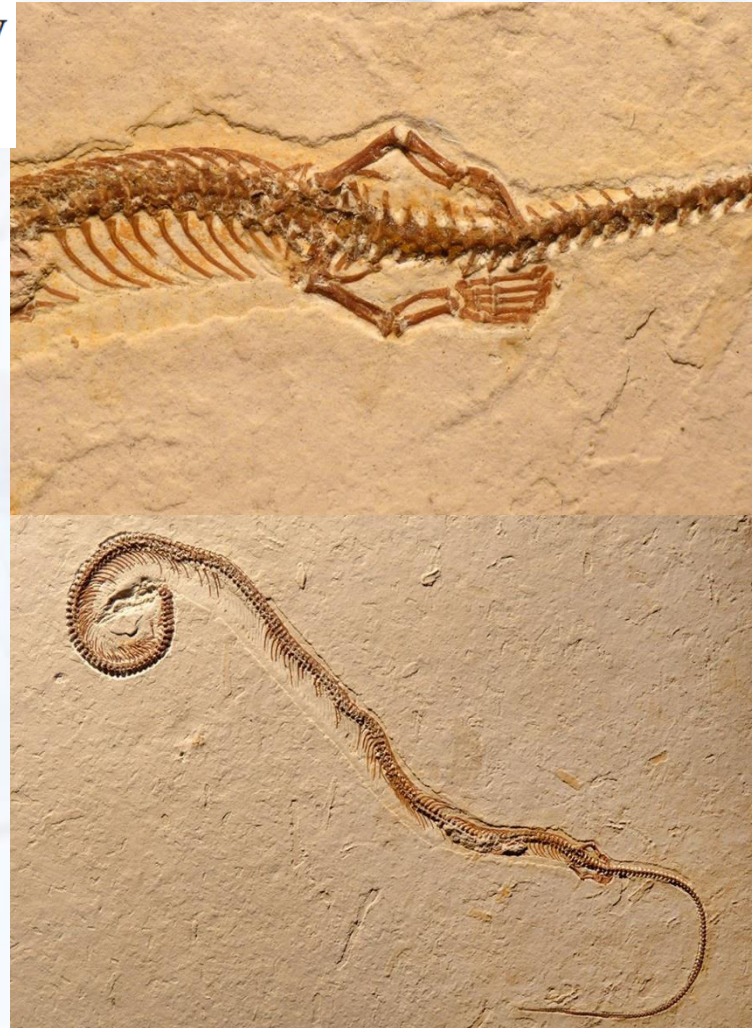
Evolution & Paleontology

A four-legged snake from the Early Cretaceous of Gondwana

David M. Martill,¹ Helmut Tischlinger,² Nicholas R. Longrich³

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Finding Missing Links in the Fossil Record



Evolution & Comparative Anatomy

- **Comparative anatomy is the study of similarities and differences in the anatomy of different species.**
- **It is closely related to evolutionary biology and phylogeny (the evolution of species).**

Evolution & Comparative Anatomy

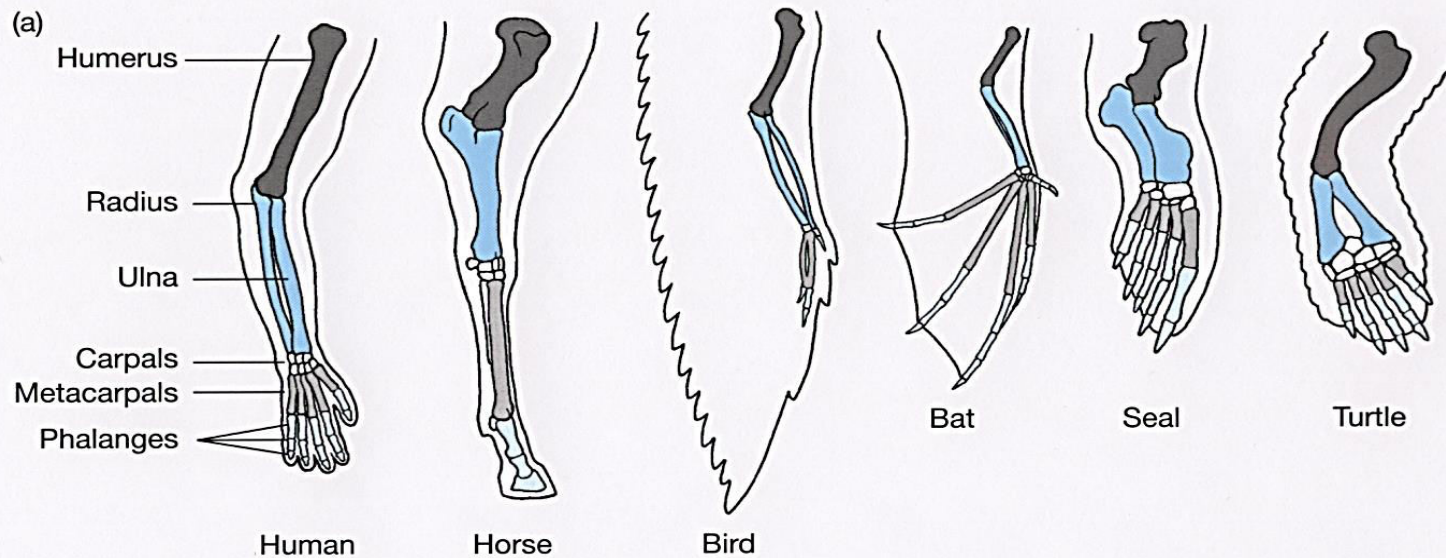
Homology and Common Descent

- Divergent evolution
- Existence of shared ancestry between a pair of structures, or genes, in different taxa
- Once related lineages are reproductively isolated, evolution can lead to modifications of the basic plan.
- Forelimbs of vertebrates

Evolution & Comparative Anatomy

T-6

Fig. 2.1 Structural homologies (a)



Evolutionary Analysis, 2nd Edition
Scott Freeman and Jon C. Herron

© 2001 by Prentice-Hall, Inc.
Pearson Education
Upper Saddle River, NJ 07458

[https://en.wikipedia.org/wiki/Homology_\(biology\)](https://en.wikipedia.org/wiki/Homology_(biology))

Evolution & Comparative Anatomy

Homology and Common Descent

- Evolution can be viewed as a series of bifurcations in a phylogenetic tree.
- All life can be traced back to a common ancestor.

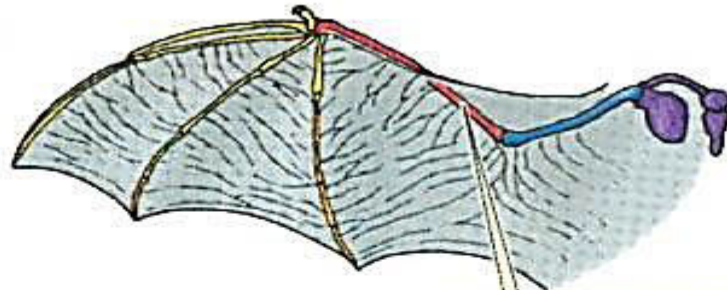
Evolution & Comparative Anatomy

Analogy

- **Convergent Evolution**
- **Independent evolution of similar features in species of different lineages**
- **Creates analogous structures that have similar form or function but were not present in the last common ancestor of those groups**

Evolution & Comparative Anatomy

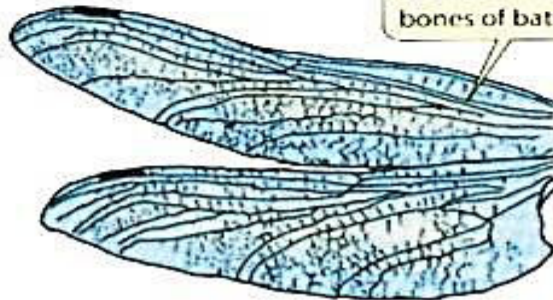
Bat wing



Bird wing



Insect wing



Bones shown in the same color are homologous.

The supports for insect wings are not homologous with the bones of bat and bird wings.

https://en.wikipedia.org/wiki/Convergent_evolution#Flight

Evolution & Comparative Anatomy

Vestigial Organs

- Retention during the process of sexual reproduction of genetically determined structures or attributes
- That have lost some or all of their ancestral function in a given species.

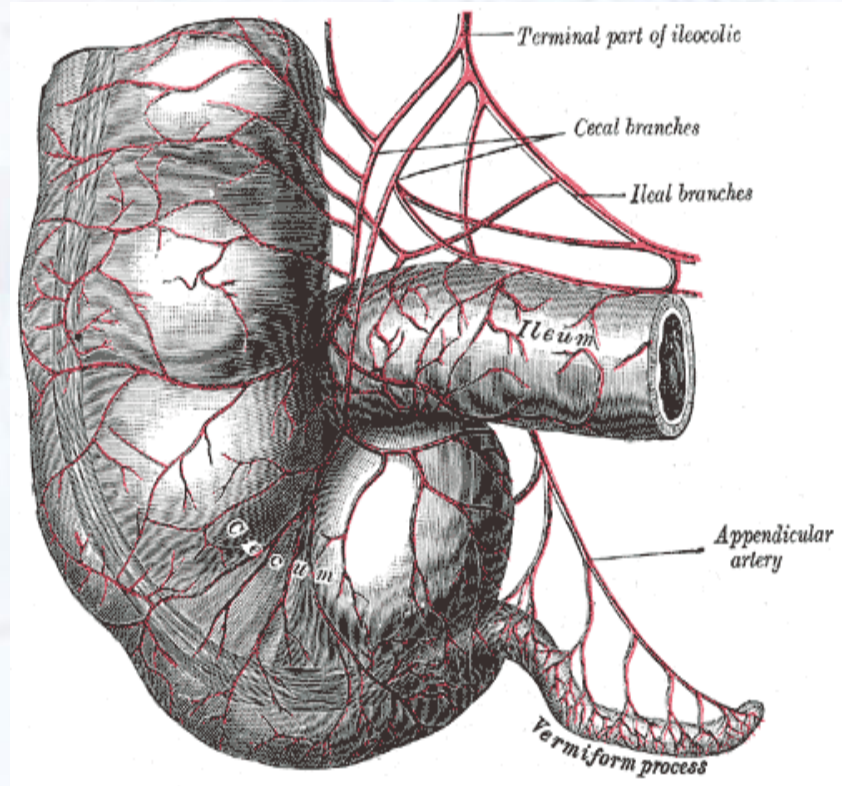
Evolution & Comparative Anatomy

Vestigial Organs

- **Vestigiality is loss of feature due to loss of its value in changed environment.**

Evolution & Comparative Anatomy

- The human vermiform appendix on the vestigial caecum.



https://en.wikipedia.org/wiki/Human_vestigiality#/media/File:Gray536.png

Evolution & Comparative Anatomy

- Vestigial hind limbs in boa constrictor



https://en.wikipedia.org/wiki/Vestigiality#/media/File:Rudimentary_hindlegs_spurs_in_Boa_constrictor_snake.jpg

Evolution & Comparative Anatomy

- Blind cave fish has no eye
- Genes for functional eyes are still in their DNA



END

<https://www.google.com/search?q=blind+cave+fish>

Evolution & Molecular Biology

- **Molecular biology has yielded a wealth of information on evolutionary relationships.**
- **Related animals have similar DNA derived from their common ancestor.**
- **Molecular biologists can estimate the elapsed time since divergence from a common ancestral molecule by looking for:**

Evolution & Molecular Biology

1. **Extraction and analysis of the structure of proteins from animal tissue, and compare the DNA of different animals**
2. **Dissimilarities in the structure of related proteins and DNA**
3. **By assuming relatively constant mutation rates**

Evolution & Molecular Biology

Genetic code of mRNA is universal.

		Second letter				
		U	C	A	G	
First letter	U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA Stop UAG Stop	UGU } Cys UGC } UGA Stop UGG Trp	U C A G
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } CGC } Arg CGA } CGG }	U C A G
	A	AUU } AUC } Ile AUA } AUG Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U C A G
	G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } GGC } Gly GGA } GGG }	U C A G

Third letter

<https://www.google.com/search?q=genetic+code&rlz>

Evolution & Molecular Biology

- Sequencing of DNA and proteins help to accurately locate the similarities between different taxa.
- Transplantation of lost genes and other techniques like cloning help to understand evolution better.

END

Lamarckism

- **Jean Baptiste LAMARCK**
1744 – 1829
- Lamarck rejected fixity
- He proposed a theory of evolution which is attractive.



<https://en.wikipedia.org/wiki/Lamarckism>

Lamarckism

- However, it was eventually rejected because of the way inheritance works.
- But it is still considered a breakthrough.
- Theory of “inheritance of acquired characteristics or soft inheritance”

Lamarckism

- Lamarck noticed that organisms adapted to a particular niche had well developed specialized organs
- For example a carnivore will have long canine teeth to grip its prey.
- Small non-functional organs (vestigial organs)

Lamarckism

- e.g. the appendix in humans, the internal hind limbs of whales and the internal legs of some species of snakes
- Comparative anatomy showed that these organs resembled those which were much more developed, with particular functions, in other species

Lamarckism

The Law of Use and Disuse

- He proposed that if an organ is used a lot it will develop and strengthen.
- If it is not used it will atrophy.
- He called this the law of use and disuse.

Lamarckism

The Inheritance of Acquired Characteristics

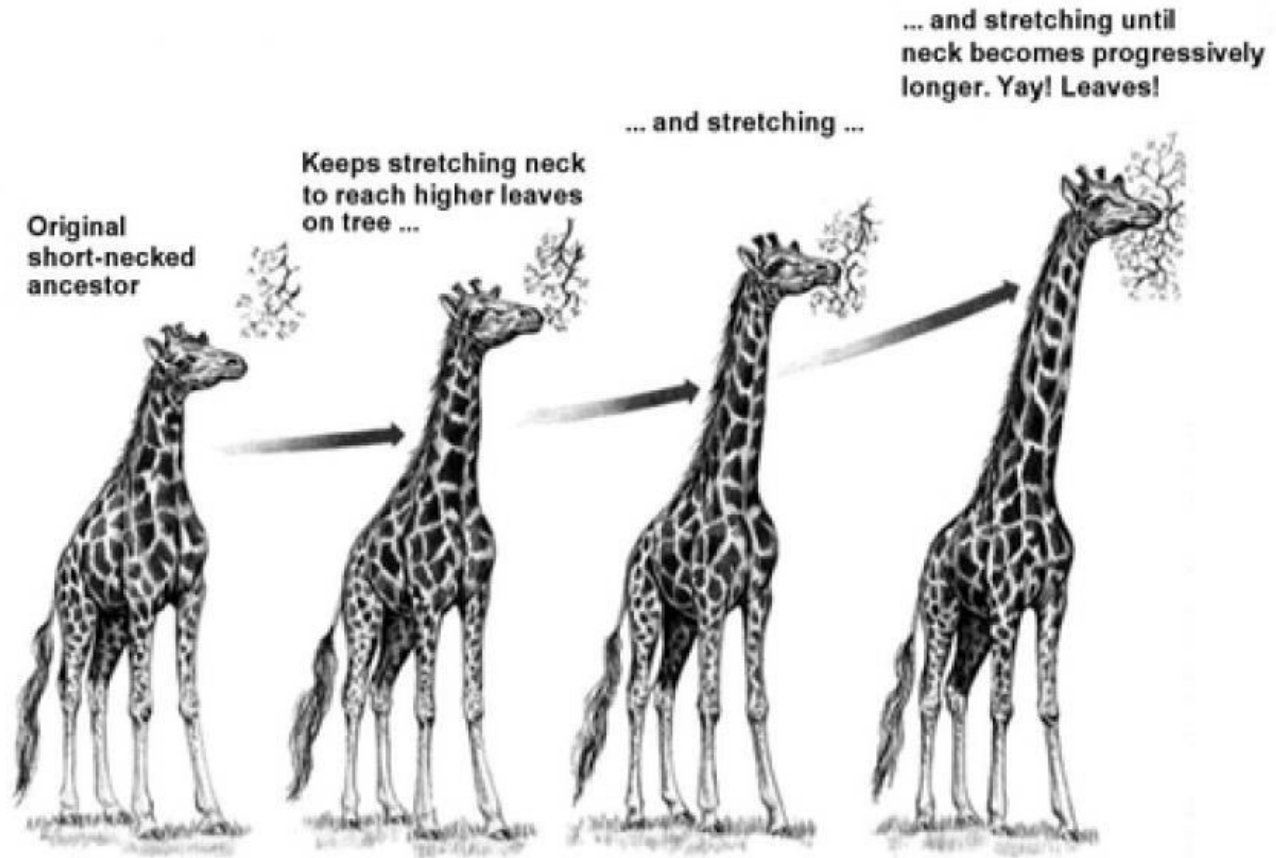
- If an organism developed a characteristic feature through adapting to a new way of life during its lifetime, it would pass this on to its offspring
- The classic example given is that of the giraffe's neck.

Lamarckism

The Inheritance of Acquired Characteristics

- As the giraffe's ancestors searched for a richer food supply they stretched to reach higher branches in trees.
- Thus their stretched bodies were passed onto their offspring.

Lamarckism



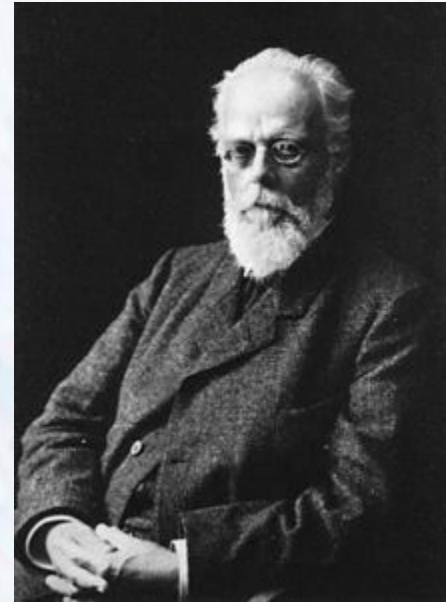
<https://en.wikipedia.org/wiki/Lamarckism>

Lamarck's Giraffe

Lamarckism

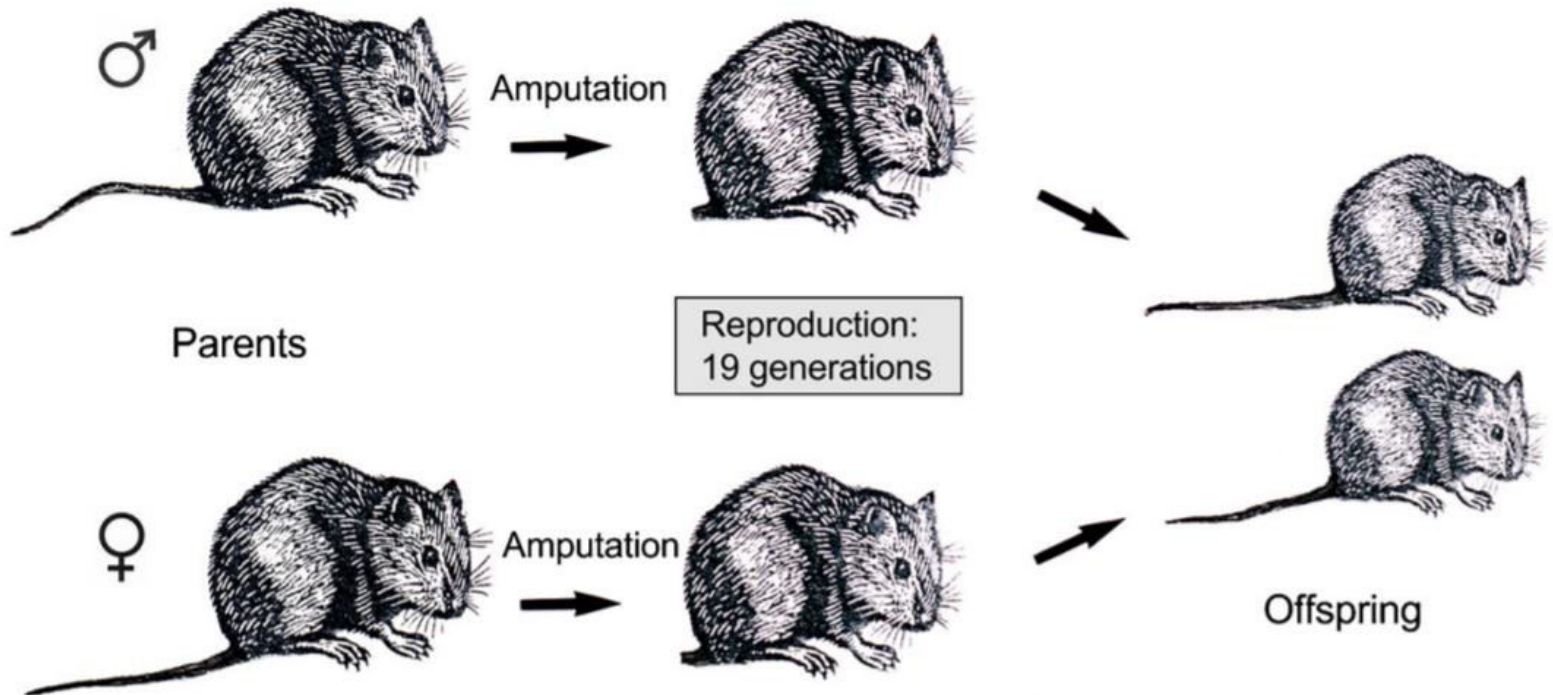
Rejection of Lamarckism

- Weismann's theory of germ plasm (1896–1910)



https://en.wikipedia.org/wiki/August_Weismann

Lamarckism



https://en.wikipedia.org/wiki/August_Weismann

Lamarckism

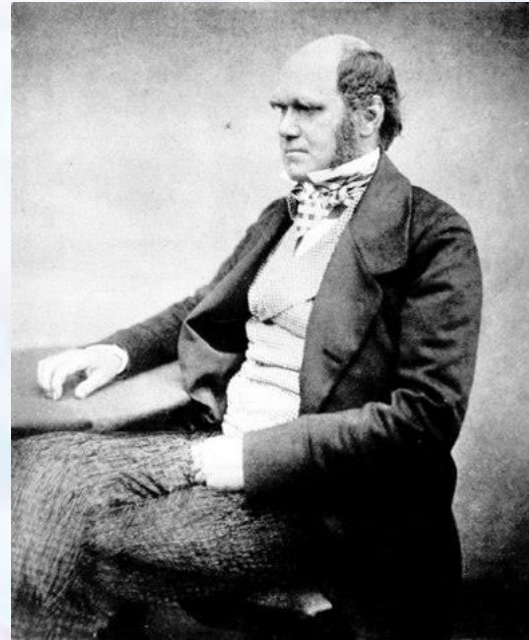
Rejection of Lamarckism

- Characteristics acquired during the lifetime of a parent are not passed onto the offspring.
- Changes only in the germ cells/DNA are transferred to next generation.
- Modern genetics confirmed germ plasm theory.

END

History of Darwin & Voyage

- **Charles Robert Darwin (1809–1882) in England.**



https://en.wikipedia.org/wiki/Natural_selection

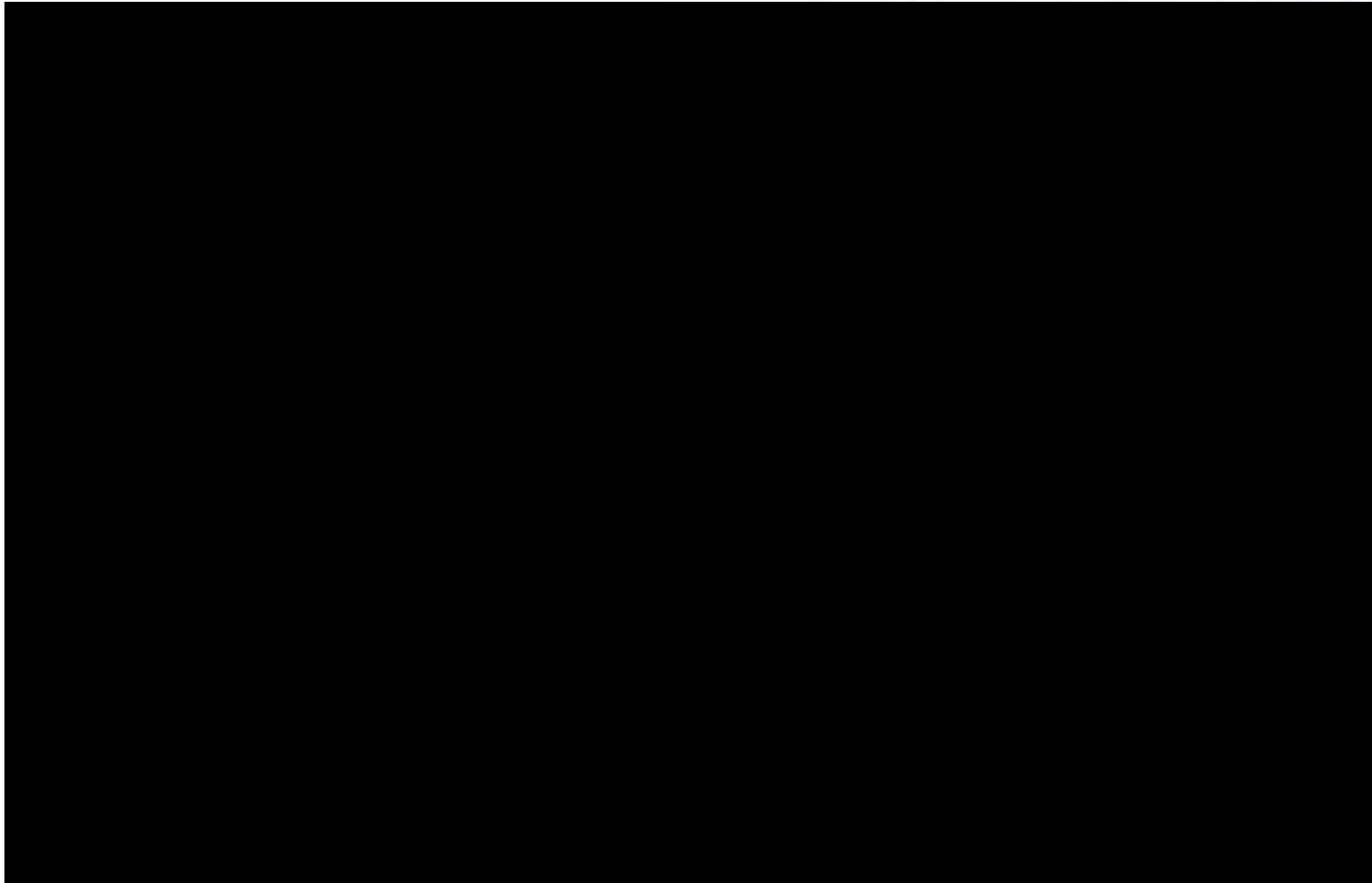
History of Darwin & Voyage

- He joined school of medicine, Scotland in 1825 but he was not interested in being physician.
- Later he joined, Christ's College in Cambridge and graduated with honors in 1831.

History of Darwin & Voyage

- Instead of clergy training, he had been more involved in work of Cambridge's scientists, especially taxonomists.
- He made valuable contributions to beetle taxonomy.
- John Henslow nominated Darwin to serve as a naturalist on a mapping expedition that was to travel around the world.

History of Darwin & Voyage



<http://youtube.com/watch?v=sOcQiljuTdg>

History of Darwin & Voyage

- **HMS Beagle set sail on December 27, 1831 on a 5-year voyage.**
- **Darwin helped with routine seafaring tasks and made numerous collections.**
- **Opportunity to explore tropical rain forests, fossil beds, the volcanic peaks of South America, and the coral atolls of the South Pacific.**

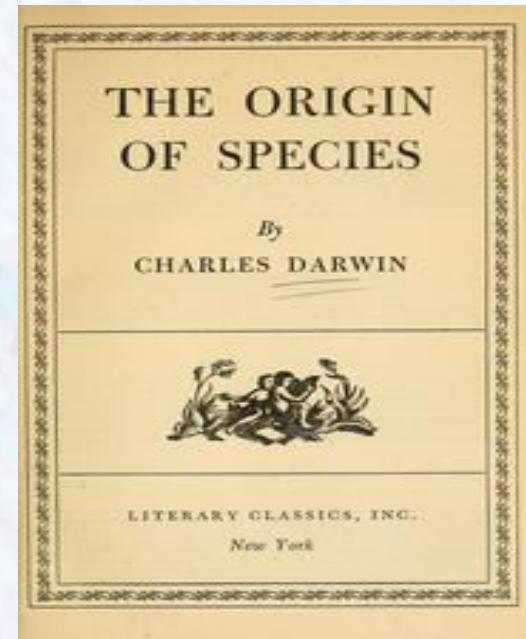
History of Darwin & Voyage

- Most importantly, Darwin spent 5 weeks on the Galápagos Islands.
- A group of volcanic islands 900 km off the coast of Ecuador
- Most revolutionary ideas came from his observations of plant and animal life on these islands.

History of Darwin & Voyage

- On his return, it took "him 22 years to publish his history changing book "Origin of Species" in 1859.

END



Evidences of Theory of Natural Selection - 1

- Darwin based his theory on many solid evidences from his observations and collected samples.
- Three major evidences:
 1. Geology
 2. Fossil Evidences
 3. Galápagos islands

Evidences of Theory of Natural Selection - 1

Geology

- Darwin was impressed by Charles Lyell and James Hutton's ideas of geology.
- Theory of uniformitarianism

Theory states:

- Changes in the earth's crust throughout history have resulted from the action of uniform, continuous processes.

Evidences of Theory of Natural Selection - 1

Geology

- Theory of uniformitarianism planted two ideas in Darwin's mind:
 - (1) The earth could be much older than 6,000 years.
 - (2) If the face of the earth changed gradually over long periods, living forms also change during that time?

Evidences of Theory of Natural Selection - 1

Geology

- Gradual changes over the time with changing environment was foundation of theory of natural selection.

END

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Effort by

AMMAN KHAN